GAGA FOR PROPER SCHEMES OVER THE FARGUES–FONTAINE CURVE

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ABSTRACT. Let X be the schematic Fargues–Fontaine curve and $\mathcal{X}$ its adic counterpart. For every proper finitely presented X-scheme Y, we prove a pseudocoherent GAGA equivalence with its Li–Huerta relative analytic realization over $\mathcal{X}$, and deduce coherent, bounded-derived, perfect, vector-bundle, and proper-pushforward GAGA for Huber analytification. The proof globalizes Li–Huerta's affinoid solid Grauert theorem using a two-annulus Mayer–Vietoris presentation of $\mathcal{X}$, varying-base relative analytic realizations, and a finite fpqc family that reflects pseudocoherence in the boundary-divisor argument.

1. INTRODUCTION

1.1. **Set-up and statements.** Let E be a nonarchimedean local field with residue field $\mathbf{F}_q$, and let F be a complete algebraically closed perfectoid field of characteristic p containing $\mathbf{F}_q$. Write

$$X = X^{\mathrm{sch}}_{F,E}, \qquad \mathcal{X} = X^{\mathrm{ad}}_{F,E},$$

and let $\lambda : \mathcal{X} \to X$ be the canonical morphism of locally ringed spaces. The scheme X is noetherian, regular, separated, integral, and one-dimensional. Kedlaya–Liu's Theorem 4.7.3 identifies the standard schematic coordinate rings as Dedekind domains and, when F is algebraically closed as assumed here, as principal ideal domains. Their analytic period rings are strongly noetherian. Theorems 4.7.4–4.7.5 give the exact tensor equivalence on coherent sheaves induced by λ and the coherent-cohomology comparison [KL19, Theorems 4.7.3–4.7.5 and Remark 4.7.6]. By a *standard affine open* of X we mean one of the affine opens arising from the usual Proj presentation of the schematic curve.

For an analytic stack $\mathcal{Z}$, write $\mathcal{C}_{\square}(\mathcal{Z})$ for its presentable stable ∞-category of solid quasicoherent complexes. For $\mathcal{Z} = \mathcal{X}$ and for every relative analytic realization $T^{\mathrm{An}}_{\mathcal{X}}$ occurring below, write

$$\mathsf{D}^{\square}_{\mathrm{pc}}(\mathcal{Z}) \subset \mathcal{C}_{\square}(\mathcal{Z})$$

for the following full subcategory. After restriction to every rational-affinoid base chart $V = \mathrm{Spa}(R, R^+) \subset \mathcal{X}$, the object must belong to Li–Huerta's nuclear-pseudocompact category on the corresponding fixed-base analytic realization, and these local objects must form a cartesian section for rational localization. Propositions 2.11 and 3.5 prove that this condition is independent of the chosen rational basis and of the bounded-chart presentation and that it is effective for descent. When T is proper and the analytic-locus theorem identifies $T^{\mathrm{An}}_{\mathcal{X}}$ with $T^{\mathrm{Alg}}_{\mathcal{X}}$, this notation is transported across that fixed equivalence. The adjective "solid" always modifies coefficients, complexes, or functors; the geometric objects are analytic stacks. For a noetherian scheme T, write

$$\mathrm{D}_{\mathrm{pc}}(T) = \mathrm{D}^{-}_{\mathrm{coh}}(T).$$

Warning 1.1 (No descended t-structure on the ambient solid quasicoherent category). We do not endow $\mathcal{C}_{\square}(\mathcal{Z})$ with a global t-structure: analytic restriction functors are not generally t-exact [CS26, Warning 12.4]. The coherent t-structure on $\mathsf{D}^{\square}_{\mathrm{pc}}(\mathcal{X})$ is constructed in Proposition 2.12; for a proper relative realization it is transported through stable GAGA in Section 6. All truncations below occur in one of these pseudocoherent structures or in an ordinary derived category.

For a separated finitely presented X-scheme T, Section 3 constructs relative algebraic and analytic realizations

$$T_{\mathcal{X}}^{\mathrm{Alg}}, \qquad T_{\mathcal{X}}^{\mathrm{An}},$$

and the ordinary relative Huber analytification $T_{\mathcal{X}}^{\mathrm{ad}}$. For proper T, Section 6 compares the transported coherent heart and bounded part on $T_{\mathcal{X}}^{\mathrm{An}}$ with the usual Huber-adic categories.

Theorem 1.2 (Stable pseudocoherent GAGA)**.** *Let $f : Y \to X$ be proper and of finite presentation. Relative analytic realization induces an equivalence of stable ∞-categories*

$$\alpha_Y : \mathrm{D}_{\mathrm{pc}}(Y) \xrightarrow{\sim} \mathsf{D}_{\mathrm{pc}}^{\square}(Y_{\mathcal{X}}^{\mathrm{An}}).$$

For every $M \in \mathrm{D}_{\mathrm{pc}}(Y)$, the canonical comparison morphism

$$\alpha_X(Rf_*M) \longrightarrow Rf_*^{\mathrm{An}}(\alpha_Y M)$$

is an equivalence in $\mathcal{C}_{\square}(\mathcal{X})$. Consequently,

$$R\Gamma(Y, M) \simeq R\Gamma(Y_{\mathcal{X}}^{\mathrm{An}}, \alpha_Y M).$$

Theorem 1.3 (Classical GAGA)**.** *Let $f : Y \to X$ be proper and of finite presentation, and let*

$$c_Y : Y_{\mathcal{X}}^{\mathrm{ad}} \longrightarrow Y$$

be its ordinary relative Huber analytification. The stable equivalence of Theorem 1.2 equips $\mathsf{D}_{\mathrm{pc}}^{\square}(Y_{\mathcal{X}}^{\mathrm{An}})$ with the coherent t-structure transported from $\mathrm{D}_{\mathrm{pc}}(Y)$. There is a canonical exact symmetric monoidal equivalence

$$\mathsf{D}_{\mathrm{pc}}^{\square}(Y_{\mathcal{X}}^{\mathrm{An}})^{\heartsuit} \xrightarrow{\sim} \mathrm{Coh}(Y_{\mathcal{X}}^{\mathrm{ad}}),$$

and a canonical exact equivalence of stable categories

$$\mathsf{D}_{\mathrm{pc}}^{\square}(Y_{\mathcal{X}}^{\mathrm{An}})^{b} \xrightarrow{\sim} \mathrm{D}_{\mathrm{coh}}^{b}(Y_{\mathcal{X}}^{\mathrm{ad}}),$$

under which the restriction of α_Y to coherent sheaves and bounded coherent complexes identifies with c_Y^* and Lc_Y^*, respectively. Consequently analytification induces equivalences

$$\mathrm{Coh}(Y) \xrightarrow{\sim} \mathrm{Coh}(Y_{\mathcal{X}}^{\mathrm{ad}}), \qquad \mathrm{D}_{\mathrm{coh}}^{b}(Y) \xrightarrow{\sim} \mathrm{D}_{\mathrm{coh}}^{b}(Y_{\mathcal{X}}^{\mathrm{ad}}),$$

and

$$\mathrm{Perf}(Y) \xrightarrow{\sim} \mathrm{Perf}(Y_{\mathcal{X}}^{\mathrm{ad}}), \qquad \mathrm{Vect}(Y) \xrightarrow{\sim} \mathrm{Vect}(Y_{\mathcal{X}}^{\mathrm{ad}}).$$

For every $\mathcal{F} \in \mathrm{Coh}(Y)$ and every $i \geq 0$, the canonical ordinary adic comparison maps are isomorphisms

$$(R^i f_* \mathcal{F})^{\mathrm{ad}} \xrightarrow{\sim} R^i f_*^{\mathrm{ad}}(\mathcal{F}^{\mathrm{ad}}), \qquad H^i(Y, \mathcal{F}) \xrightarrow{\sim} H^i(Y_{\mathcal{X}}^{\mathrm{ad}}, \mathcal{F}^{\mathrm{ad}}).$$

Li–Huerta states local proper GAGA over one complete Tate Huber pair, and Wang proves a related relative theorem over Fredholm analytic rings [LH26, Theorem A.18][Wan26, Theorem 4.4.4]. Because the posted proof of Theorem A.18 contains a sign error in its boundary-localization telescope, Proposition 4.15 supplies a corrected fixed-base proof over the strongly noetherian Tate bases needed here. Hall provides a general GAGA framework and, in particular, a full-faithfulness result for vector bundles on the Fargues–Fontaine curve [Hal22]. The contribution here is the globalization to proper schemes over the nonaffinoid schematic curve: enhanced base full faithfulness, varying-base analytic realizations, global solid Grauert, and finite fpqc reflection of pseudocoherence.

1.2. **Strategy and organization.** Section 2 enhances coherent GAGA on the base curve to a fully faithful functor

$$\alpha_X : \mathrm{D}_{\mathrm{qc}}(X) \longrightarrow \mathcal{C}_{\square}(\mathcal{X})$$

with colimit-closed essential image. The proof uses compact generation and a finite two-annulus Mayer–Vietoris presentation; the same cover yields a finite fpqc family of affine charts that later reflects pseudocoherence.

Section 3 constructs the relative algebraic, analytic, and Huber realizations and proves relative full faithfulness by module categories and Zariski descent. Section 4 globalizes Li–Huerta's solid Grauert theorem, establishes the Cartier-localization telescope, and proves the fixed-base comparison with Huber analytification used in the corrected local GAGA argument. Section 5 combines these results with the boundary modification and finite fpqc reflection to prove stable GAGA. Section 6 transports the coherent structure and globalizes the comparison of its heart and bounded part with Huber analytification.

2. The base curve and its solid quasicoherent category

2.1. **Finite Mayer–Vietoris descent.**

Definition (Finite Mayer–Vietoris presentation)**.** A *finite Mayer–Vietoris presentation* of an analytic stack $\mathcal{Z}$ consists of a finite rooted tree $\mathcal{T}$ together with an analytic stack $\mathcal{Z}_v$ attached to every vertex v of $\mathcal{T}$, subject to the following conditions.

(i) If r is the root, then $\mathcal{Z}_r = \mathcal{Z}$.

(ii) Every nonterminal vertex v is equipped with an open cover

$$\mathcal{Z}_v = \mathcal{U}_v \cup \mathcal{V}_v,$$

and has three children, labelled by

$$\mathcal{Z}_{v_{\mathcal{U}}} = \mathcal{U}_v, \qquad \mathcal{Z}_{v_{\mathcal{V}}} = \mathcal{V}_v, \qquad \mathcal{Z}_{v_{\mathcal{U}\mathcal{V}}} = \mathcal{U}_v \cap \mathcal{V}_v.$$

(iii) Every terminal vertex v is labelled either by the empty stack or by an analytic affine.

A terminal vertex of $\mathcal{T}$ is called a *leaf* of the presentation, and the corresponding stack $\mathcal{Z}_v$ is also called a leaf. Thus every nonempty leaf is an analytic affine. We call $\mathcal{Z}$ *finitely Mayer–Vietoris* if it admits such a presentation. The rooted tree and all of its chosen open covers are part of the datum.

Let $\mathcal{Y}$ be the adic Fargues–Fontaine precurve, let

$$\pi : \mathcal{Y} \longrightarrow \mathcal{X} = \mathcal{Y}/\varphi^{\mathbf{Z}}$$

be the quotient map, and let $\mathrm{rad} : |\mathcal{Y}| \to (0, \infty)$ be the radius function. Choose compatible pseudo-uniformizers ϖ_E and ϖ_F. For a rational interval $I = [a, b] \subset (0, \infty)$, Fargues–Scholze define the rational affinoid

$$\mathcal{Y}_I = \left\{ x \in \mathcal{Y} : |\varpi_E(x)|^b \leq |[\varpi_F](x)| \leq |\varpi_E(x)|^a \right\} \subset \mathrm{rad}^{-1}(I),$$

and prove

$$\mathrm{rad} \circ \varphi = q \ \mathrm{rad}$$

[FS24, Definition II.1.15 and Proposition II.1.16]. Since $\varphi(\varpi_E) = \varpi_E$ and $\varphi([\varpi_F]) = [\varpi_F]^q$, the same defining inequalities give

$$\varphi(\mathcal{Y}_I) = \mathcal{Y}_{qI}.$$

The inclusion in the radius inverse image is not used as an equality; all intersection calculations below are made from the displayed defining inequalities.

Lemma 2.1 (Calculus of rational annuli)**.** *Let $I = [a, b]$ and $J = [c, d]$ be rational intervals in $(0, \infty)$. Then*

$$\mathcal{Y}_I \cap \mathcal{Y}_J = \begin{cases} \mathcal{Y}_{[\max(a,c),\, \min(b,d)]}, & \max(a, c) \leq \min(b, d), \\ \varnothing, & \max(a, c) > \min(b, d). \end{cases}$$

Moreover, for every $n \in \mathbf{Z}$,

$$\varphi^n(\mathcal{Y}_I) = \mathcal{Y}_{q^n I}.$$

In particular, disjoint intervals define disjoint rational subspaces.

Proof. Imposing simultaneously the four inequalities defining $\mathcal{Y}_I$ and $\mathcal{Y}_J$ gives the inequalities for $[\max(a, c), \min(b, d)]$. If the resulting interval is empty, the rational domain is empty. The Frobenius identity is the iterated form of Fargues–Scholze's normalization. The final assertion follows. □

Lemma 2.2 (Open subsets in the Fargues–Fontaine quotient)**.** *Let $U \subset \mathcal{Y}$ be an open analytic subspace satisfying*

$$U \cap \varphi^n(U) = \varnothing \qquad (n \in \mathbf{Z} \setminus \{0\}).$$

Then the restriction $\pi|_U : U \to \mathcal{X}$ is an open immersion.

Proof. Definition II.1.15 and Proposition II.1.16 of Fargues–Scholze construct $\mathcal{X}$ as the analytic quotient of the free, totally discontinuous $\varphi^{\mathbf{Z}}$-action on $\mathcal{Y}$ [FS24, Definition II.1.15 and Proposition II.1.16]. Put

$$U^{\mathrm{sat}} = \bigcup_{n \in \mathbf{Z}} \varphi^n(U).$$

The assumed disjointness identifies this union with the coproduct $\coprod_n \varphi^n(U)$, so it is open, and

$$\pi^{-1}(\pi(U)) = U^{\mathrm{sat}}.$$

The quotient topology makes $\pi(U)$ open and identifies $|U|$ homeomorphically with it. Restricting the action groupoid

$$\mathcal{Y} \times_{\mathcal{X}} \mathcal{Y} \rightrightarrows \mathcal{Y}$$

to U^{sat} and using U as a fundamental domain gives the trivial groupoid $U \rightrightarrows U$. Effective descent of the analytic structure along the quotient groupoid therefore identifies U with the open analytic subspace of $\mathcal{X}$ supported on $\pi(U)$. □

Proposition 2.3 (A two-annulus presentation)**.** *Choose rational numbers*

$$1 < a' < a < b < b' < q.$$

Put

$$I_0 = [b/q, a], \qquad I_1 = [a', b'], \qquad \mathcal{U} = \pi(\mathcal{Y}_{I_0}), \qquad \mathcal{V} = \pi(\mathcal{Y}_{I_1}).$$

Then π restricts to open immersions on $\mathcal{Y}_{I_0}$ and $\mathcal{Y}_{I_1}$, the two images cover $\mathcal{X}$, and

$$\mathcal{U} \cap \mathcal{V} \simeq \mathcal{Y}_{[a',a]} \amalg \mathcal{Y}_{[b,b']}.$$

In the second component one uses φ to identify the representative in $\mathcal{Y}_{I_0}$ with the interval $[b, b'] \subset I_1$. In particular, $\mathcal{X}$ is finitely Mayer–Vietoris.

Proof. The multiplicative widths of I_0 and I_1 are strictly smaller than q:

$$\frac{a}{b/q} = q\frac{a}{b} < q, \qquad \frac{b'}{a'} < q.$$

Thus $I_k \cap q^n I_k = \varnothing$ for $k = 0, 1$ and $n \neq 0$. Lemma 2.1 and Lemma 2.2 give the open immersions.

The quotient is covered by the images. Indeed, every $\varphi^{\mathbf{Z}}$-orbit has a representative in the fundamental annulus $\mathcal{Y}_{[1,q]}$. We claim that

$$\mathcal{Y}_{[1,q]} = \mathcal{Y}_{[1,a]} \cup \mathcal{Y}_{[a',b']} \cup \mathcal{Y}_{[b,q]}. \tag{1}$$

The inclusion from right to left follows from Lemma 2.1. Conversely, let $x \in \mathcal{Y}_{[1,q]}$ and put

$$y = |[\varpi_F](x)|.$$

If $x \notin \mathcal{Y}_{[1,a]}$ and $x \notin \mathcal{Y}_{[b,q]}$, then the inequalities already supplied by $x \in \mathcal{Y}_{[1,q]}$ show that

$$|\varpi_E(x)|^b < y < |\varpi_E(x)|^a.$$

Since $a' < a < b < b'$ and $0 < |\varpi_E(x)| < 1$, one has

$$|\varpi_E(x)|^{b'} < |\varpi_E(x)|^b < y < |\varpi_E(x)|^a < |\varpi_E(x)|^{a'}.$$

Thus $x \in \mathcal{Y}_{[a',b']}$, proving (1). Now

$$\mathcal{Y}_{[1,a]} = \mathcal{Y}_{I_0} \cap \mathcal{Y}_{[1,q]}, \qquad \mathcal{Y}_{[b,q]} = \varphi(\mathcal{Y}_{I_0}) \cap \mathcal{Y}_{[1,q]},$$

while $\mathcal{Y}_{[a',b']} = \mathcal{Y}_{I_1}$. Hence (1) shows that $\mathcal{U} \cup \mathcal{V} = \mathcal{X}$.

For the intersection, a representative in $\mathcal{Y}_{I_1}$ can meet a translate $\varphi^n(\mathcal{Y}_{I_0}) = \mathcal{Y}_{q^n I_0}$ only for $n = 0$ or $n = 1$: for $n \le -1$ the upper endpoint is at most $a/q < a'$, and for $n \ge 2$ the lower endpoint is at least $qb > b'$. Lemma 2.1 gives

$$\mathcal{Y}_{I_1} \cap \mathcal{Y}_{I_0} = \mathcal{Y}_{[a',a]}, \qquad \mathcal{Y}_{I_1} \cap \varphi(\mathcal{Y}_{I_0}) = \mathcal{Y}_{[b,b']}.$$

The two intervals are disjoint because $a < b$. Since both source annuli embed in the quotient, these are exactly the two components of $\mathcal{U} \cap \mathcal{V}$. □

Lemma 2.4 (Binary Mayer–Vietoris for mapping spectra)**.** *Let $\mathcal{Z} = \mathcal{U} \cup \mathcal{V}$ be an open cover by analytic substacks for which solid quasicoherent complexes satisfy descent. For $K, L \in \mathcal{C}_\square(\mathcal{Z})$, there is a natural cartesian square of spectra*

$$\begin{array}{ccc} \mathrm{Map}_{\mathcal{Z}}(K, L) & \longrightarrow & \mathrm{Map}_{\mathcal{U}}(K|_{\mathcal{U}}, L|_{\mathcal{U}}) \\ \downarrow & & \downarrow \\ \mathrm{Map}_{\mathcal{V}}(K|_{\mathcal{V}}, L|_{\mathcal{V}}) & \longrightarrow & \mathrm{Map}_{\mathcal{U}\cap\mathcal{V}}(K|_{\mathcal{U}\cap\mathcal{V}}, L|_{\mathcal{U}\cap\mathcal{V}}). \end{array}$$

Equivalently, the first term is the fiber of the difference of the two restriction maps from the direct sum of the middle terms to the last term.

Proof. Descent gives an equivalence of stable ∞-categories

$$\mathcal{C}_\square(\mathcal{Z}) \simeq \mathcal{C}_\square(\mathcal{U}) \times_{\mathcal{C}_\square(\mathcal{U}\cap\mathcal{V})} \mathcal{C}_\square(\mathcal{V}).$$

Mapping spaces in a limit of ∞-categories are the corresponding limits of mapping spaces. Applying this to the two descent data defined by K and L gives the displayed cartesian square. Since spectra form a stable category, a cartesian square is also cocartesian, and the equivalent fiber-sequence formulation follows. □

Proposition 2.5 (Finite descent for filtered-colimit compactness)**.** *Let $\mathcal{Z}$ be equipped with a finite Mayer–Vietoris presentation whose nonempty leaves are analytic affines* $\mathrm{AnSpec}(R, R^+)^\square$ *attached to complete Huber pairs. Then:*

(i) *$R\Gamma(\mathcal{Z}, -)$ commutes with filtered colimits;*
(ii) *if $K \in \mathcal{C}_\square(\mathcal{Z})$ restricts to a compact object on every analytic-affine leaf of the presentation, then K is compact in $\mathcal{C}_\square(\mathcal{Z})$.*

Proof. We use structural induction on the chosen finite Mayer–Vietoris presentation. On a nonempty leaf $\mathrm{AnSpec}(R, R^+)^{\square}$, global sections are mapping spectra out of the analytic unit. The unit is compact: it is the measure object attached to the one-point profinite set, so this is Andreychev's Proposition 3.23. Hence global sections commute with filtered colimits [And21, Proposition 3.23]. This proves both assertions at every nonempty leaf; the empty leaf is immediate.

Suppose now that the root of the presentation is

$$\mathcal{Z} = \mathcal{U} \cup \mathcal{V}$$

and that the assertions have already been proved for the three strictly smaller presentations of $\mathcal{U}$, $\mathcal{V}$, and $\mathcal{U} \cap \mathcal{V}$. The morphism $\mathcal{U} \amalg \mathcal{V} \to \mathcal{Z}$ is an effective epimorphism of analytic stacks, so Proposition 3.6 verifies the descent hypothesis in Lemma 2.4. That lemma, applied with first variable $\mathcal{O}_{\mathcal{Z}}$, expresses $R\Gamma(\mathcal{Z}, L)$ as the finite limit of

$$R\Gamma(\mathcal{U}, L|_{\mathcal{U}}), \qquad R\Gamma(\mathcal{V}, L|_{\mathcal{V}}), \qquad R\Gamma(\mathcal{U} \cap \mathcal{V}, L|_{\mathcal{U}\cap\mathcal{V}}).$$

Each term commutes with filtered colimits by the inductive hypothesis, and filtered colimits commute with finite limits in spectra. This proves (i).

For (ii), apply the same cartesian square to $\mathrm{Map}(K, -)$. The restrictions of K satisfy the leafwise compactness hypothesis for the three subpresentations, so their mapping functors commute with filtered colimits by induction. Their finite limit therefore does as well. Thus K is compact. □

Lemma 2.6 (Descent of compatible functors and transformations)**.** *Let $\mathcal{C}^\bullet$ and $\mathcal{D}^\bullet$ be cosimplicial diagrams of ∞-categories and let*

$$F^\bullet : \mathcal{C}^\bullet \longrightarrow \mathcal{D}^\bullet$$

be a morphism of diagrams. It induces a functor on limits. If every F^n is fully faithful, respectively an equivalence, then the induced functor is fully faithful, respectively an equivalence. If every F^n is fully faithful, a cartesian section of $\mathcal{D}^\bullet$ lies in the essential image whenever each of its components lies in the essential image of F^n. Natural transformations descend to limits. If the diagrams are presentable, the transition functors preserve colimits, and colimits in the limits are computed degreewise, then degreewise colimit preservation descends as well.

In addition, if $F : \mathcal{C} \to \mathcal{D}$ is fully faithful and I is any small ∞-category, then

$$\mathrm{Fun}(I, \mathcal{C}) \longrightarrow \mathrm{Fun}(I, \mathcal{D})$$

is fully faithful, with essential image the diagrams whose values lie in the essential image of F.

Proof. Mapping spaces in a limit of ∞-categories are limits of the degreewise mapping spaces, which proves full faithfulness. Let $\mathcal{E}^n \subset \mathcal{D}^n$ be the full essential-image subcategory of F^n. Naturality makes the $\mathcal{E}^n$ a cosimplicial full subdiagram, and $F^n : \mathcal{C}^n \to \mathcal{E}^n$ is an equivalence. Hence

$$\lim \mathcal{C}^\bullet \xrightarrow{\sim} \lim \mathcal{E}^\bullet.$$

A cartesian section of $\mathcal{D}^\bullet$ whose components lie in the $\mathcal{E}^n$ is precisely a cartesian section of $\mathcal{E}^\bullet$, giving the essential-image assertion including all higher coherences. Natural transformations and degreewise colimits descend by evaluation in each cosimplicial degree.

For the functor-category assertion, let $\mathcal{E} \subset \mathcal{D}$ be the full essential image of F. The equivalence $\mathcal{C} \simeq \mathcal{E}$ induces $\mathrm{Fun}(I, \mathcal{C}) \simeq \mathrm{Fun}(I, \mathcal{E})$, and the latter is the full subcategory of $\mathrm{Fun}(I, \mathcal{D})$ consisting of diagrams valued in $\mathcal{E}$. □

2.2. **Enhanced GAGA on the base.** Let $\mathrm{RatAff}_X(\mathcal{X})$ denote the basis of strongly noetherian rational affinoids $V \subset \lambda^{-1}(W)$, where W is a standard affine open of X. It is stable under rational localization, and every intersection of two members admits a finite cover by members of the same basis.

Fix a finite standard affine cover $(W_j)_{j=1}^r$ of X. Refine the annular presentation of Proposition 2.3 so that every analytic-affine leaf lies in some $\lambda^{-1}(W_j)$. Each original leaf is quasicompact and therefore has a finite rational cover subordinate to the $\lambda^{-1}(W_j)$.

We explain the binary refinement in a form stable under the induction. Suppose that a quasicompact analytic open Z is the union of finitely many analytic affines $V_1, \dots, V_N$ and that every nonempty finite intersection of the V_i is an analytic affine. Then Z admits a finite Mayer–Vietoris presentation with leaves among those finite intersections. We prove this simultaneously for all such finite families by induction on N. For $N > 1$, put

$$Z' = V_1 \cup \cdots \cup V_{N-1}.$$

The family $(V_i)_{i<N}$ satisfies the same intersection hypothesis, so Z' has a presentation by induction. Moreover

$$Z' \cap V_N = \bigcup_{i<N} (V_i \cap V_N),$$

and the family of nonempty opens $V_i \cap V_N$ again has analytic-affine finite intersections, namely the nonempty intersections $V_{i_1} \cap \cdots \cap V_{i_k} \cap V_N$. Hence $Z' \cap V_N$ has a presentation by the same induction. Attach the presentations of Z', V_N, and $Z' \cap V_N$ below the root corresponding to $Z = Z' \cup V_N$. Rational affinoids and all of their nonempty finite intersections are analytic affines, so the criterion applies to each subordinate rational cover above. Replacing every original leaf in this way preserves finiteness. Throughout this subsection, "the fixed finite Mayer–Vietoris presentation" means this subordinate refinement.

We now construct the comparison functor

$$\alpha_X : \mathrm{D}_{\mathrm{qc}}(X) \longrightarrow \mathcal{C}_{\square}(\mathcal{X}).$$

Let $M \in \mathrm{D}_{\mathrm{qc}}(X)$. For an affine $W = \operatorname{Spec} A \subset X$ and an affinoid $V = \operatorname{Spa}(R, R^+) \subset \lambda^{-1}(W)$, set

$$\alpha_{W,V}(M) = \mathrm{Cond}_R\big(R \otimes^{\mathbf{L}}_A R\Gamma(W, M|_W)\big) \in \mathsf{D}_{\square}(R, R^+).$$

Equivalently, under the usual equivalence $\mathrm{D}_{\mathrm{qc}}(W) \simeq \mathrm{D}(A)$, this is the composite

$$\mathrm{D}(A) \xrightarrow{-\otimes^{\mathbf{L}}_A R} \mathrm{D}(R) \xrightarrow{\mathrm{Cond}_R} \mathsf{D}_{\square}(R, R^+).$$

This object is independent of the affine neighborhood W. Indeed, if V lies over affine opens $W_1 = \operatorname{Spec} A_1$ and $W_2 = \operatorname{Spec} A_2$, then $W_{12} = W_1 \cap W_2$ is affine because X is separated. Quasicoherent restriction identifies

$$R\Gamma(W_{12}, M|_{W_{12}}) \simeq A_{12} \otimes^{\mathbf{L}}_{A_i} R\Gamma(W_i, M|_{W_i}),$$

where $A_{12} = \Gamma(W_{12}, \mathcal{O}_X)$. Since $A_i \to R$ factors through A_{12}, associativity of derived tensor products gives canonical equivalences from both definitions to

$$\mathrm{Cond}_R\big(R \otimes^{\mathbf{L}}_{A_{12}} R\Gamma(W_{12}, M|_{W_{12}})\big).$$

On three affine neighborhoods the two comparison composites are the standard associativity composites for the same iterated tensor product and agree by the pentagon coherence. Thus the local definition is canonically independent of W, with coherent comparison maps.

Consider a rational inclusion $V' = \operatorname{Spa}(R', R'^+) \subset V$. Choose any affine neighborhood $W' \subset W$ through which the image of V' factors; one may simply take $W' = W$. Restricting M first from W to W' and then extending scalars to R' gives

$$R' \otimes^{\mathbf{L}}_{\Gamma(W', \mathcal{O}_X)} R\Gamma(W', M|_{W'}).$$

Restricting $\alpha_{W,V}(M)$ from V to V' gives

$$R' \otimes^{\mathbf{L}}_R \big(R \otimes^{\mathbf{L}}_A R\Gamma(W, M|_W)\big).$$

The canonical restriction equivalence for quasicoherent complexes on X, associativity of derived tensor products, and scalar-extension compatibility of condensification identify these two objects. The resulting comparison is independent of the auxiliary choice of W': two choices admit the affine common refinement $W_1' \cap W_2'$, and both comparisons restrict there to the same associativity isomorphism. For a chain $V'' \subset V' \subset V$, the two composites are the two standard reassociations of the same iterated tensor product; the associativity coherence in the symmetric monoidal derived category identifies them. Thus the objects $\alpha_{W,V}(M)$, their restriction equivalences, and these coherences form a cartesian section on the rational-affinoid basis.

Solid quasicoherent complexes satisfy rational descent, i.e.

$$\mathcal{C}_\square(\mathcal{X}) \simeq \lim_{V \in \mathrm{RatAff}_X(\mathcal{X})^{\mathrm{op}}} \mathcal{C}_\square(V),$$

where the right side is the category of cartesian sections. The section just constructed therefore corresponds to a unique object $\alpha_X(M) \in \mathcal{C}_\square(\mathcal{X})$. A morphism $M \to N$ gives a morphism of the two cartesian sections in every chart, and the scalar-extension compatibilities make these local maps a natural transformation of the basis diagrams. Passing to the limit gives the global morphism $\alpha_X(M) \to \alpha_X(N)$. The identity and composition laws hold because their restrictions do so in every chart and restriction to the basis is conservative. This defines the functor α_X. The local tensor-product comparison maps commute with rational restriction and satisfy the associativity, unit, and symmetry coherences before descent; they consequently define a strong symmetric monoidal structure on the global functor.

Theorem 5.9 and Lemma 5.10 of Andreychev show that every local functor is exact and preserves filtered colimits. Exactness gives preservation of finite colimits. For any family $(M_i)_{i \in I}$, the coproduct $\bigoplus_{i \in I} M_i$ is the filtered colimit of its finite partial sums, so every local functor preserves arbitrary coproducts. Since restriction to the rational-affinoid basis is conservative and colimits of solid quasicoherent complexes are computed locally, the global functor has the same properties. Finally, every small colimit in a stable ∞-category is assembled from coproducts and cofibers, so α_X preserves all small colimits.

Lemma 2.7 (Solid and ordinary coherent cohomology)**.** *For every bounded coherent complex K on X, the natural maps*

$$R\Gamma_\square(\mathcal{X}, \alpha_X K) \xrightarrow{\sim} R\Gamma_{\mathrm{adic}}(\mathcal{X}, \lambda^* K) \xleftarrow{\sim} R\Gamma(X, K)$$

are equivalences.

Proof. The second map is the Kedlaya–Liu coherent cohomology comparison [KL19, Theorem 4.7.5], extended from coherent sheaves to bounded coherent complexes by finite Postnikov induction.

For the first map, use the fixed finite Mayer–Vietoris presentation of $\mathcal{X}$. Let $V = \mathrm{Spa}(R, R^+) \subset \lambda^{-1}(W)$ be one of its affinoid leaves, with $W = \mathrm{Spec}\, A$. Then

$$K_R = R \otimes^{\mathbf{L}}_A R\Gamma(W, K|_W).$$

Lemma 2.9, whose proof is independent of the present lemma, shows that $A \to R$ is flat. Hence K_R is a bounded complex with finite R-cohomology modules, and the restriction of $\alpha_X K$ is its condensification. Solid global sections are mapping out of the analytic unit, so full faithfulness of condensification [And21, Theorem 5.9] gives

$$R\Gamma_\square(V, \mathrm{Cond}_R K_R) \simeq K_R.$$

The finite-module theorem and affinoid acyclicity for the strongly noetherian chart give

$$R\Gamma_{\mathrm{adic}}(V, K_R) \simeq K_R.$$

Thus the two cohomology functors agree on every analytic-affine leaf. We next verify that these leafwise equivalences respect the restriction maps in the Mayer–Vietoris tree. For a rational

inclusion $V' \subset V$, both cohomology complexes restrict by the ordinary scalar-extension morphism

$$K_R \longrightarrow R' \otimes_R^{\mathbf{L}} K_R,$$

and the equivalence furnished by full faithfulness of condensification is natural for this scalar-extension map. Consequently the leafwise comparisons define a natural transformation between the two finite diagrams obtained from the fixed Mayer–Vietoris presentation.

Starting at the leaves and proceeding upward through the finite rooted tree, compare the solid and ordinary cohomology objects vertex by vertex. At a nonterminal vertex, Lemma 2.4 identifies the solid cohomology object with the pullback of the three solid cohomology objects at its children. The two-open Mayer–Vietoris distinguished triangle for ordinary derived sheaf cohomology [Stacks, Tag 08BX] gives the analogous pullback description on the ordinary adic side. If the comparison is an equivalence at the three children, it is therefore an equivalence at their parent. The leafwise equivalences and their naturality under every restriction edge start this induction, which reaches the root and proves the first comparison. □

Theorem 2.8 (Enhanced base full faithfulness)**.** *The functor*

$$\alpha_X : \mathrm{D}_{\mathrm{qc}}(X) \longrightarrow \mathcal{C}_{\square}(\mathcal{X})$$

is fully faithful. It sends perfect complexes to compact objects, and its essential image is closed under arbitrary colimits.

Proof. Let $P, Q \in \mathrm{Perf}(X)$. Since P is dualizable and α_X is strong symmetric monoidal,

$$\begin{aligned} \mathrm{Map}_{\mathcal{C}_{\square}(\mathcal{X})}(\alpha_X P, \alpha_X Q) &\simeq R\Gamma_{\square}\big(\mathcal{X}, \alpha_X(P^{\vee} \otimes Q)\big) \\ &\simeq R\Gamma(X, P^{\vee} \otimes Q) \\ &\simeq \mathrm{Map}_{\mathrm{D}_{\mathrm{qc}}(X)}(P, Q), \end{aligned}$$

where the middle equivalence is Lemma 2.7.

We next prove that $\alpha_X P$ is compact. Dualizability gives

$$\mathrm{Map}(\alpha_X P, -) \simeq R\Gamma_{\square}\big(\mathcal{X}, \alpha_X(P^{\vee}) \otimes -\big).$$

Tensoring with a fixed dualizable object commutes with all colimits, and Proposition 2.5(i) shows that global sections commute with filtered colimits. Hence the displayed mapping functor commutes with filtered colimits.

The category $\mathrm{D}_{\mathrm{qc}}(X)$ has a perfect generator, and its compact objects are precisely the perfect complexes; hence it is canonically the Ind-completion of $\mathrm{Perf}(X)$ [Stacks, Theorem 36.15.3, Tag 09IS, and Proposition 36.17.1, Tag 09M0]. The preceding calculation defines a fully faithful functor on $\mathrm{Perf}(X)$ whose values are compact. By the universal property of Ind-completion, its colimit-preserving extension is fully faithful. This extension is α_X, because both functors agree on perfect generators and preserve colimits.

Finally, full faithfulness lifts any diagram in the essential image, including all of its coherences, to a diagram (M_i) in $\mathrm{D}_{\mathrm{qc}}(X)$. Since α_X preserves colimits,

$$\operatorname*{colim}_i \alpha_X M_i \simeq \alpha_X(\operatorname*{colim}_i M_i).$$

Thus every colimit of objects in the essential image remains in the essential image. □

2.3. **Flatness and a finite fpqc affinoid family.**

Lemma 2.9 (Flatness on the base curve)**.** *Let $W = \operatorname{Spec} A \subset X$ be a standard affine open, and let*

$$V = \operatorname{Spa}(R, R^+) \subset \lambda^{-1}(W)$$

be a strongly noetherian affinoid open. Then $A \to R$ is flat. Consequently, the morphism of locally ringed spaces $\lambda : \mathcal{X} \to X$ is flat on local rings.

Proof. Because A is noetherian, it suffices by the ideal criterion to prove that

$$I \otimes_A R \longrightarrow R$$

is injective for every finitely generated ideal $I \subset A$.

Let $I \subset A$ be such an ideal. Since $W \hookrightarrow X$ is a quasi-compact open immersion and X is noetherian, the coherent ideal $\widetilde{I} \subset \mathcal{O}_W$ extends to a coherent ideal $\mathcal{J} \subset \mathcal{O}_X$ with $\mathcal{J}|_W = \widetilde{I}$; see [Stacks, Section 28.23, especially Lemma 28.23.2]. Apply the exact Kedlaya–Liu pullback functor [KL19, Theorem 4.7.4] to

$$0 \longrightarrow \mathcal{J} \longrightarrow \mathcal{O}_X.$$

After restriction to V this remains an injection of coherent sheaves. Compatibility of the Kedlaya–Liu equivalence with the standard affine/analytic charts identifies

$$(\lambda^*\mathcal{J})|_V \simeq \widetilde{I \otimes_A R}, \qquad (\lambda^*\mathcal{O}_X)|_V \simeq \mathcal{O}_V.$$

The global-sections functor is left exact, and the finite-module theorem on the strongly noetherian affinoid V identifies the induced map on global sections with

$$I \otimes_A R \longrightarrow R.$$

Thus R is flat over A.

To obtain the assertion on local rings, let $x \in V$. Apply Lemma 6.7 of Zavyalov's *Some Foundational Results in Adic Geometry* [Zav25, Lemma 6.7] to the relative analytification of $\operatorname{Spec} R$ over $\operatorname{Spa}(R, R^+)$; it gives a flat map $R \to \mathcal{O}_{V,x}$. Localizing the already flat map $A \to R$ at the image of x and composing shows that

$$\mathcal{O}_{X,\lambda(x)} \longrightarrow \mathcal{O}_{\mathcal{X},x}$$

is flat. □

Lemma 2.10 (A finite fpqc family of analytic charts)**.** *There are finitely many nonempty strongly noetherian rational affinoid opens*

$$V_i = \operatorname{Spa}(R_i, R_i^+) \subset \mathcal{X} \qquad (1 \le i \le m),$$

and standard affine opens $W_i = \operatorname{Spec} A_i \subset X$ with $V_i \subset \lambda^{-1}(W_i)$, such that the induced morphism

$$s : \coprod_{i=1}^{m} \operatorname{Spec} R_i \longrightarrow X$$

is fpqc.

Proof. Proposition 2.3 makes $\mathcal{X}$ quasicompact. Choose a finite standard affine cover (W_j) of X. The opens $\lambda^{-1}(W_j)$ cover $\mathcal{X}$. Refining the finite annular cover by the rational-affinoid basis and using quasicompactness gives a finite rational-affinoid cover (V_i) such that each V_i is contained in some $\lambda^{-1}(W_i)$.

The map $A_i \to R_i$ is flat by Lemma 2.9. Since X is separated, the inclusion $W_i \hookrightarrow X$ is affine; consequently $\operatorname{Spec} R_i \to X$ is affine and flat. It remains to prove joint surjectivity on spectra.

Let η be the generic point of X. A nonempty affinoid has nonzero coordinate ring, so $R_i \neq 0$. Since A_i is a domain and R_i is flat over A_i, multiplication by every nonzero $a \in A_i$ remains injective on R_i; in particular $A_i \to R_i$ is injective and R_i is torsion-free. Therefore

$$\operatorname{Frac}(A_i) \otimes_{A_i} R_i \neq 0.$$

Any prime of this nonzero localization pulls back to a prime of R_i contracting to (0), hence to η.

Now let x be a closed point of X. The skyscraper $\kappa(x)$ is a nonzero coherent sheaf. Kedlaya–Liu coherent GAGA is an exact equivalence and therefore conservative, so its analytic pullback is

nonzero. Since the V_i cover $\mathcal{X}$, its restriction to some V_i is nonzero. Necessarily $x \in W_i$, and on that chart the restriction is the coherent module

$$\kappa(x) \otimes_{A_i} R_i.$$

It is nonzero, hence has a prime; the inverse image of that prime in A_i is the maximal ideal of x. As X is noetherian, integral, and one-dimensional, every point is either generic or closed. Thus s is surjective. It is a finite disjoint union of affine flat morphisms and hence is quasi-compact. Therefore it is fpqc. □

Proposition 2.11 (Rational descent for pseudocoherent objects on the base)**.** *For $V = \mathrm{Spa}(R, R^+) \in \mathrm{RatAff}_X(\mathcal{X})$, regard $\mathrm{D}_{\mathrm{pc}}(R)$ as the full nuclear-pseudocompact subcategory of $\mathsf{D}_\square(R, R^+)$ through condensification. Restriction induces an equivalence*

$$\mathsf{D}^\square_{\mathrm{pc}}(\mathcal{X}) \xrightarrow{\sim} \mathrm{Cart}_{V \in \mathrm{RatAff}_X(\mathcal{X})} \mathrm{D}_{\mathrm{pc}}(R).$$

The right side denotes cartesian sections for derived scalar extension along rational localizations. The resulting full subcategory of $\mathcal{C}_\square(\mathcal{X})$ is unchanged after replacing the displayed rational basis or any of the bounded-chart presentations by a cofinal refinement.

Proof. Solid quasicoherent complexes satisfy rational descent, so restriction identifies $\mathcal{C}_\square(\mathcal{X})$ with the category of cartesian sections of the diagram $V \mapsto \mathsf{D}_\square(R, R^+)$. On each chart, Andreychev's relative-discreteness theorem and Li–Huerta's bounded-chart description identify the local nuclear-pseudocompact category with the essential image of

$$\mathrm{Cond}_R : \mathrm{D}_{\mathrm{pc}}(R) \hookrightarrow \mathsf{D}_\square(R, R^+)$$

[And21, Theorems 5.44 and 5.50][LH26, A.15]. Rational localization is a steady localization on the solid side and is flat on underlying rings; Andreychev's scalar-extension compatibility identifies its restriction on relatively discrete objects with ordinary derived scalar extension. Consequently the local essential-image subcategories form a cartesian full subdiagram.

A global object belongs to $\mathsf{D}^\square_{\mathrm{pc}}(\mathcal{X})$ exactly when its cartesian section takes values in that full subdiagram. Conversely, a cartesian section of the local pseudocoherent subcategories is in particular a cartesian section of the ambient solid categories and therefore descends to a unique object of $\mathcal{C}_\square(\mathcal{X})$; its local restrictions show that this object belongs to $\mathsf{D}^\square_{\mathrm{pc}}(\mathcal{X})$. This proves the equivalence.

Any two rational-affinoid bases admit common rational refinements. Likewise, Li–Huerta's bounded-chart systems associated with two choices of pseudo-uniformizer or generators admit the cofinal common refinements constructed in Lemma 3.2. Since membership in the displayed cartesian full subdiagram may be tested after such a jointly conservative refinement, the full subcategory is independent of all these choices. □

Proposition 2.12 (Base pseudocoherent GAGA)**.** *The pseudocoherent subcategory $\mathsf{D}^\square_{\mathrm{pc}}(\mathcal{X})$ carries the t-structure obtained by rational descent from ordinary pseudocoherent complexes on the strongly noetherian affinoid charts. Analytification restricts to a t-exact equivalence*

$$\alpha_X : \mathrm{D}_{\mathrm{pc}}(X) \xrightarrow{\sim} \mathsf{D}^\square_{\mathrm{pc}}(\mathcal{X}).$$

Proof. For a rational localization $\mathrm{Spa}(R', R'^+) \to \mathrm{Spa}(R, R^+)$ between the charts under consideration, the map $R \to R'$ is flat. Ordinary derived extension of scalars therefore commutes with all cohomology objects of a pseudocoherent complex and is t-exact. Define full subcategories of $\mathsf{D}^\square_{\mathrm{pc}}(\mathcal{X})$ by

$$\mathsf{D}^\square_{\mathrm{pc}}(\mathcal{X})^{\leq 0} = \{C : \ C|_V \text{ is represented by an object of } \mathrm{D}_{\mathrm{pc}}(R)^{\leq 0} \text{ for every } V = \mathrm{Spa}(R, R^+)\},$$
$$\mathsf{D}^\square_{\mathrm{pc}}(\mathcal{X})^{\geq 0} = \{C : \ C|_V \text{ is represented by an object of } \mathrm{D}_{\mathrm{pc}}(R)^{\geq 0} \text{ for every } V = \mathrm{Spa}(R, R^+)\}.$$

Relative discreteness makes the representing ordinary complex unique, and flatness shows that the two conditions are preserved under rational restriction. They are therefore independent of the chosen affinoid cover.

Let $C \in \mathsf{D}^{\square}_{\mathrm{pc}}(\mathcal{X})$. On every chart choose its ordinary pseudocoherent representative C_V. The local truncation triangles

$$\tau_{\leq 0}C_V \longrightarrow C_V \longrightarrow \tau_{\geq 1}C_V$$

are compatible with rational restriction because those restriction functors are t-exact. More precisely, for every rational inclusion $u : V' \subset V$ there are canonical equivalences

$$u^*\tau_{\leq 0}C_V \simeq \tau_{\leq 0}u^*C_V, \qquad u^*\tau_{\geq 1}C_V \simeq \tau_{\geq 1}u^*C_V.$$

These equivalences are the unique ones carrying the restricted truncation maps to the truncation maps of u^*C_V. For a chain $V'' \subset V' \subset V$, both composites have this same defining property; uniqueness of truncations therefore gives the cocycle equality. Thus the two families of truncations, together with their arrows to and from $(C_V)_V$, are cartesian sections of the rational-localization diagrams. Andreychev's effective descent theorem for pseudocoherent complexes produces global objects $C^{\leq 0}$ and $C^{\geq 1}$ and global maps

$$C^{\leq 0} \longrightarrow C \longrightarrow C^{\geq 1}$$

whose restrictions are the displayed local maps. Cofibers are computed after restriction, so this is a cofiber triangle because it is one on a jointly conservative rational-affinoid basis. Orthogonality is checked after restriction to the same basis, where it is the ordinary orthogonality in $\mathrm{D}_{\mathrm{pc}}(R)$. Functoriality of truncation and the identities $\tau^2_{\leq 0} = \tau_{\leq 0}$ and $\tau^2_{\geq 1} = \tau_{\geq 1}$ are likewise local and therefore descend. This verifies the t-structure axioms and constructs the asserted t-structure; it is the chartwise structure used in Li–Huerta's A.15 [LH26, A.15][And21, Theorems 5.44 and 5.50].

An object lies in the heart precisely when every affinoid representative is a finite module in degree zero. Finite modules satisfy rational descent on a strongly noetherian adic space, so this heart is canonically $\mathrm{Coh}(\mathcal{X})$.

The functor α_X sends $\mathrm{D}_{\mathrm{pc}}(X)$ into $\mathsf{D}^{\square}_{\mathrm{pc}}(\mathcal{X})$. Indeed, on a chart $W = \operatorname{Spec} A$ and $V = \operatorname{Spa}(R, R^+) \subset \lambda^{-1}(W)$, it is condensification of $- \otimes^{\mathbf{L}}_A R$. The map $A \to R$ is flat by Lemma 2.9, and condensification identifies ordinary pseudocoherent complexes with the relatively discrete nuclear-pseudocompact objects. This also proves t-exactness of the forward functor. Full faithfulness is inherited from Theorem 2.8.

It remains to prove essential surjectivity. Let $C \in \mathsf{D}^{\square}_{\mathrm{pc}}(\mathcal{X})$. The finite annular Mayer–Vietoris cover of Proposition 2.3 gives a finite affinoid cover on which C is represented by bounded-above pseudocoherent complexes. Taking the maximum of their upper bounds, we obtain an integer b such that $C \in \mathsf{D}^{\square}_{\mathrm{pc}}(\mathcal{X})^{\leq b}$. For $n \geq 0$ put

$$C_n = \tau_{\geq -n}C.$$

Each C_n is a bounded object of $\mathsf{D}^{\square}_{\mathrm{pc}}(\mathcal{X})$.

We algebraize C_n by finite Postnikov induction. Each cohomology object $\mathcal{H}^j(C_n)$ belongs to $\mathrm{Coh}(\mathcal{X})$, so Kedlaya–Liu coherent GAGA gives a coherent sheaf H^j_n on X together with an equivalence $\alpha_X(H^j_n) \simeq \mathcal{H}^j(C_n)$. Assume inductively that $\tau_{\geq j+1}C_n$ is the image of an object $M_{n,j+1} \in \mathrm{D}^b_{\mathrm{coh}}(X)$. The standard truncation triangle is

$$\tau_{\geq j+1}C_n \longrightarrow \tau_{\geq j}C_n \longrightarrow \mathcal{H}^j(C_n)[-j] \xrightarrow{\delta_j} \tau_{\geq j+1}C_n[1].$$

The connecting morphism δ_j is a map between objects in the essential image of the fully faithful functor α_X. Theorem 2.8 therefore gives a unique map

$$H^j_n[-j] \longrightarrow M_{n,j+1}[1]$$

whose image is δ_j. Define $M_{n,j}$ by the corresponding fiber, or equivalently by the rotated cone triangle. Exactness of α_X identifies $\alpha_X(M_{n,j})$ with $\tau_{\geq j}C_n$. Starting above the largest nonzero cohomological degree and descending to $j = -n$ produces $M_n = M_{n,-n} \in \mathrm{D}^b_{\mathrm{coh}}(X)$ and an equivalence

$$\alpha_X(M_n) \simeq C_n.$$

The truncation maps $C_n \to C_{n+1}$ form a functor from the ordered set $\mathbf{N}$ to $\mathcal{C}_\square(\mathcal{X})$. Every value C_n lies in the essential image of α_X. The functor-category clause of Lemma 2.6 therefore shows that this entire sequential diagram, including its composition homotopies and all higher coherences, lies in the essential image of

$$\mathrm{Fun}(\mathbf{N}, \mathrm{D}_{\mathrm{qc}}(X)) \longrightarrow \mathrm{Fun}(\mathbf{N}, \mathcal{C}_\square(\mathcal{X})).$$

Choose the corresponding diagram (M_n); full faithfulness makes the space of such lifts contractible. Set

$$M = \operatorname*{colim}_n M_n \in \mathrm{D}_{\mathrm{qc}}(X).$$

Since α_X preserves colimits,

$$\alpha_X(M) \simeq \operatorname*{colim}_n C_n \simeq C.$$

The last equivalence is checked on each affinoid chart, where it is the standard identity expressing a bounded-above complex as the filtered colimit of its lower truncations.

We finally prove that M is pseudocoherent. Let

$$s : \coprod_i \operatorname{Spec} R_i \longrightarrow X$$

be the finite fpqc morphism of Lemma 2.10, and write $M_i = Ls_i^* M$. On $V_i = \mathrm{Spa}(R_i, R_i^+)$, the equivalence $\alpha_X(M) \simeq C$ reads

$$\mathrm{Cond}_{R_i}(M_i) \simeq C|_{V_i}.$$

By the definition of $\mathsf{D}^{\square}_{\mathrm{pc}}$, the right side is $\mathrm{Cond}_{R_i}(P_i)$ for some $P_i \in \mathrm{D}_{\mathrm{pc}}(R_i)$. Full faithfulness of condensification gives $M_i \simeq P_i$, so every fpqc pullback M_i is pseudocoherent. Pseudocoherence is fpqc local on schemes. More explicitly, for an affine open $W = \operatorname{Spec} A \subset X$, the base change of s is a finite affine fpqc cover of W, hence is represented by one faithfully flat map $A \to A'$ (the finite product of the coordinate rings of its members). The complex $M|_W \otimes^{\mathbf{L}}_A A'$ is pseudocoherent by the preceding paragraph, so [Stacks, Tag 068R] gives $M|_W \in \mathrm{D}_{\mathrm{pc}}(A)$. Varying W proves $M \in \mathrm{D}_{\mathrm{pc}}(X)$.

Thus α_X is essentially surjective. Since it is a t-exact equivalence, its inverse is t-exact as well. □

3. Relative realizations and relative full faithfulness

3.1. **Relative algebraic and analytic realizations.** Let $\mathrm{RatAff}_X(\mathcal{X})$ denote the rational-affinoid basis consisting of strongly noetherian rational affinoids

$$V \subset \lambda^{-1}(W)$$

for some standard affine open $W \subset X$. This is a basis because the opens $\lambda^{-1}(W)$ cover $\mathcal{X}$ and rational affinoids form a basis on each of them. It is stable under rational localization; intersections are covered by members of the same basis. All varying-base constructions below are first made on this basis and then descended to $\mathcal{X}$.

Let T be a separated finitely presented X-scheme. Choose an affinoid

$$V = \mathrm{Spa}(R, R^+) \subset \mathcal{X}$$

whose image is contained in an affine $W = \operatorname{Spec} A \subset X$, and set

$$T_R = T \times_X \operatorname{Spec} R.$$

Li–Huerta attaches to T_R an algebraic realization $(T_R)^{\mathrm{Alg}}$, which is an analytic stack over $(\operatorname{Spec} R)^{\mathrm{Alg}}$, and an analytic realization $(T_R)^{\mathrm{An}}$ over $V = (\operatorname{Spec} R)^{\mathrm{An}}$ [LH26, Theorem A.13]. Define locally

$$(T_{\mathcal{X}}^{\mathrm{Alg}})|_V = (T_R)^{\mathrm{Alg}} \times_{(\operatorname{Spec} R)^{\mathrm{Alg}}} V, \tag{2}$$

$$(T_{\mathcal{X}}^{\mathrm{An}})|_V = (T_R)^{\mathrm{An}}. \tag{3}$$

Lemma 3.1 (Integral structures under rational base change)**.** *Let $(R, R^+) \to (R', R'^+)$ be a rational localization of strongly noetherian complete Tate Huber pairs, let B be a finitely presented R-algebra, and put $B' = B \otimes_R R'$. Let B^+ be the integral closure of the image of R^+ in B, and let B'^+ be the integral closure of the image of R'^+ in B'. Then there is a canonical equivalence of normalized discrete analytic rings*

$$(R'_{\mathrm{disc}}, R'^+_{\mathrm{disc}})^{\square} \otimes^{\mathrm{AnRing}}_{(R_{\mathrm{disc}}, R^+_{\mathrm{disc}})^{\square}} (B_{\mathrm{disc}}, B^+_{\mathrm{disc}})^{\square} \simeq (B'_{\mathrm{disc}}, B'^+_{\mathrm{disc}})^{\square}.$$

For composable rational localizations, these equivalences commute with the associativity identifications for analytic-ring pushouts.

Proof. Put

$$\mathcal{R}_0 = (R_{\mathrm{disc}}, R^+_{\mathrm{disc}})^{\square}, \quad \mathcal{R}'_0 = (R'_{\mathrm{disc}}, R'^+_{\mathrm{disc}})^{\square}, \quad \mathcal{B}_0 = (B_{\mathrm{disc}}, B^+_{\mathrm{disc}})^{\square}$$

and let $\mathcal{P} = \mathcal{R}'_0 \otimes^{\mathrm{AnRing}}_{\mathcal{R}_0} \mathcal{B}_0$.

Underlying derived ring. The pre-normalized pushout has underlying connective derived ring

$$D = B \otimes^{\mathbf{L}}_R R'.$$

Rational localization is flat, so $D \simeq B'[0]$.

Complete connective objects. Proposition 12.12 of Scholze's *Lectures on Analytic Geometry* identifies the connective $\mathcal{P}$-complete B'-complexes with those whose restrictions are simultaneously $\mathcal{R}'_0$-complete and $\mathcal{B}_0$-complete [Sch26a, Proposition 12.12]. By Andreychev's restriction criterion, these two conditions can be tested on the minimal analytic structures generated by R'^+ and B^+, respectively [And21, Proposition 3.21]. Hence simultaneous completeness is completeness for the union of generators

$$R'^+ \cup B^+ \subset B'.$$

Let $C^+ \subset B'$ be the subring generated by that union. The image of B^+ is integral over R'^+, so C^+ is integral over R'^+ and its integral closure in B' is B'^+. Andreychev's generator criterion and invariance under integral closure therefore identify the connective complete subcategory of $\mathcal{P}$ with

$$\mathrm{D}_{\mathrm{conn}}\big((B'_{\mathrm{disc}}, B'^+_{\mathrm{disc}})^{\square}\big) \subset \mathrm{D}_{\mathrm{conn}}(B')$$

[And21, Propositions 3.22 and 3.32].

Normalization. The unit $B'[0]$ is complete for the analytic structure on the right [And21, Lemma 3.24]; by the preceding identification it is also $\mathcal{P}$-complete. Thus the normalization completion map from the pre-normalized ring $D \simeq B'[0]$ to the underlying animated ring of $\mathcal{P}$ is an equivalence. The two normalized analytic rings now have the same underlying animated ring $B'[0]$ and the same connective complete subcategory. Scholze's uniqueness theorem for normalized analytic structures gives the asserted equivalence [Sch26a, Proposition 12.20].

For $R \to R' \to R''$, both the direct and iterated constructions are normalizations of the same threefold pushout and have underlying ring $B \otimes_R R''$. Their connective complete objects are characterized by the same final integral subring, namely the integral closure of R''^+ in $B \otimes_R R''$. Proposition 12.20 gives uniqueness of the normalized analytic structure with these data. To make the coherence assertion precise, each such structure is determined by the reflector onto the fixed full subcategory of connective complete objects. A left adjoint to a fixed fully faithful inclusion is unique up to a contractible space of choices. Hence the direct and iterated normalization

equivalences are canonically identified, and their identity, composition, and higher coherence relations are forced by that contractible uniqueness. □

Lemma 3.2 (Bounded-chart rational base change)**.** *Let B be a finitely presented R-algebra, choose a topologically nilpotent unit $\varpi \in R$ and a finite set of R-algebra generators $\mathbf{b} = (b_1, \dots, b_m)$, and put $B' = B \otimes_R R'$ for a rational localization $(R, R^+) \to (R', R'^+)$. Then:*

(i) *the algebraic analytic rings satisfy a canonical base-change equivalence*

$$B_R^{\mathrm{alg}} \otimes^{\mathrm{AnRing}}_{(R,R^+)^\square} (R', R'^+)^\square \simeq (B')^{\mathrm{alg}}_{R'};$$

(ii) *for every $n \geq 0$, the simultaneous bounded chart defined by $|\varpi^n b_i| \leq 1$ base changes to the chart defined by $|\varpi'^n b'_i| \leq 1$, compatibly with the transition maps in n;*

(iii) *the increasing union of these charts is $(\operatorname{Spec} B)^{\mathrm{An}}$, and the equivalences in (ii) induce*

$$(\operatorname{Spec} B)^{\mathrm{An}} \times_{\operatorname{Spa}(R,R^+)} \operatorname{Spa}(R', R'^+) \simeq (\operatorname{Spec} B')^{\mathrm{An}}.$$

These constructions are functorial in morphisms of affine schemes and compatible with a second rational localization. Replacing ϖ or $\mathbf{b}$ gives the same map after passage to the analytic-locus union.

Proof. Write

$$\mathcal{R}_0 = (R_{\mathrm{disc}}, R^+_{\mathrm{disc}})^\square, \quad \mathcal{B}_0 = (B_{\mathrm{disc}}, B^+_{\mathrm{disc}})^\square, \quad \mathcal{R} = (R, R^+)^\square,$$

and similarly after base change. Li–Huerta's algebraic analytic ring is

$$B_R^{\mathrm{alg}} = \mathcal{R} \otimes^{\mathrm{AnRing}}_{\mathcal{R}_0} \mathcal{B}_0.$$

Associativity of analytic-ring pushouts and Lemma 3.1 give

$$\begin{aligned} B_R^{\mathrm{alg}} \otimes^{\mathrm{AnRing}}_{\mathcal{R}} \mathcal{R}' &\simeq \mathcal{R}' \otimes^{\mathrm{AnRing}}_{\mathcal{R}'_0} \left(\mathcal{R}'_0 \otimes^{\mathrm{AnRing}}_{\mathcal{R}_0} \mathcal{B}_0 \right) \\ &\simeq \mathcal{R}' \otimes^{\mathrm{AnRing}}_{\mathcal{R}'_0} \mathcal{B}'_0 = (B')^{\mathrm{alg}}_{R'}, \end{aligned}$$

which proves (i).

For each n, let $B^{\mathrm{univ}}_{|\varpi^n \mathbf{b}| \leq 1}$ be the pushout

$$\begin{array}{ccc} (\mathbf{Z}[T_1, \dots, T_m], \mathbf{Z})^\square & \longrightarrow & \mathbf{Z}[T_1, \dots, T_m]^\square \\ {\scriptstyle T_i \mapsto \varpi^n b_i}\downarrow & & \downarrow \\ B_R^{\mathrm{alg}} & \longrightarrow & B^{\mathrm{univ}}_{|\varpi^n \mathbf{b}| \leq 1}. \end{array} \tag{4}$$

The upper row is independent of the Tate base. Base changing this pushout and using (i) therefore gives

$$B^{\mathrm{univ}}_{|\varpi^n \mathbf{b}| \leq 1} \otimes^{\mathrm{AnRing}}_{(R,R^+)^\square} (R', R'^+)^\square \simeq (B')^{\mathrm{univ}}_{|\varpi'^n \mathbf{b}'| \leq 1}. \tag{5}$$

The transition from n to $n+1$ is induced by the identity $\varpi^{n+1} b_i = \varpi(\varpi^n b_i)$; functoriality of the pushout makes the equivalences compatible with these transitions.

Li–Huerta A.11–A.15 identifies the analytic locus with the filtered union

$$(\operatorname{Spec} B)^{\mathrm{An}} \simeq \operatorname*{colim}^{\{\text{analytic stacks}\}}_{n \geq 0} \operatorname{AnSpec} B^{\mathrm{univ}}_{|\varpi^n \mathbf{b}| \leq 1}. \tag{6}$$

The diagonal $\mathbf{N} \to \mathbf{N}^m$ is cofinal in the simultaneous bounded-chart system. Pullback in the slice ∞-topos of analytic stacks preserves colimits, so the chartwise equivalences give (iii).

All maps above are induced by universal pushout squares. For a second rational localization, the direct and iterated maps are already the two reassociations of the same iterated pushout in every bounded chart; the associativity coherence in AnRing identifies them compatibly with the

transition maps. Thus their induced maps on the filtered union satisfy the composition cocycle, including the higher coherences inherited from the associator.

We finally make independence of the choices explicit. Given two finite generating tuples, enlarge both to their union $\mathbf{c}$. If $\mathbf{b}$ is one of the original tuples, every extra member of $\mathbf{c}$ is a polynomial in $\mathbf{b}$ with coefficients in R. After multiplying by a sufficiently large power of a fixed topologically nilpotent unit, all coefficients become power-bounded. It follows that for every n there is an m such that

$$U_{\varpi,n}(\mathbf{b}) \subset U_{\varpi,m}(\mathbf{c}),$$

where $U_{\varpi,n}(\mathbf{b})$ denotes the chart $|\varpi^n b_i| \leq 1$; the reverse comparison is immediate after forgetting the extra inequalities. Thus adjoining generators gives a cofinal refinement of the bounded-chart union.

Now let $\widetilde{\varpi}$ be a second topologically nilpotent unit of R. Since ϖ and $\widetilde{\varpi}$ define the same Tate topology, after choosing a common ring of definition there are positive integers r, s for which $\widetilde{\varpi}^r/\varpi$ and $\varpi^s/\widetilde{\varpi}$ are power-bounded. Consequently

$$U_{\varpi,n}(\mathbf{c}) \subset U_{\widetilde{\varpi},rn}(\mathbf{c}), \qquad U_{\widetilde{\varpi},n}(\mathbf{c}) \subset U_{\varpi,sn}(\mathbf{c}).$$

These inclusions show that the two one-parameter systems have cofinal common refinements. On every chart in such a refinement, both rational-base-change maps are induced by the base change of the same universal pushout square (4); hence they agree before passage to the colimit. Taking the union of three generating tuples and common domination powers for three pseudo-uniformizers proves the transitivity and higher composition coherences in the same way. □

Lemma 3.3 (Cartesian rational base change)**.** *Let*

$$u : V' = \mathrm{Spa}(R', R'^{+}) \longrightarrow V = \mathrm{Spa}(R, R^{+})$$

be a rational localization between members of $\mathrm{RatAff}_X(\mathcal{X})$*. For every separated finitely presented* X*-scheme* T*, there are natural equivalences*

$$u^*\big((T_R)^{\mathrm{Alg}} \times_{(\operatorname{Spec} R)^{\mathrm{Alg}}} V\big) \simeq (T_{R'})^{\mathrm{Alg}} \times_{(\operatorname{Spec} R')^{\mathrm{Alg}}} V', \tag{7}$$

$$u^*(T_R)^{\mathrm{An}} \simeq (T_{R'})^{\mathrm{An}}. \tag{8}$$

They are functorial in T*, compatible with Zariski restriction, and satisfy the identity and composition cocycles for chains of rational localizations.*

Proof. The scheme $T_R = T \times_X \operatorname{Spec} R$ is intrinsic. If the structural map $\operatorname{Spec} R \to X$ factors through two affine opens, it factors uniquely through their affine intersection; associativity of fiber products identifies the two descriptions. The same applies to R'.

Assume first that $T_R = \operatorname{Spec} B$ is affine. Lemma 3.2(i), followed by the contravariant functor AnSpec, gives (7). Lemma 3.2(iii) gives (8). The functoriality and cocycles in the affine case are part of that lemma.

For general T_R, choose a finite affine Zariski cover (T_a). Because T_R is separated, every multiple intersection

$$T_{a_0 \ldots a_n} = T_{a_0} \cap \cdots \cap T_{a_n}$$

is affine. The affine comparison maps give degreewise equivalences between the simplicial analytic-stack diagrams attached to the full Čech nerve. Li–Huerta's fixed-base construction identifies both realizations with the corresponding geometric realizations:

$$(T_R)^{\mathrm{Alg}} \simeq \operatorname*{colim}_{[n]\in\Delta^{\mathrm{op}}} \coprod_{(a_0,\ldots,a_n)} (T_{a_0\ldots a_n})^{\mathrm{Alg}},$$

$$(T_R)^{\mathrm{An}} \simeq \operatorname*{colim}_{[n]\in\Delta^{\mathrm{op}}} \coprod_{(a_0,\ldots,a_n)} (T_{a_0\ldots a_n})^{\mathrm{An}}.$$

Pullback in a slice ∞-topos preserves colimits. Taking geometric realizations of the degreewise affine equivalences proves both displayed base-change equivalences. Naturality in T, compatibility with Zariski restriction, and the identity and composition cocycles are inherited degreewise from the affine universal pushouts and therefore persist after geometric realization. A common affine refinement proves independence of the chosen cover. □

Proposition 3.4 (Construction and rational base change)**.** *Let T be a separated finitely presented X-scheme. The local objects in* (2) *and* (3) *glue to analytic stacks over $\mathcal{X}$, denoted*

$$T_{\mathcal{X}}^{\mathrm{Alg}}, \qquad T_{\mathcal{X}}^{\mathrm{An}}.$$

The construction is functorial in T, commutes with rational open base change on $\mathcal{X}$, and is compatible with Zariski descent in T. If $T \hookrightarrow X$ is an open immersion, $T_{\mathcal{X}}^{\mathrm{An}}$ identifies with the analytic open $\lambda^{-1}(T) \subset \mathcal{X}$. No analogous analytic-openness assertion is made for $T_{\mathcal{X}}^{\mathrm{Alg}}$. The local analytic-locus maps glue to a natural morphism

$$T_{\mathcal{X}}^{\mathrm{An}} \longrightarrow T_{\mathcal{X}}^{\mathrm{Alg}},$$

which is an isomorphism when $T \to X$ is proper.

Proof. Lemma 3.3 makes each assignment

$$V \longmapsto (T_R)^{\mathrm{Alg}} \times_{(\operatorname{Spec} R)^{\mathrm{Alg}}} V, \qquad V \longmapsto (T_R)^{\mathrm{An}}$$

a cartesian section of the stack of analytic stacks on the rational-affinoid basis. Effective descent for analytic stacks along rational covers therefore produces the two global analytic stacks. The same descent applied to morphisms gives functoriality in T, while the fixed-base Zariski gluing of Li–Huerta and Lemma 3.3 give compatibility with Zariski descent.

For an open immersion $T \hookrightarrow X$, the fixed-base analytic realization is the inverse-image analytic open; the cartesian base-change identifications glue these opens to $\lambda^{-1}(T)$. On every affine fixed-base realization, Li–Huerta's bounded charts carry their natural maps to the algebraic realization. These maps are defined by the same pushout diagrams used in Lemma 3.2. For a general fixed-base scheme T_R, choose a finite affine Zariski cover. Functoriality of the affine maps makes them a morphism between the two simplicial analytic-stack gluing diagrams attached to the full Čech nerve. Taking geometric realizations in the ∞-topos of analytic stacks descends this morphism to $(T_R)^{\mathrm{An}} \to (T_R)^{\mathrm{Alg}}$, exactly as in Li–Huerta's Zariski-gluing construction [LH26, Theorem A.13]. The affine construction commutes with rational base change, so these fixed-base morphisms form a morphism of cartesian sections on $\mathrm{RatAff}_X(\mathcal{X})$ and hence glue to the global analytic-locus morphism.

If $T \to X$ is proper, then $T_R \to \operatorname{Spec} R$ is proper for every base chart. Li–Huerta's proper analytic-locus theorem makes each local morphism an equivalence [LH26, Theorem A.13]. Equivalences are local on the target, so the global analytic-locus morphism is an equivalence. □

Proposition 3.5 (Rational descent for relative pseudocoherent objects)**.** *Let T be a separated finitely presented X-scheme. Restriction induces an equivalence*

$$\mathsf{D}_{\mathrm{pc}}^{\square}(T_{\mathcal{X}}^{\mathrm{An}}) \xrightarrow{\sim} \mathrm{Cart}_{V \in \mathrm{RatAff}_X(\mathcal{X})}\, \mathsf{D}_{\mathrm{pc}}^{\square}((T_R)^{\mathrm{An}}),$$

where the categories on the right are Li–Huerta's fixed-base nuclear-pseudocompact categories and the transition functors are the rational pullbacks of Lemma 3.3. This description is independent of the rational basis and of all bounded-chart presentations. If T is proper, the same description is transported to $T_{\mathcal{X}}^{\mathrm{Alg}}$ through the analytic-locus equivalence.

Proof. Rational descent for solid quasicoherent complexes identifies $\mathcal{C}_{\square}(T_{\mathcal{X}}^{\mathrm{An}})$ with cartesian sections of the ambient fixed-base categories. Li–Huerta defines the local nuclear-pseudocompact category

by descent along the bounded steady-localization charts. Lemma 3.2 identifies those chart systems under rational localization, including their transition maps and common refinements.

We make the nonaffine step explicit. For a fixed base chart $V = \mathrm{Spa}(R, R^+)$, choose a finite affine Zariski cover of the separated R-scheme T_R. Every multiple intersection is affine. On each such intersection, any two bounded-chart presentations admit the common refinements constructed in Lemma 3.2; functoriality of the universal pushout squares makes these refinements compatible with every face and degeneracy map of the Zariski Čech nerve. Li–Huerta's Theorem A.13 sends this Zariski gluing diagram to a simplicial analytic-stack gluing diagram whose analytic maps are cohomologically étale steady localizations, while A.15 identifies the chartwise pseudocompact and nuclear full subcategories and shows that they glue [LH26, Theorem A.13 and A.15]. Applying Proposition 3.6 to the ambient categories and then restricting to these chartwise full subcategories identifies $\mathsf{D}^{\square}_{\mathrm{pc}}((T_R)^{\mathrm{An}})$ with the limit of the affine nuclear-pseudocompact categories on the Čech nerve. Applying the affine rational-base-change comparisons degreewise and taking this limit shows that rational pullback preserves the local nuclear-pseudocompact subcategories and that the result is independent of the affine cover and bounded-chart presentation.

Thus the local pseudocoherent categories form a cartesian full subdiagram of the ambient rational-localization diagram. Exactly as in Proposition 2.11, an ambient cartesian section belongs to the global full subcategory precisely when all its values belong to this local subdiagram; conversely every such cartesian local section descends to a global ambient object. Common rational, Zariski, and bounded-chart refinements prove independence of all choices. The final assertion follows from Proposition 3.4. □

Proposition 3.6 (Descent of solid quasicoherent categories on analytic stacks)**.** *Let*

$$\mathcal{C} = \mathrm{AnRing}^{\mathrm{op}}$$

with the topology $\mathcal{D}$ generated by finite families of jointly conservative steady localizations, and let

$$\mathrm{Shv}_{\mathcal{D}}(\mathcal{C})$$

be the resulting ∞-topos of analytic stacks. Let $a : \mathcal{U}_0 \to \mathcal{Z}$ be an effective epimorphism in this topos, and let $\mathcal{U}_\bullet$ be its Čech nerve. Then restriction is an equivalence in symmetric monoidal presentable stable ∞-categories

$$\mathcal{C}_{\square}(\mathcal{Z}) \xrightarrow{\sim} \lim_{[n]\in\Delta} \mathcal{C}_{\square}(\mathcal{U}_n). \tag{9}$$

More generally, for every small diagram $(\mathcal{Z}_i)_{i\in I}$ of analytic stacks, there is a natural equivalence

$$\mathcal{C}_{\square}\left(\operatorname*{colim}^{\mathrm{Shv}_{\mathcal{D}}}_{i\in I} \mathcal{Z}_i\right) \xrightarrow{\sim} \lim_{i\in I} \mathcal{C}_{\square}(\mathcal{Z}_i). \tag{10}$$

In particular, for every simplicial analytic stack $\mathcal{V}_\bullet$,

$$\mathcal{C}_{\square}(|\mathcal{V}_\bullet|) \simeq \lim_{[n]\in\Delta} \mathcal{C}_{\square}(\mathcal{V}_n).$$

Colimits in each category on the right are computed degreewise. For every $A \in \mathrm{CAlg}(\mathcal{C}_{\square}(\mathcal{Z}))$, restriction also gives an equivalence

$$\mathrm{Mod}_A(\mathcal{C}_{\square}(\mathcal{Z})) \xrightarrow{\sim} \lim_{[n]\in\Delta} \mathrm{Mod}_{A|_{\mathcal{U}_n}}(\mathcal{C}_{\square}(\mathcal{U}_n)). \tag{11}$$

In particular, the geometric-realization statement applies to the simplicial analytic-stack gluing diagrams by which Li–Huerta constructs algebraic realizations from schematic Zariski covers.

Proof. Li–Huerta A.2 endows $\mathcal{C} = \mathrm{AnRing}^{\mathrm{op}}$ with the topology $\mathcal{D}$ above and equips analytic affines with the symmetric monoidal presentable stable formalism

$$D_0 : \mathcal{C}^{\mathrm{op}} \longrightarrow \mathrm{CAlg}(\mathrm{Pr}^L_{\mathrm{st}}), \qquad \mathrm{AnSpec}(A, M) \longmapsto \mathrm{D}(A, M).$$

The affine formalism is sheafy for $\mathcal{D}$ [LH26, A.2][Sch26a, Proposition 12.18].

Let

$$y : \mathcal{C} \longrightarrow \mathrm{PSh}(\mathcal{C}) = \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathcal{S})$$

be the Yoneda embedding. Right Kan extension along y^{op} gives

$$\overline{D} : \mathrm{PSh}(\mathcal{C})^{\mathrm{op}} \longrightarrow \mathrm{CAlg}(\mathrm{Pr}^L_{\mathrm{st}}), \qquad \overline{D}(F) = \lim_{(y(U)\to F)} D_0(U),$$

where the limit ranges over affine representables mapping to F. Since the presheaf category is the free cocompletion of $\mathcal{C}$, this right-Kan extension is the unique extension of D_0 that carries small colimits of presheaves to limits. This is the categorical content of Scholze's right-Kan-extension formula in Proposition 5.16 [Sch26b, Proposition 5.16].

Let

$$L : \mathrm{PSh}(\mathcal{C}) \longrightarrow \mathrm{Shv}_{\mathcal{D}}(\mathcal{C})$$

be sheafification. Because D_0 satisfies $\mathcal{D}$-descent, $\overline{D}$ sends the Čech-realization map of every $\mathcal{D}$-cover to an equivalence. The morphisms inverted by sheafification form the strongly saturated class generated by these Čech-realization maps. The class of morphisms inverted by $\overline{D}$ is strongly saturated as well: equivalences and retracts are immediate, while pushouts and transfinite compositions in the presheaf category are carried to pullbacks and inverse limits, respectively, and equivalences are stable under both operations. Hence $\overline{D}$ inverts every sheafification equivalence and factors uniquely through a functor

$$D : \mathrm{Shv}_{\mathcal{D}}(\mathcal{C})^{\mathrm{op}} \longrightarrow \mathrm{CAlg}(\mathrm{Pr}^L_{\mathrm{st}}).$$

Equivalently, for every presheaf F there is a natural equivalence

$$\overline{D}(F) \simeq D(LF).$$

Scholze's Proposition 5.16 identifies this descended functor with the stack-level six-functor formalism and retains the affine-limit formula. Its value on an analytic stack is precisely the category denoted $\mathcal{C}_{\square}$ [Sch26b, Proposition 5.16].

Now let $(F_i)_{i\in I}$ be any small diagram of analytic stacks, regarded as presheaves. Colimits in the sheaf topos are obtained by sheafifying presheaf colimits, and the preceding factorization gives

$$\begin{aligned} D\left(\operatorname*{colim}^{\mathrm{Shv}}_i F_i\right) &\simeq D\left(L\left(\operatorname*{colim}^{\mathrm{PSh}}_i F_i\right)\right) \\ &\simeq \overline{D}\left(\operatorname*{colim}^{\mathrm{PSh}}_i F_i\right) \\ &\simeq \lim_i \overline{D}(F_i) \\ &\simeq \lim_i D(F_i). \end{aligned}$$

This proves the arbitrary-colimit equivalence (10); in particular, the stack-level functor sends geometric realizations of simplicial analytic stacks to totalizations of symmetric monoidal presentable stable categories.

Since a is an effective epimorphism,

$$\mathcal{Z} \simeq |\mathcal{U}_\bullet|$$

in $\mathrm{Shv}_{\mathcal{D}}(\mathcal{C})$. Applying the preceding colimit-to-limit formula gives (9). All transition functors in the totalization are pullbacks and hence preserve colimits. Limits of such diagrams in $\mathrm{Pr}^L_{\mathrm{st}}$ are created on the underlying ∞-categories, and colimits in the limit are computed degreewise. Finally, formation of commutative algebra objects and of modules over a cartesian commutative algebra commutes with limits of symmetric monoidal presentable categories [Lur17, Sections 3.2.2 and 4.8.5]. This proves (11). □

Lemma 3.7 (Zariski descent for the algebraic realization)**.** *Let T be a separated finitely presented X-scheme and let $(U_i)_{i\in I}$ be a finite Zariski open cover. Put $U_{i_0\dots i_n} = U_{i_0} \cap \cdots \cap U_{i_n}$. Restriction induces canonical equivalences*

$$\mathrm{D}_{\mathrm{qc}}(T) \simeq \lim_{[n]\in\Delta} \prod_{(i_0,\dots,i_n)\in I^{n+1}} \mathrm{D}_{\mathrm{qc}}(U_{i_0\dots i_n}), \tag{12}$$

$$\mathcal{C}_{\square}(T^{\mathrm{Alg}}_{\mathcal{X}}) \simeq \lim_{[n]\in\Delta} \prod_{(i_0,\dots,i_n)\in I^{n+1}} \mathcal{C}_{\square}(U^{\mathrm{Alg}}_{i_0\dots i_n}\mathcal{X}). \tag{13}$$

The relative algebraization functors form a natural transformation between these cosimplicial diagrams. Degreewise full faithfulness, equivalence, or colimit preservation therefore descends to the two limits. Symmetric monoidal structures, module objects, and pullback transformations descend through the same diagrams.

Proof. The first equivalence is ordinary Zariski descent for quasicoherent complexes. For the second, Li–Huerta's algebraic realization is the geometric realization of the simplicial analytic-stack gluing diagram obtained by applying the affine algebraic realization to the full schematic Čech nerve. The varying-base construction preserves this simplicial presentation by Lemma 3.3 and Proposition 3.4. The arbitrary geometric-realization clause of Proposition 3.6 therefore identifies the solid quasicoherent category of the realization with the displayed totalization.

The local algebraization functors commute with every face and degeneracy map, so they form a morphism of cosimplicial diagrams. Lemma 2.6 gives the assertions about full faithfulness and equivalences. Proposition 3.6 shows that colimits are computed degreewise and that the symmetric monoidal and module structures are the corresponding limits; this proves the remaining claims. □

3.2. **Ordinary relative Huber analytification.**

Proposition 3.8 (Global Huber realization)**.** *Let T be a finitely presented X-scheme. The local relative Huber analytifications*

$$(T_R)^{\mathrm{ad}/V} \longrightarrow V = \mathrm{Spa}(R, R^+), \qquad T_R = T \times_X \operatorname{Spec} R,$$

glue canonically to a locally strongly noetherian adic space $T^{\mathrm{ad}}_{\mathcal{X}}$ over $\mathcal{X}$, equipped with a morphism of locally ringed spaces

$$c_T : T^{\mathrm{ad}}_{\mathcal{X}} \longrightarrow T.$$

The construction commutes with rational base change on $\mathcal{X}$ and Zariski gluing in T. If $T \to X$ is proper, then $T^{\mathrm{ad}}_{\mathcal{X}} \to \mathcal{X}$ is proper.

Proof. Over a member $V = \mathrm{Spa}(R, R^+)$ of $\mathrm{RatAff}_X(\mathcal{X})$, Huber's Proposition 3.8 constructs the relative analytification of every locally finite type R-scheme and characterizes it by a mapping property [Hub94, Proposition 3.8]. This local object is independent of the affine neighborhood used to describe the map $\operatorname{Spec} R \to X$: if the map factors through two affine opens, it factors uniquely through their affine intersection, the two schemes denoted T_R are identified by associativity of fiber products, and Huber's representing property identifies their analytifications. Let $u : V' = \mathrm{Spa}(R', R'^+) \subset V$ be rational. Base changing the local analytification over V to V' has the same mapping property as the relative analytification of $T_{R'} = T_R \times_R R'$ over V'. The representing property therefore gives a unique isomorphism

$$(T_R)^{\mathrm{ad}/V} \times_V V' \xrightarrow{\sim} (T_{R'})^{\mathrm{ad}/V'}.$$

For a chain $V'' \subset V' \subset V$, both composites represent the identity natural transformation of the same mapping functor over V''; uniqueness of representing objects forces the cocycle equality.

Choose a rational-affinoid cover (V_i) of $\mathcal{X}$. For every multiple intersection $V_{i_0\dots i_n}$, the representing-property isomorphisms identify all restrictions of the local analytifications with the analytification

of T after base change to that intersection. The uniqueness part of the representing property implies that the face and degeneracy maps satisfy every Čech identity: after evaluating either composite on a test adic space over $V_{i_0\dots i_n}$, both are the identity transformation of the same represented mapping functor. Thus the local adic spaces form an effective descent datum for open gluing. Explicitly, their underlying spaces are glued along the specified open subsets; the sheaves of topological rings and the valuation maps glue because the overlap isomorphisms are isomorphisms of locally ringed adic spaces and satisfy the cocycle. The resulting adic space is denoted $T^{\mathrm{ad}}_{\mathcal{X}}$. The local maps to T agree after every restriction, again by uniqueness in the local mapping property, and hence define $c_T : T^{\mathrm{ad}}_{\mathcal{X}} \to T$. Any two such glued objects have locally identical restrictions and a unique compatible family of local isomorphisms; effective open descent then gives a unique global isomorphism. Applying this construction to the full Čech nerve of a Zariski affine cover of T, and using the same uniqueness argument on all multiple intersections, proves Zariski gluing in T.

If $T_R \to \operatorname{Spec} R$ is proper, then its relative Huber analytification $T_R^{\mathrm{ad}/V} \to V$ is proper by Zavyalov's Remark 6.3, which records Huber's properness theorem for relative analytification [Zav25, Remark 6.3]. The pullback of $T^{\mathrm{ad}}_{\mathcal{X}} \to \mathcal{X}$ to every V_i is therefore proper. Properness of a morphism of adic spaces is local on the target for open covers, so the global morphism is proper. □

3.3. **Module categories and relative full faithfulness.**

Lemma 3.9 (Module essential image)**.** *Let $F : \mathcal{C} \to \mathcal{D}$ be a fully faithful strong symmetric monoidal functor between presentable symmetric monoidal stable ∞-categories, and let $A \in \mathrm{CAlg}(\mathcal{C})$. Then*

$$\mathrm{Mod}_A(\mathcal{C}) \longrightarrow \mathrm{Mod}_{F(A)}(\mathcal{D})$$

is fully faithful. Its essential image consists exactly of the modules whose underlying object lies in the essential image of F.

Proof. Let $\mathcal{E} \subset \mathcal{D}$ be the full essential image of F. Strong monoidality makes $\mathcal{E}$ a full symmetric monoidal subcategory, and $F : \mathcal{C} \xrightarrow{\sim} \mathcal{E}$ is a symmetric monoidal equivalence. An $F(A)$-module structure on an object of $\mathcal{E}$ is encoded by a map from the associative operad to the corresponding colored endomorphism operad. Full faithfulness identifies this endomorphism operad with the one computed in $\mathcal{C}$. Thus module structures, their coherences, and module maps lift uniquely up to equivalence. Equivalently, the induced functor on module categories identifies $\mathrm{Mod}_A(\mathcal{C})$ with the full subcategory of $\mathrm{Mod}_{F(A)}(\mathcal{D})$ whose underlying objects belong to $\mathcal{E}$; compare [Lur17, Section 4.2.3]. □

Let $q : U \to X$ be affine and of finite presentation, and put

$$\mathcal{A} = q_*\mathcal{O}_U \in \mathrm{CAlg}(\mathrm{D}_{\mathrm{qc}}(X)).$$

Then

$$\mathrm{D}_{\mathrm{qc}}(U) \simeq \mathrm{Mod}_{\mathcal{A}}(\mathrm{D}_{\mathrm{qc}}(X)).$$

Lemma 3.10 (Integral induced analytic structure)**.** *Let $R \to B$ be a ring map, let $R^+ \subset R$ be an integrally closed subring, and let B^+ be the integral closure of the image of R^+ in B. Put*

$$\mathcal{R}_0 = (R_{\mathrm{disc}}, R^+_{\mathrm{disc}})^{\square}, \qquad \mathcal{B}_0 = (B_{\mathrm{disc}}, B^+_{\mathrm{disc}})^{\square}.$$

Then B, completed as a commutative algebra over $\mathcal{R}_0$, is already complete on its underlying module, and the analytic ring induced from $\mathcal{R}_0$ along $R \to B$ is canonically equivalent to $\mathcal{B}_0$. Consequently restriction of scalars gives a canonical symmetric monoidal equivalence

$$\mathsf{D}(\mathcal{B}_0) \xrightarrow{\sim} \mathrm{Mod}_B\big(\mathsf{D}(\mathcal{R}_0)\big).$$

Under the forgetful functors to $\mathrm{D}(B)$*, both sides are the full subcategory of* B*-complexes whose underlying* R*-complex is* $\mathcal{R}_0$*-complete. The construction is functorial in maps of such integral pairs.*

Proof. Let $\mathcal{B}_{\mathrm{ind}}$ be the normalized analytic ring induced from $\mathcal{R}_0$ along $R \to B$ in the sense of Scholze's Proposition 12.8. Equivalently, first complete the ordinary commutative R-algebra B in $\mathsf{D}(\mathcal{R}_0)$ and use the resulting commutative algebra object. Li–Huerta's Lemma A.1 and the discussion in A.3 identify its module category as

$$\mathsf{D}(\mathcal{B}_{\mathrm{ind}}) \simeq \mathrm{Mod}_B\big(\mathsf{D}(\mathcal{R}_0)\big) \tag{14}$$

[Sch26a, Proposition 12.8][LH26, A.1–A.3]. Under the forgetful functor to $\mathrm{D}(B)$, the right side consists precisely of those B-complexes whose underlying R-complex is $\mathcal{R}_0$-complete.

On the other hand, Andreychev's restriction criterion identifies $\mathcal{B}_0$-completeness of a B-complex with completeness for the minimal analytic structure on B^+. Since B^+ is integral over the image of R^+, invariance under integral extension identifies this with $\mathcal{R}_0$-completeness of the underlying R-complex [And21, Propositions 3.21 and 3.22]. Applied to the unit B, this also shows that the completion used to define the algebra object in the first paragraph has underlying module B itself.

Thus $\mathcal{B}_{\mathrm{ind}}$ and $\mathcal{B}_0$ have the same underlying condensed ring B and the same connective complete subcategory of $\mathrm{D}(B)$. The uniqueness theorem for normalized analytic structures gives a canonical equivalence

$$\mathcal{B}_{\mathrm{ind}} \simeq \mathcal{B}_0$$

[Sch26a, Proposition 12.20]. Combining this with (14) proves the displayed module equivalence. The proof of Proposition 12.20 reconstructs the normalized analytic structure from the reflector onto the prescribed full subcategory of complete connective modules. A left adjoint to a fixed fully faithful inclusion is unique up to a contractible space of choices. Therefore the identity on the underlying ring determines the displayed equivalence uniquely up to contractible choice. For a map of integral pairs, the two possible comparison composites are induced by the same underlying ring map and the same restriction functor on complete objects; the same uniqueness identifies them and supplies the identity, composition, and higher coherences. □

Lemma 3.11 (Affine relative algebraization is a module category)**.** *There is a canonical equivalence*

$$\mathcal{C}_{\square}(U_{\mathcal{X}}^{\mathrm{Alg}}) \simeq \mathrm{Mod}_{\alpha_X \mathcal{A}}(\mathcal{C}_{\square}(\mathcal{X})),$$

compatible with rational restriction on $\mathcal{X}$*. Under this equivalence,* Rq_*^{Alg} *is the forgetful functor to* $\mathcal{C}_{\square}(\mathcal{X})$*, and relative algebraization of an object of* $\mathrm{D}_{\mathrm{qc}}(U)$ *is the module functor induced by* α_X*.*

Proof. The assertion is local on $\mathcal{X}$. Let $V = \mathrm{Spa}(R, R^+) \subset \lambda^{-1}(W)$ and write

$$U_R = U \times_X \mathrm{Spec}\, R = \mathrm{Spec}\, B.$$

Set

$$\mathcal{R}_0 = (R_{\mathrm{disc}}, R^+_{\mathrm{disc}})^{\square}, \quad \mathcal{B}_0 = (B_{\mathrm{disc}}, B^+_{\mathrm{disc}})^{\square}, \quad \mathcal{R} = (R, R^+)^{\square},$$

where B^+ is the integral closure of R^+ in B. Li–Huerta A.9 gives

$$B_R^{\mathrm{alg}} = \mathcal{R} \otimes_{\mathcal{R}_0}^{\mathrm{AnRing}} \mathcal{B}_0.$$

By Lemma 3.10,

$$\mathsf{D}(\mathcal{B}_0) \simeq \mathrm{Mod}_B\big(\mathsf{D}(\mathcal{R}_0)\big)$$

as symmetric monoidal presentable categories over $\mathsf{D}(\mathcal{R}_0)$. Li–Huerta's Lemma A.1 identifies the module category of an analytic-ring pushout with the relative tensor product of the module

categories. The standard base-change formula for module categories therefore gives

$$\begin{aligned}\mathsf{D}(B_R^{\mathrm{alg}}) &\simeq \mathsf{D}(\mathcal{R}) \otimes_{\mathsf{D}(\mathcal{R}_0)} \mathsf{D}(\mathcal{B}_0)\\ &\simeq \mathsf{D}(\mathcal{R}) \otimes_{\mathsf{D}(\mathcal{R}_0)} \mathrm{Mod}_B\big(\mathsf{D}(\mathcal{R}_0)\big)\\ &\simeq \mathrm{Mod}_{\widehat{B}}\big(\mathsf{D}_\square(R,R^+)\big),\end{aligned}$$

where, if

$$F_{\mathcal{R}/\mathcal{R}_0} : \mathsf{D}(\mathcal{R}_0) \longrightarrow \mathsf{D}(\mathcal{R})$$

denotes the symmetric monoidal extension-of-scalars functor, then

$$\widehat{B} := F_{\mathcal{R}/\mathcal{R}_0}(B) \in \mathrm{CAlg}(\mathsf{D}_\square(R,R^+)).$$

Thus $\widehat{B}$ is the image of the commutative algebra object $B \in \mathrm{CAlg}(\mathsf{D}(\mathcal{R}_0))$, not a tensor product of analytic rings [LH26, Lemma A.1][Lur17, Section 4.8.5].

We identify this algebra object. On $W = \operatorname{Spec} A$, the algebra $\mathcal{A} = q_*\mathcal{O}_U$ corresponds to the A-algebra defining U. Restricting $\alpha_X\mathcal{A}$ to V applies derived scalar extension $R \otimes^{\mathbf{L}}_A -$ and condensification. Since $U_R = \operatorname{Spec} B$, this is precisely $\widehat{B}$ as a commutative algebra object of $\mathsf{D}_\square(R,R^+)$. Thus the displayed local equivalence is the required module-category description.

For a rational localization $V' \subset V$, Lemma 3.1 and associativity of analytic-ring pushouts identify the base change of this calculation with the same calculation for $B \otimes_R R'$. The induced equivalences of module categories are the reassociations of one relative tensor product and satisfy the composition cocycle. Proposition 3.6, applied to rational descent and to modules over the cartesian algebra $\alpha_X\mathcal{A}$, therefore glues the local equivalences to the asserted global equivalence.

Locally, the structural direct image is restriction of scalars and hence the forgetful functor. These forgetful functors commute with rational restriction, so descent identifies Rq_*^{Alg} with the global forgetful functor. The same calculation identifies relative algebraization with extension of scalars along α_X, compatibly with all rational restrictions. □

Lemma 3.12 (Fixed-base affine module category)**.** *Let (R,R^+) be a complete Tate Huber pair and let A be a finitely presented R-algebra. Write $(\operatorname{Spec} A)^{\mathrm{Alg}}$ for Li–Huerta's fixed-base algebraic realization and*

$$q_A^{\mathrm{Alg}} : (\operatorname{Spec} A)^{\mathrm{Alg}} \longrightarrow \mathrm{AnSpec}(R,R^+)^\square$$

for its structural morphism. There is a canonical symmetric monoidal equivalence

$$\mathcal{C}_\square((\operatorname{Spec} A)^{\mathrm{Alg}}) \simeq \mathrm{Mod}_{\mathrm{Cond}_R(A)}\big(\mathsf{D}_\square(R,R^+)\big). \tag{15}$$

Under this equivalence, $Rq_{A,*}^{\mathrm{Alg}}$ is the forgetful functor and fixed-base algebraization is the module functor induced by

$$\mathrm{Cond}_R : \mathrm{D}(R) \longrightarrow \mathsf{D}_\square(R,R^+).$$

Consequently

$$\alpha_A^{\mathrm{Alg}} : \mathrm{D}(A) \longrightarrow \mathcal{C}_\square((\operatorname{Spec} A)^{\mathrm{Alg}})$$

is fully faithful. Its essential image consists exactly of the modules whose underlying object of $\mathsf{D}_\square(R,R^+)$ belongs to the essential image of Cond_R.

Proof. Put

$$\mathcal{R}_0 = (R_{\mathrm{disc}}, R^+_{\mathrm{disc}})^\square, \qquad \mathcal{A}_0 = (A_{\mathrm{disc}}, A^+_{\mathrm{disc}})^\square, \qquad \mathcal{R} = (R,R^+)^\square,$$

where A^+ is the integral closure of the image of R^+ in A. Li–Huerta's definition gives

$$A_R^{\mathrm{alg}} = \mathcal{R} \otimes^{\mathrm{AnRing}}_{\mathcal{R}_0} \mathcal{A}_0.$$

Lemma 3.10, Li–Huerta's Lemma A.1, and the base-change formula for module categories give

$$\begin{aligned}\mathsf{D}(A_R^{\mathrm{alg}}) &\simeq \mathsf{D}(\mathcal{R}) \otimes_{\mathsf{D}(\mathcal{R}_0)} \mathsf{D}(\mathcal{A}_0) \\ &\simeq \mathsf{D}(\mathcal{R}) \otimes_{\mathsf{D}(\mathcal{R}_0)} \mathrm{Mod}_A\big(\mathsf{D}(\mathcal{R}_0)\big) \\ &\simeq \mathrm{Mod}_{\mathrm{Cond}_R(A)}\big(\mathsf{D}_\square(R, R^+)\big).\end{aligned}$$

The last algebra object is the image of $A \in \mathrm{CAlg}(\mathrm{D}(R))$ under the strong symmetric monoidal condensification functor. This proves (15); restriction of scalars is visibly the forgetful functor.

The functor Cond_R is fully faithful, strong symmetric monoidal, exact, and preserves filtered colimits and direct sums [And21, Theorem 5.9 and Lemma 5.10]. Lemma 3.9 therefore proves full faithfulness and the asserted essential-image description for α_A^{Alg}.

□

Proposition 3.13 (Fixed-base full faithfulness)**.** *Let (R, R^+) be a complete Tate Huber pair and let S be a separated finitely presented R-scheme. Fixed-base algebraization induces a fully faithful colimit-preserving strong symmetric monoidal functor*

$$\alpha_S^{\mathrm{Alg}} : \mathrm{D}_{\mathrm{qc}}(S) \longrightarrow \mathcal{C}_\square(S^{\mathrm{Alg}}).$$

It is functorial in S and compatible with pullback.

Proof. Choose a finite affine Zariski cover (S_a) of S. Since S is separated, every multiple intersection is affine. Lemma 3.12 gives a fully faithful colimit-preserving strong symmetric monoidal algebraization functor on every term of the full Čech nerve, and the affine scalar-extension comparisons commute with every face and degeneracy map. Ordinary Zariski descent for quasicoherent complexes and Proposition 3.6 for Li–Huerta's simplicial algebraic-realization diagram identify the global source and target with the corresponding totalizations. Lemma 2.6 therefore gives full faithfulness and colimit preservation; the monoidal structure and pullback transformations descend degreewise. A common affine refinement proves independence of the cover. □

Proposition 3.14 (Relative full faithfulness)**.** *For every separated finitely presented X-scheme T, relative algebraization induces a fully faithful colimit-preserving functor*

$$\alpha_T : \mathrm{D}_{\mathrm{qc}}(T) \longrightarrow \mathcal{C}_\square(T_{\mathcal{X}}^{\mathrm{Alg}}).$$

It is strong symmetric monoidal and compatible with pullback. If $T \to X$ is proper, we use Proposition 3.4 to regard its target as $\mathcal{C}_\square(T_{\mathcal{X}}^{\mathrm{An}})$.

Proof. If $T = U$ is affine over X, the assertion follows from Theorem 2.8, the algebraic module description, Lemma 3.11, and Lemma 3.9.

For general separated T, first choose a finite affine open cover (W_a) of X and then a finite affine cover of each quasi-compact open $T \times_X W_a$. We thereby obtain a finite affine cover (U_i) of T such that every composite $U_i \to W_a \to X$ is affine: the first map is a morphism between affine schemes, and the second is affine because X is separated. Since T is separated, every finite intersection of the U_i is affine. If the opens involved lie over $W_{a_1}, \dots, W_{a_k}$, their intersection maps into the affine open $W_{a_1} \cap \cdots \cap W_{a_k}$; the resulting morphism between affine schemes is affine, and the inclusion of that intersection into X is affine. Hence every iterated intersection is affine over X.

Apply Lemma 3.7 to this cover. In cosimplicial degree n, the comparison functor is the product over $(i_0, \dots, i_n)$ of the affine-over-X comparison functors for $U_{i_0 \dots i_n}$. Each factor is fully faithful by Lemmas 3.11 and 3.9, so the product is fully faithful. Lemma 2.6, together with the mapping-space formula for limits, then proves full faithfulness of the induced functor on T.

Each degreewise comparison preserves colimits. The restriction functors in the two Čech diagrams preserve colimits, so the descent categories are limits in presentable categories in which colimits

are computed degreewise. The final clause of Lemma 2.6 therefore proves colimit preservation globally.

On an affine intersection, strong monoidality is the canonical monoidal structure on extension of scalars. On a further intersection, the two restricted monoidal structures agree by associativity of tensor product; hence they form a monoidal natural transformation of the Čech diagrams. The same reasoning applies to the canonical base-change transformations expressing compatibility with pullback. Descent of these transformations supplies the global strong symmetric monoidal structure and the global pullback compatibility. This argument uses category-valued Zariski descent for the algebraic realization, not analytic openness of the maps $U_i^{\mathrm{Alg}}{}_{\mathcal{X}} \to T_{\mathcal{X}}^{\mathrm{Alg}}$. □

Proposition 3.15 (Forward preservation of pseudocoherence)**.** *Let T be a separated finitely presented X-scheme, and let*

$$\jmath_T : T_{\mathcal{X}}^{\mathrm{An}} \longrightarrow T_{\mathcal{X}}^{\mathrm{Alg}}$$

be the analytic-locus morphism. The composite

$$\alpha_T^{\mathrm{An}} := \jmath_T^* \alpha_T : \mathrm{D}_{\mathrm{pc}}(T) \longrightarrow \mathcal{C}_{\square}(T_{\mathcal{X}}^{\mathrm{An}})$$

has image in $\mathsf{D}_{\mathrm{pc}}^{\square}(T_{\mathcal{X}}^{\mathrm{An}})$. If T is proper, $\jmath_T$ is an isomorphism, so α_T^{An} is the restriction of the fully faithful functor of Proposition 3.14; in that case we again write it simply as α_T.

Proof. Let $M \in \mathrm{D}_{\mathrm{pc}}(T)$. Choose a finite Zariski affine cover of T and, over each member of the rational-affinoid basis of $\mathcal{X}$, refine the analytic realization by Li–Huerta's bounded analytic-affine charts. It is enough to verify the defining nuclearity and pseudocompactness conditions after restriction to this covering family, because $\mathsf{D}_{\mathrm{pc}}^{\square}$ is defined by descent for those steady localization covers.

On one such chart, write the algebraic coordinate ring as a finite type R-algebra B. Zariski locally on $\operatorname{Spec} B$, the restriction of M is represented by a bounded-above complex of finite free B-modules. Derived extension of scalars to the underlying ring of the bounded analytic chart carries this resolution to a bounded-above complex of finite free modules. Its condensification is nuclear and pseudocompact by the affine forward-preservation result preceding Li–Huerta's Theorem A.18, using Andreychev's relative-discreteness theorem and stability of nuclear objects [LH26, A.15 and the first paragraph of the proof of Theorem A.18].

For two bounded charts with one contained in the other, the two local objects are related by the scalar-extension isomorphism used to construct α_T^{An}; hence the local nuclear-pseudocompact objects and their descent maps are the restrictions of a single object of $\mathcal{C}_{\square}(T_{\mathcal{X}}^{\mathrm{An}})$. Descent of nuclearity and pseudocompactness along the covering family therefore places $\jmath_T^* \alpha_T(M)$ in $\mathsf{D}_{\mathrm{pc}}^{\square}(T_{\mathcal{X}}^{\mathrm{An}})$. □

3.4. **Flatness of global relative analytification.**

Proposition 3.16 (Flatness of global relative analytification)**.** *For every finitely presented X-scheme T, the morphism*

$$c_T : T_{\mathcal{X}}^{\mathrm{ad}} \longrightarrow T$$

is flat.

Proof. Flatness is local on source and target. Use a standard affine $W = \operatorname{Spec} A \subset X$, an affinoid $V = \operatorname{Spa}(R, R^+) \subset \lambda^{-1}(W)$, and an affine $U \subset T \times_X W$. Over V, the map factors as

$$U_R^{\mathrm{ad}/V} \xrightarrow{c_{U_R/V}} U_R \longrightarrow U, \qquad U_R = U \times_A \operatorname{Spec} R.$$

The second arrow is flat by Lemma 2.9. The first arrow is flat on local rings by Lemma 6.7 of Zavyalov's *Some Foundational Results in Adic Geometry* [Zav25, Lemma 6.7]. Their composite is flat. □

4. Proper pushforward and localization at the boundary

4.1. **Globalizing solid Grauert.**

4.1.1. *The fixed-base input.* The following is Li–Huerta's Corollary A.17, with notation aligned to the present paper [LH26, Corollary A.17, pp. 58–59].

Let (R, R^+) be a complete Tate Huber pair. For a scheme S locally of finite type over R, write S^{An} for Li–Huerta's analytic realization and

$$\pi : S^{\mathrm{An}} \longrightarrow \mathrm{AnSpec}(R, R^+)^{\square}$$

for its structural morphism. Li–Huerta writes $D_{\mathrm{pc}}(S^{\mathrm{An}})$ for the full subcategory of $\mathcal{C}_{\square}(S^{\mathrm{An}})$ consisting of objects that are both nuclear and pseudocompact; this is the category denoted $\mathsf{D}^{\square}_{\mathrm{pc}}(S^{\mathrm{An}})$ in the main text. We view

$$\mathrm{Cond}_R : \mathrm{D}_{\mathrm{pc}}(R) \hookrightarrow \mathsf{D}_{\square}(R, R^+)$$

as a fully faithful functor, as furnished by Andreychev's comparison theorem.

Theorem 4.1 (Local solid Grauert; Li–Huerta)**.** *Let (R, R^+) be a complete Tate Huber pair, and let S be a proper scheme of finite presentation over R. Then the solid direct-image functor*

$$R\pi_* : \mathcal{C}_{\square}(S^{\mathrm{An}}) \longrightarrow \mathsf{D}_{\square}(R, R^+)$$

sends $\mathsf{D}^{\square}_{\mathrm{pc}}(S^{\mathrm{An}})$ into the essential image of

$$\mathrm{Cond}_R : \mathrm{D}_{\mathrm{pc}}(R) \hookrightarrow \mathsf{D}_{\square}(R, R^+).$$

Equivalently, for every $K \in \mathsf{D}^{\square}_{\mathrm{pc}}(S^{\mathrm{An}})$ there exists an object $M \in \mathrm{D}_{\mathrm{pc}}(R)$ and an equivalence

$$\mathrm{Cond}_R(M) \simeq R\pi_* K$$

in $\mathsf{D}_{\square}(R, R^+)$. The object M is unique up to a contractible space of choices.

Corollary 4.2 (Colimit preservation used in the global argument)**.** *In the setting of Theorem 4.1, the functor*

$$R\pi_* : \mathcal{C}_{\square}(S^{\mathrm{An}}) \longrightarrow \mathsf{D}_{\square}(R, R^+)$$

commutes with arbitrary colimits.

Proof. Li–Huerta's proof identifies $R\pi_*$ with $R\pi_!$, which is left adjoint to $\pi^!$ and therefore preserves colimits. □

4.1.2. *Globalization.* Let $g : Z \to X$ be proper and of finite presentation. Its relative analytification will be denoted

$$g^{\mathrm{An}} : Z^{\mathrm{An}}_{\mathcal{X}} \longrightarrow \mathcal{X}.$$

Lemma 4.3 (Open Beck–Chevalley)**.** *Let $i : V \hookrightarrow \mathcal{X}$ be a quasi-compact morphism in Li–Huerta's class I of analytic open embeddings. In particular, i may be any rational-affinoid open immersion used below. Form the cartesian square*

$$\begin{array}{ccc} Z^{\mathrm{An}}_V & \xrightarrow{i'} & Z^{\mathrm{An}}_{\mathcal{X}} \\ {\scriptstyle g^{\mathrm{An}}_V}\downarrow & & \downarrow{\scriptstyle g^{\mathrm{An}}} \\ V & \xrightarrow{i} & \mathcal{X}. \end{array}$$

Then the canonical transformation

$$i^* R g^{\mathrm{An}}_* \longrightarrow R g^{\mathrm{An}}_{V,*}\, i'^*$$

is an equivalence on $\mathcal{C}_{\square}(Z^{\mathrm{An}}_{\mathcal{X}})$.

Proof. Li–Huerta A.2 defines class I by the condition that pullback on solid module categories is an open embedding. Rodríguez Camargo's Proposition 2.3.2 proves that restriction to any analytic open of a Tate adic space is an open localization on solid module categories; in particular, every rational-affinoid open immersion used here belongs to I [RC24, Proposition 2.3.2]. The suitable-decomposition axioms recorded by Li–Huerta imply that I is stable under base change, so the morphism i' in the displayed square also lies in I [LH26, A.2]. Scholze's Proposition 5.16 and Theorem 5.19 extend the affine formalism and the class of allowed morphisms to analytic stacks [Sch26b, Proposition 5.16 and Theorem 5.19]. For a class-I morphism, pullback admits an exact fully faithful left adjoint given by extension by zero, and the Beck–Chevalley transformations are part of the correspondence formalism [Sch26b, Definition 2.5]. Thus there is an exact functor

$$i_! : \mathcal{C}_\square(V) \longrightarrow \mathcal{C}_\square(\mathcal{X}), \qquad i_! \dashv i^*,$$

and the exceptional base-change equivalence

$$(g^{\mathrm{An}})^* i_! \simeq i'_!(g_V^{\mathrm{An}})^*. \tag{16}$$

The corresponding assertions for i' hold as well.

Let $A \in \mathcal{C}_\square(Z_{\mathcal{X}}^{\mathrm{An}})$ and $B \in \mathcal{C}_\square(V)$. Applying the adjunctions and (16) gives

$$\begin{aligned}
\mathrm{Map}_V(B, i^* R g_*^{\mathrm{An}} A) &\simeq \mathrm{Map}_{\mathcal{X}}(i_! B, R g_*^{\mathrm{An}} A) \\
&\simeq \mathrm{Map}_{Z_{\mathcal{X}}^{\mathrm{An}}}((g^{\mathrm{An}})^* i_! B, A) \\
&\simeq \mathrm{Map}_{Z_{\mathcal{X}}^{\mathrm{An}}}(i'_!(g_V^{\mathrm{An}})^* B, A) \\
&\simeq \mathrm{Map}_{Z_V^{\mathrm{An}}}((g_V^{\mathrm{An}})^* B, i'^* A) \\
&\simeq \mathrm{Map}_V(B, R g_{V,*}^{\mathrm{An}} i'^* A).
\end{aligned}$$

The equivalence is natural in A and B. Under the adjunctions it is the mate of the cartesian square, hence agrees with the canonical Beck–Chevalley transformation. Yoneda proves the claim. □

Proposition 4.4 (Solid Grauert over the Fargues–Fontaine curve)**.** *Let $g : Z \to X$ be proper and of finite presentation. Then*

$$R g_*^{\mathrm{An}} : \mathsf{D}_{\mathrm{pc}}^{\square}(Z_{\mathcal{X}}^{\mathrm{An}}) \longrightarrow \mathsf{D}_{\mathrm{pc}}^{\square}(\mathcal{X})$$

preserves chartwise pseudocoherent objects. Moreover, the functor

$$R g_*^{\mathrm{An}} : \mathcal{C}_\square(Z_{\mathcal{X}}^{\mathrm{An}}) \longrightarrow \mathcal{C}_\square(\mathcal{X})$$

commutes with arbitrary colimits.

Proof. Choose a member

$$V = \mathrm{Spa}(R, R^+) \in \mathrm{RatAff}_X(\mathcal{X}).$$

By Proposition 3.4, the pullback of $Z_{\mathcal{X}}^{\mathrm{An}}$ to V is the Li–Huerta analytification of the proper finitely presented R-scheme

$$Z_R = Z \times_X \mathrm{Spec}\, R.$$

Lemma 4.3 identifies the restriction of $R g_*^{\mathrm{An}}$ to V with

$$R(g_R)_*^{\mathrm{An}}.$$

Theorem 4.1, which records Li–Huerta's Corollary A.17, states that this local functor sends nuclear pseudocompact objects to the essential image of $\mathrm{D}_{\mathrm{pc}}(R)$ in $\mathsf{D}_\square(R, R^+)$. Let $K \in \mathsf{D}_{\mathrm{pc}}^{\square}(Z_{\mathcal{X}}^{\mathrm{An}})$. Its restriction to Z_R^{An} is nuclear and pseudocompact by definition. The local theorem therefore shows that $(R g_*^{\mathrm{An}} K)|_V$ is pseudocoherent for every member V of the rational-affinoid basis. On a rational inclusion $V' \subset V$, the identifications are related by the open Beck–Chevalley equivalence of Lemma 4.3; hence they are the restrictions of the single global object $R g_*^{\mathrm{An}} K$. The defining descent condition for $\mathsf{D}_{\mathrm{pc}}^{\square}(\mathcal{X})$ is therefore satisfied, proving the first assertion.

For colimits, let (K_a) be any diagram in $\mathcal{C}_\square(Z_{\mathcal{X}}^{\mathrm{An}})$. There is a canonical morphism

$$\operatorname*{colim}_a Rg_*^{\mathrm{An}} K_a \longrightarrow Rg_*^{\mathrm{An}}\Big(\operatorname*{colim}_a K_a\Big).$$

After restriction to V, open base change identifies this map with the corresponding map for $(g_R)^{\mathrm{An}}$. Corollary 4.2 records the consequence of Li–Huerta's identification

$$R(g_R)_*^{\mathrm{An}} \simeq R(g_R)_!^{\mathrm{An}},$$

namely that the local direct image preserves arbitrary colimits. Thus the restriction of the comparison morphism to every member of a rational-affinoid cover is an equivalence. Pullback to that cover is jointly conservative in $\mathcal{C}_\square(\mathcal{X})$: the fiber of a morphism restricts to zero on every covering member only when it is zero by descent. Applying this to the fiber of the comparison morphism proves that the comparison is an equivalence globally. □

Lemma 4.5 (Right adjoints and categorical descent)**.** *Let $\mathcal{C}^\bullet$ and $\mathcal{D}^\bullet$ be cosimplicial diagrams of presentable stable ∞-categories, and let*

$$F^\bullet : \mathcal{C}^\bullet \longrightarrow \mathcal{D}^\bullet$$

be a morphism of diagrams. Suppose that every F^n admits a right adjoint G^n and that every square determined by a morphism of Δ is right-adjointable. Then the G^n form a morphism of the opposite descent diagrams, and the induced functor

$$F : \lim_{[n]\in\Delta} \mathcal{C}^n \longrightarrow \lim_{[n]\in\Delta} \mathcal{D}^n$$

admits the degreewise right adjoint

$$G : \lim_{[n]\in\Delta} \mathcal{D}^n \longrightarrow \lim_{[n]\in\Delta} \mathcal{C}^n.$$

Proof. Let $\theta : [m] \to [n]$ be a morphism of Δ, and denote the corresponding transition functors by $c_\theta : \mathcal{C}^m \to \mathcal{C}^n$ and $d_\theta : \mathcal{D}^m \to \mathcal{D}^n$. The morphism of diagrams gives an equivalence $d_\theta F^m \simeq F^n c_\theta$. Right-adjointability means that its right-adjoint mate

$$c_\theta G^m \xrightarrow{\sim} G^n d_\theta$$

is an equivalence. If $y = (y_n)$ is cartesian, these mates transport its cartesian structure maps to equivalences relating the objects $G^n y_n$. The compatibility of mates with composition supplies the simplicial cocycles, so $(G^n y_n)_n$ is a cartesian section. This defines G on objects and morphisms.

For cartesian sections $x = (x_n)$ and $y = (y_n)$, mapping spectra in a limit of stable ∞-categories are computed degreewise. Hence

$$\begin{aligned}\mathrm{Map}_{\lim \mathcal{D}^\bullet}(Fx, y) &\simeq \lim_{[n]\in\Delta} \mathrm{Map}_{\mathcal{D}^n}(F^n x_n, y_n)\\ &\simeq \lim_{[n]\in\Delta} \mathrm{Map}_{\mathcal{C}^n}(x_n, G^n y_n)\\ &\simeq \mathrm{Map}_{\lim \mathcal{C}^\bullet}(x, Gy).\end{aligned}$$

These equivalences are natural in both variables and therefore exhibit G as the right adjoint of F. □

4.2. **Algebraic Cartier localization.**

Lemma 4.6 (Fixed-base localization by a line-bundle section)**.** *Let (R, R^+) be a complete Tate Huber pair, let S be a separated finitely presented R-scheme, let $\mathcal{L}$ be a line bundle on S, and let*

$$s : \mathcal{O}_S \longrightarrow \mathcal{L}$$

be a section. Write $U = S_s \subset S$ for its nonvanishing locus and $j : U \hookrightarrow S$ for the open immersion. Use the same symbols $\mathcal{L}$ and s for their fixed-base algebraizations. Then

$$j^{\mathrm{Alg},*} : \mathcal{C}_\square(S^{\mathrm{Alg}}) \longrightarrow \mathcal{C}_\square(U^{\mathrm{Alg}})$$

admits a right adjoint Rj_*^{Alg}, and for every $K \in \mathcal{C}_\square(S^{\mathrm{Alg}})$ there is a natural equivalence

$$Rj_*^{\mathrm{Alg}} j^{\mathrm{Alg},*} K \simeq \operatorname*{colim}_{n\geq 0}\big(K \otimes \mathcal{L}^{\otimes n}\big), \tag{17}$$

whose transition maps are induced by s. No regularity assumption on the section is required.

Proof. Choose a finite affine Zariski cover (S_a) on which $\mathcal{L}$ is trivial. Since S is separated, every multiple intersection is affine. On an intersection $\operatorname{Spec} A$, write the section as an element $t \in A$; then its nonvanishing locus is $\operatorname{Spec} A[t^{-1}]$, even when t is a zero divisor. Lemma 3.12 identifies restriction to this open with

$$- \otimes_A^{\mathbf{L}} A[t^{-1}],$$

whose right adjoint is restriction of scalars.

For a further Zariski restriction $A \to A'$, the right-adjoint mate is

$$A' \otimes_A^{\mathbf{L}} \operatorname{Res}_{A[t^{-1}]/A} M \longrightarrow \operatorname{Res}_{A'[t^{-1}]/A'} \left(A'[t^{-1}] \otimes_{A[t^{-1}]}^{\mathbf{L}} M\right).$$

The localization identity

$$A'[t^{-1}] \simeq A' \otimes_A^{\mathbf{L}} A[t^{-1}]$$

and associativity of derived tensor product make this mate an equivalence. The two identifications for a chain $A \to A' \to A''$ are the two reassociations of the same iterated tensor product and therefore satisfy all composition coherences. Li–Huerta's Zariski construction of S^{Alg}, Proposition 3.6, and Lemma 4.5 consequently glue the affine right adjoints to Rj_*^{Alg}.

On $\operatorname{Spec} A$ one has

$$A[t^{-1}] \simeq \operatorname{colim}\big(A \xrightarrow{t} A \xrightarrow{t} A \xrightarrow{t} \cdots\big).$$

Tensoring with K gives the local form of (17). Under a trivialization of $\mathcal{L}$, multiplication by the section is multiplication by t. Changing the trivialization multiplies t by a unit and gives the canonically equivalent telescope characterized by universally inverting s. The local equivalences therefore agree on all Čech intersections and descend to the asserted global equivalence. □

Lemma 4.7 (Algebraic localization along an effective Cartier divisor)**.** *Let Z be a separated finitely presented X-scheme, let $D \subset Z$ be an effective Cartier divisor, and let*

$$j : U = Z \setminus D \hookrightarrow Z, \qquad \mathcal{I} = \mathcal{O}_Z(-D), \qquad \mathcal{L} = \mathcal{I}^{-1} = \mathcal{O}_Z(D).$$

Let $s : \mathcal{O}_Z \to \mathcal{L}$ be the canonical section, and use the same symbols $\mathcal{L}$ and s for their images under relative algebraization. The algebraic extension-of-scalars functor

$$j^{\mathrm{Alg},*} : \mathcal{C}_\square(Z_{\mathcal{X}}^{\mathrm{Alg}}) \longrightarrow \mathcal{C}_\square(U_{\mathcal{X}}^{\mathrm{Alg}})$$

admits a right adjoint Rj_*^{Alg}. For every $K \in \mathcal{C}_\square(Z_{\mathcal{X}}^{\mathrm{Alg}})$ there is a natural equivalence

$$Rj_*^{\mathrm{Alg}} j^{\mathrm{Alg},*} K \simeq \operatorname*{colim}_{n\geq 0}\big(K \otimes \mathcal{L}^{\otimes n}\big), \tag{18}$$

where the transition maps are multiplication by s. This is an algebraic-localization statement in module categories; it does not assert that $U_{\mathcal{X}}^{\mathrm{Alg}} \to Z_{\mathcal{X}}^{\mathrm{Alg}}$ is an analytic open immersion.

Proof. Choose a finite affine cover (Z_a) of Z, subordinate to affine opens of X, such that $D|_{Z_a}$ is principal. On an iterated intersection $Z_{a_0\dots a_n} = \operatorname{Spec} A$, write $D = (t)$; then

$$U \cap Z_{a_0\dots a_n} = \operatorname{Spec} A[t^{-1}].$$

Under Lemma 3.11, restriction to the complement is the extension-of-scalars functor

$$- \otimes^{\mathbf{L}}_{A} A[t^{-1}],$$

whose right adjoint is restriction of scalars. For a further Zariski restriction $A \to A'$, consider the square formed by localization and derived base change. Its right-adjoint mate, evaluated on an $A[t^{-1}]$-module M, is

$$A' \otimes^{\mathbf{L}}_{A} \mathrm{Res}_{A[t^{-1}]/A}\, M \longrightarrow \mathrm{Res}_{A'[t^{-1}]/A'} \left(A'[t^{-1}] \otimes^{\mathbf{L}}_{A[t^{-1}]} M \right).$$

The canonical equivalence

$$A'[t^{-1}] \simeq A' \otimes^{\mathbf{L}}_{A} A[t^{-1}]$$

and associativity of derived tensor products identify this mate with an equivalence. For $A \to A' \to A''$, the two resulting identifications are the two standard reassociations of the same derived tensor product, so they satisfy the composition cocycle. Thus the affine right adjoints form a morphism of the Čech descent diagrams.

Lemma 3.7 identifies the global source and target categories with the limits of these diagrams. Lemma 4.5 therefore constructs the global right adjoint Rj_*^{Alg} and computes it degreewise by restriction of scalars.

On an affine intersection, one consequently has

$$Rj_*^{\mathrm{Alg}} j^{\mathrm{Alg},*} K \simeq K \otimes^{\mathbf{L}}_{A} A[t^{-1}]. \tag{19}$$

The canonical localization identity

$$A[t^{-1}] \simeq \mathrm{colim}\big(A \xrightarrow{t} A \xrightarrow{t} A \xrightarrow{t} \cdots\big)$$

holds in the derived category of A-modules. Relative algebraization is symmetric monoidal and preserves colimits, so tensoring with K identifies the right side of (19) with

$$\mathrm{colim}\big(K \xrightarrow{t} K \xrightarrow{t} K \xrightarrow{t} \cdots\big).$$

Locally $\mathcal{L} = \mathcal{O}_Z(D)$ is trivialized by the generator t^{-1}, and under this trivialization the canonical section $s : \mathcal{O}_Z \to \mathcal{O}_Z(D)$ is multiplication by t. Hence the displayed telescope is exactly the local form of (18). Changing the local equation multiplies t by a unit, and the resulting telescope is canonically equivalent; equivalently, it is characterized by universally inverting the section s. The affine equivalences are therefore compatible on every Čech intersection. Passing to the limit gives

$$Rj_*^{\mathrm{Alg}} j^{\mathrm{Alg},*} K \simeq \operatorname*{colim}_{n \geq 0} (K \otimes \mathcal{L}^{\otimes n})$$

with transition maps induced by s. Thus the sign is fixed unambiguously by the local telescope $A \xrightarrow{t} A \xrightarrow{t} \cdots$, and the global terms are $K \otimes \mathcal{O}_Z(nD)$. The posted v1 proof of Li–Huerta's Theorem A.18 writes $\mathcal{O}_Z(-nD)$. The correct localization terms are the positive twists above; Proposition 4.15 uses the same corrected telescope in the fixed-base proof [LH26, proof of Theorem A.18]. □

Lemma 4.8 (Making the boundary Cartier)**.** *Let Y be noetherian and let $j : U \hookrightarrow Y$ be a quasi-compact open immersion. There is a coherent ideal $\mathcal{J} \subset \mathcal{O}_Y$ with vanishing locus $Y \setminus U$ such that, for*

$$p : \widetilde{Y} = \mathrm{Bl}_{\mathcal{J}}(Y) \longrightarrow Y,$$

the following hold:

(i) *p is projective and of finite presentation (hence proper), and is an isomorphism over U;*
(ii) *$\mathcal{J}\mathcal{O}_{\widetilde{Y}} = \mathcal{O}_{\widetilde{Y}}(-D)$ is an invertible ideal;*
(iii) *the complement of U in $\widetilde{Y}$ is the effective Cartier divisor D.*

If $Y \to X$ is proper and of finite presentation, then $\widetilde{Y} \to X$ is proper and of finite presentation.

Proof. Because Y is noetherian and $Y \setminus U$ is closed, it is the vanishing locus of a coherent, hence finitely generated, ideal $\mathcal{J}$. The standard properties of the blowup [Stacks, Section 31.33, especially Lemma 31.33.4] show that p is projective and of finite presentation, is an isomorphism where $\mathcal{J}$ is the unit ideal, and makes $\mathcal{J}\mathcal{O}_{\widetilde{Y}}$ invertible. Its zero locus is the inverse image of $Y \setminus U$, hence is exactly the complement of U. An invertible ideal defines an effective Cartier divisor. If $Y \to X$ is proper and of finite presentation, composition shows that $\widetilde{Y} \to X$ is again proper and of finite presentation. □

Lemma 4.9 (A finitely presented boundary modification)**.** *Let R be a ring, let S be a proper finitely presented R-scheme, and let $j : U \hookrightarrow S$ be a quasi-compact open immersion. There exist a proper finitely presented morphism*

$$p : \widetilde{S} \longrightarrow S$$

that is an isomorphism over U, a line bundle $\mathcal{L}$ on $\widetilde{S}$, and a section

$$s : \mathcal{O}_{\widetilde{S}} \longrightarrow \mathcal{L}$$

whose nonvanishing locus is the copy of U in $\widetilde{S}$.

Proof. Because S is quasi-compact and quasi-separated and U is quasi-compact, there is a finite type quasi-coherent ideal $\mathcal{J} \subset \mathcal{O}_S$ whose vanishing locus is $S \setminus U$ [Stacks, Tag 01PH]. The existence proof for almost blow-up squares in Stacks Project, Lemma 38.37.4, writes the pair $(S, V(\mathcal{J}))$ as the base change of a pair over a noetherian scheme, blows up there, and then base changes the resulting diagram [Stacks, Tag 0EVA]. Let $p : \widetilde{S} \to S$ be the resulting upper horizontal modification. By construction, p is proper and of finite presentation and is an isomorphism over U.

On the noetherian model, the standard blowup theorem identifies the inverse image of the center with an effective Cartier divisor D_0 and identifies its ideal with $\mathcal{O}(-D_0)$ [Stacks, Tag 02OS]. Thus $\mathcal{O}(D_0)$ carries its canonical section. Base changing this line bundle and section gives a line bundle $\mathcal{L}$ and a section s on $\widetilde{S}$. Formation of the nonvanishing locus of a section commutes with base change, and on the noetherian model that locus is the complement of the exceptional divisor. Hence the nonvanishing locus of s is precisely $p^{-1}(U) \simeq U$. Notice that the base-changed section need not remain regular; Lemma 4.6 was formulated so that no regularity is needed. □

4.3. **Fixed-base comparison with Huber analytification.** Recall from Li–Huerta's A.19 that the assignment

$$\mathrm{Spa}(A, A^+) \longmapsto \mathrm{AnSpec}(A, A^+)^{\square}$$

from Tate affinoid adic spaces to analytic stacks sends rational covers to steady-localization covers and therefore glues to a functor

$$(-)^{\square} : \{\text{analytic adic spaces}\} \longrightarrow \{\text{analytic stacks}\}.$$

Lemma 4.10 (Open gluing and increasing unions for solid avatars)**.** *Let Z be a quasiseparated analytic adic space and let $(Z_i)_{i \in I}$ be an arbitrary open cover by analytic open subspaces. Put*

$$Z_{i_0 \ldots i_n} = Z_{i_0} \cap \cdots \cap Z_{i_n}.$$

Then the canonical morphism of analytic stacks

$$\left| [n] \longmapsto \coprod_{(i_0, \ldots, i_n) \in I^{n+1}} Z^{\square}_{i_0 \ldots i_n} \right| \longrightarrow Z^{\square}$$

is an equivalence. In particular, if

$$Z_0 \subset Z_1 \subset \cdots \subset Z$$

is an increasing open cover by affinoid subspaces and every $Z_n \hookrightarrow Z_{n+1}$ is a rational open immersion, then

$$\operatorname*{colim}_{n\geq 0} Z_n^{\square} \xrightarrow{\sim} Z^{\square}.$$

Moreover, every quasicompact affinoid open $W \subset Z$ is contained in some Z_N and admits a finite affinoid cover whose members are rational subdomains of both W and Z_N.

Proof. Li–Huerta's A.19 constructs $(-)^{\square}$ by gluing the affinoid assignment along rational covers. Choose, for every i, a rational-affinoid cover of Z_i, and choose these covers together as a rational-affinoid cover of Z subordinate to $(Z_i)_{i\in I}$. Because Z is quasiseparated, every finite intersection of members of this affinoid cover is quasicompact and admits a finite rational-affinoid refinement. Applying the same construction to all multiple intersections produces a bisimplicial diagram of affinoid solid avatars. Realizing first in the rational-affinoid direction computes each $Z^{\square}_{i_0\dots i_n}$ by the gluing construction of A.19, and then realizing in the Čech direction computes the left side of the displayed morphism. Realizing in the opposite order first computes the solid avatar of the rational-affinoid Čech presentation of Z, hence $Z^{\square}$. Geometric realizations commute with one another, so the two iterated realizations agree. This proves the asserted equivalence, and the same common-refinement argument proves independence of all choices and compatibility with refinements.

For the increasing cover, the maps $Z_n^{\square} \to Z_{n+1}^{\square}$ are monomorphisms because rational open immersions induce open localizations on solid module categories and hence become analytic open embeddings [RC24, Proposition 2.3.2]. In an ∞-topos, the colimit of an increasing sequence of subobjects is their union. The first assertion identifies that union with $Z^{\square}$, proving the second assertion.

Finally, the opens $W \cap Z_n$ form an increasing cover of the quasicompact space W, so a finite subcover is already contained in one $W \cap Z_N$ and hence $W \subset Z_N$. For every $x \in W$, choose a rational subdomain $W_x \subset Z_N$ with $x \in W_x \subset W$; rational subdomains form a basis on the affinoid Z_N. Rational subdomains are stable under base change, so the inclusion $W \hookrightarrow Z_N$ identifies W_x with the same rational subdomain of the affinoid W. Quasicompactness of W supplies finitely many such W_x, proving the last assertion. □

Proposition 4.11 (Li–Huerta and Huber analytifications agree)**.** *Let $V = \operatorname{Spa}(R, R^+)$ be a strongly noetherian complete Tate affinoid, and let T be a separated finitely presented R-scheme. There is a canonical equivalence of analytic stacks over V*

$$\vartheta_{T,V} : T^{\mathrm{An}} \xrightarrow{\sim} (T^{\mathrm{ad}/V})^{\square}. \tag{20}$$

It is functorial in T, compatible with Zariski restriction, and compatible with rational base change on V, including the identity, composition, and higher associativity coherences.

Under (20)*, Li–Huerta's subcategory $\mathsf{D}^{\square}_{\mathrm{pc}}(T^{\mathrm{An}})$ is identified with the full subcategory of $\mathcal{C}_{\square}((T^{\mathrm{ad}/V})^{\square})$ whose restriction to every strongly noetherian affinoid open $W = \operatorname{Spa}(C, C^+)$ belongs to the essential image of*

$$\mathrm{Cond}_C : \mathrm{D}_{\mathrm{pc}}(C) \hookrightarrow \mathsf{D}_{\square}(C, C^+).$$

Moreover, if $W \subset c_{T/V}^{-1}(U)$ with $U = \operatorname{Spec} A \subset T$ affine, then for every $M \in \mathrm{D}_{\mathrm{qc}}(T)$ the restriction of fixed-base algebraization, transported to the analytic locus, satisfies

$$\alpha_T^{\mathrm{LH}}(M)|_{W^{\square}} \simeq \mathrm{Cond}_C\big(C \otimes^{\mathbf{L}}_A R\Gamma(U, M|_U)\big). \tag{21}$$

Proof. Huber's Proposition 3.8 constructs $T^{\mathrm{ad}/V}$ for every locally finite type R-scheme, and Zavyalov's Examples 6.4–6.5 describe it by bounded affinoid charts [Hub94, Proposition 3.8][Zav25, Examples 6.4–6.5]. Li–Huerta's proof of A.19 compares an existing adic analytification with T^{An} by the same bounded charts. We spell out why that comparison applies here without smoothness.

Assume first that $T = \operatorname{Spec} B$ is affine. Choose a presentation

$$B = R[T_1, \dots, T_m]/I,$$

write b_i for the image of T_i, and choose a pseudo-uniformizer $\varpi \in R$. For the bounded chart attached to the tuple $(\varpi^n b_i)_i$, regard B as an $R[T_1, \dots, T_m]$-algebra through $T_i \mapsto \varpi^n b_i$. Since ϖ is a unit, the automorphism $T_i \mapsto \varpi^n T_i$ identifies this algebra structure with the original presentation. Below, I denotes the transported kernel under this automorphism; with this convention the completed polynomial ring is written $R\langle \varpi^n T_1, \dots, \varpi^n T_m\rangle$ and its quotient is written using the same symbol I. For $n \geq 0$, let $W_n \subset T^{\mathrm{ad}/V}$ be the affinoid subspace cut out by

$$|\varpi^n b_i| \leq 1 \qquad (1 \leq i \leq m).$$

Zavyalov's Examples 6.4–6.5 identify $T^{\mathrm{ad}/V}$ with the increasing union of the W_n: relative affine space is the union of the closed polydiscs of radii $|\varpi|^{-n}$, and $T^{\mathrm{ad}/V}$ is the common vanishing locus of a finite set of generators of I inside that union. Moreover, $W_n \hookrightarrow W_{n+1}$ is rational: inside W_{n+1} it is cut out by the finitely many inequalities $|\varpi^{n+1} b_i| \leq |\varpi|$.

Let B_n^{bd} denote the first (underlined) bounded analytic ring of Li–Huerta's A.15 for the tuple $(\varpi^n b_1, \dots, \varpi^n b_m)$. Put

$$C_{0,n} = R\langle \varpi^n T_1, \dots, \varpi^n T_m\rangle, \qquad C_n = C_{0,n}/IC_{0,n},$$

and let $C_{0,n}^+ = R^+\langle \varpi^n T_1, \dots, \varpi^n T_m\rangle$ and C_n^+ be the integral closure in C_n of the image of $C_{0,n}^+$. We first account for the possible derived structure in the analytic-ring pushout. The map

$$R[T_1, \dots, T_m] \longrightarrow C_{0,n}$$

is flat. Indeed, the corresponding morphism from the closed polydisc $\operatorname{Spa}(C_{0,n}, C_{0,n}^+)$ to affine m-space is a restriction of the relative analytification morphism, which is flat on local rings by Zavyalov's Lemma 6.7. For a finitely generated ideal $J \subset R[T_1, \dots, T_m]$, exact pullback of $0 \to \widetilde{J} \to \mathcal{O}$ and the finite-presentation calculation together with Zavyalov's Theorem 4.2 identifies the resulting injection on the polydisc with

$$J \otimes_{R[T_1,\dots,T_m]} C_{0,n} \longrightarrow C_{0,n}.$$

The ideal criterion gives flatness [Zav25, Theorem 4.2 and Lemma 6.7]. Thus the ordinary derived tensor product in $\mathrm{D}(C_{0,n})$ is concentrated in degree zero and equals C_n.

The Tate algebra $C_{0,n}$ is a noetherian Banach ring and $IC_{0,n}$ is finitely generated. The open mapping theorem and the completeness theorem for finite modules show that $IC_{0,n}$ is complete for the induced topology and hence closed in $C_{0,n}$; they also show that the quotient C_n is complete for its quotient topology [KL19, Theorem 1.2.9 and Corollaries 1.2.11–1.2.12]. Zavyalov's Example 6.5 identifies

$$W_n = \operatorname{Spa}(C_n, C_n^+).$$

In the polynomial case, Li–Huerta's Lemma A.8 identifies the first bounded analytic ring with $(C_{0,n}, C_{0,n}^+)^{\square}$. For the quotient, A.16 observes that the right vertical map in its pushout square is induced in the sense of Scholze's Proposition 12.8. We verify that the underlying quotient is the classical C_n and has no hidden condensed derived structure. Scholze's Proposition 12.12 identifies the underlying animated condensed ring before the final completion with

$$E_n = C_{0,n} \otimes^{\mathbf{L}}_{R[T_1,\dots,T_m]} B,$$

where $C_{0,n}$ carries its natural condensed topology and the polynomial algebra and B are discrete. Since $R[T_1, \dots, T_m]$ is noetherian and B is finitely presented, choose a bounded-above resolution $P^\bullet \to B$ by finite free polynomial modules. The discrete embedding of modules into condensed modules is exact, and a finite free module remains finite free (hence projective) over the discrete

condensed polynomial ring. Thus, viewed as a complex of discrete condensed modules, the bounded-above finite-free complex $P^\bullet$ is still K-flat. Term by term,

$$C_{0,n} \otimes_{R[T_1,\dots,T_m]} P^\bullet$$

as a complex of condensed $C_{0,n}$-modules is the condensification of the corresponding ordinary complex of finite free $C_{0,n}$-modules. The ordinary complex has cohomology only C_n in degree zero by the flatness just proved. Exactness of condensification and Andreychev's identification of a finite complete module with its condensification show that the canonical quotient morphism of animated condensed rings

$$E_n \longrightarrow C_n[0]$$

is an equivalence on underlying condensed modules, and hence an equivalence of animated condensed rings; here C_n has its quotient topology. Thus the target of the induced analytic structure has underlying condensed ring C_n, and the induced map is the quotient $C_{0,n} \to C_n$. Its connective complete objects are exactly the C_n-complexes whose restriction to $C_{0,n}$ is $(C_{0,n}, C_{0,n}^+)^\square$-complete. Andreychev's Proposition 3.29 first identifies this topological completeness condition with completeness for the corresponding discrete plus-ring analytic structure. Put $D_n^+ = \mathrm{im}(C_{0,n}^+ \to C_n)$. The quotient map $C_{0,n}^+ \twoheadrightarrow D_n^+$ is integral, so Proposition 3.22 identifies completeness over $C_{0,n}^+$ with completeness over D_n^+. Proposition 3.21 then identifies this with completeness for the discrete Huber pair (C_n, D_n^+), and a second application of Proposition 3.22 permits replacement of D_n^+ by its integral closure C_n^+. Andreychev's Lemma 3.24 shows in particular that the unit C_n is complete, so the final completion does not change the underlying ring. Scholze's uniqueness theorem for normalized analytic structures therefore gives

$$B_n^{\mathrm{bd}} \simeq (C_n, C_n^+)^\square.$$

Equivalently, both analytic rings classify bounded m-tuples satisfying every equation in I. Hence

$$\mathrm{AnSpec}\, B_n^{\mathrm{bd}} \simeq W_n^\square. \tag{22}$$

These equivalences commute with the transition maps in n: on both sides the transition is induced by the identity on B and the inclusion of the locus $|\varpi^n b_i| \le 1$ into $|\varpi^{n+1} b_i| \le 1$. This chartwise calculation extends the one used in the proof of Li–Huerta's A.19. That theorem treats smooth schemes over sousperfectoid bases; here Huber's Proposition 3.8 supplies the adic analytification, while the preceding flatness and condensed-derived-quotient calculation identifies the bounded charts for arbitrary finite presentation over a strongly noetherian base [LH26, Lemmas A.8, A.15–A.16, and proof of Theorem A.19][And21, Lemmas 3.24, 5.7, and 5.23, Propositions 3.21–3.22, 3.29, and 3.32, and Theorem 5.9][Sch26a, Propositions 12.8, 12.12, and 12.20].

Write

$$\underline{\mathcal{U}}_n = \mathrm{AnSpec}\, B_n^{\mathrm{bd}} \simeq W_n^\square$$

for these first, underlined bounded charts, and write $\mathcal{U}_n$ for the second bounded charts from Li–Huerta's A.15 used to define nuclearity. Applying the factorization in A.11 successively to the finite tuple $(b_1, \dots, b_m)$ gives steady open embeddings

$$\underline{\mathcal{U}}_n \longrightarrow \mathcal{U}_n \longrightarrow \underline{\mathcal{U}}_{n+1}. \tag{23}$$

Both resulting sequences are cofinal bounded-chart presentations of T^{An}. On the Huber side, the W_n form an increasing rational-open cover of $T^{\mathrm{ad}/V}$. Lemma 4.10 therefore gives

$$T^{\mathrm{An}} \simeq \operatorname*{colim}_n \underline{\mathcal{U}}_n \simeq \operatorname*{colim}_n W_n^\square \simeq (T^{\mathrm{ad}/V})^\square,$$

which proves (20) in the affine case without assuming that the solid-avatar functor preserves the increasing union a priori.

We next prove the object formula on an arbitrary affinoid open. Let $W = \mathrm{Spa}(C, C^+) \subset c_{T/V}^{-1}(U)$, with $U = \operatorname{Spec} A \subset T$ affine, and let $K = R\Gamma(U, M|_U)$. Since W is quasicompact, Lemma 4.10 places it inside some bounded chart W_N and supplies a finite cover

$$W = \bigcup_{j=1}^{s} W_j, \qquad W_j = \mathrm{Spa}(C_j, C_j^+),$$

whose members are rational in both W and W_N. On $W_N^{\square}$, relative algebraization is derived scalar extension followed by condensification. Since $W_j \subset c_{T/V}^{-1}(U)$, compatibility with Zariski restriction identifies the scalar-extension formula computed from T with the one computed from the affine open U. Restricting to $W_j^{\square}$ and using Andreychev's scalar-extension compatibility therefore gives

$$\alpha_T^{\mathrm{LH}}(M)|_{W_j^{\square}} \simeq \mathrm{Cond}_{C_j}(C_j \otimes_A^{\mathbf{L}} K).$$

The restriction of the proposed right side of (21) from $W^{\square}$ to $W_j^{\square}$ is the same object, by the two reassociations

$$C_j \otimes_C^{\mathbf{L}} (C \otimes_A^{\mathbf{L}} K) \simeq C_j \otimes_A^{\mathbf{L}} K.$$

These local equivalences agree on all rational intersections because they are induced by the same associativity transformation. Effective rational descent for solid quasicoherent complexes therefore gives (21) on W. The same construction proves independence of the presentation and of ϖ, as well as functoriality and rational-base-change compatibility: the common bounded-chart refinements of Lemma 3.2, followed by common rational refinements, reduce every comparison to the associativity coherence for one iterated analytic-ring pushout [And21, Lemma 5.10].

We now compare the pseudocoherent subcategories. Suppose first that $K \in \mathsf{D}_{\mathrm{pc}}^{\square}(T^{\mathrm{An}})$ in Li–Huerta's sense. Its restriction to $\mathcal{U}_N$ is nuclear by definition, so its further restriction along the first arrow of (23) is nuclear on $\underline{\mathcal{U}}_N$; its restriction there is pseudocompact by the defining underlined-chart presentation. Every affinoid $W = \mathrm{Spa}(C, C^+) \subset T^{\mathrm{ad}/V}$ lies in some W_N. On the finite common rational refinement constructed above, every restriction is nuclear and pseudocompact. Andreychev's Theorem 5.42, including preservation of nuclearity under localization, descends nuclearity to $W^{\square}$; Rodríguez Camargo's Proposition 2.3.2 shows that the rational localizations in the refinement are open localizations on solid module categories, so Li–Huerta's Proposition A.5 descends pseudocompactness [RC24, Proposition 2.3.2]. Viewing $W^{\square}$ as the Li–Huerta analytification of $\operatorname{Spec} C$ over the complete Tate pair (C, C^+), the last assertion of A.15 identifies this restriction with an object in the image of $\mathrm{D}_{\mathrm{pc}}(C)$.

Conversely, suppose that an ambient object restricts on every strongly noetherian affinoid $W = \mathrm{Spa}(C, C^+)$ to the image of $\mathrm{D}_{\mathrm{pc}}(C)$. In particular, its restriction to every underlined chart $\underline{\mathcal{U}}_n = W_n^{\square}$ is both nuclear and pseudocompact. The underlined-chart descent condition in A.15, together with A.5 applied to the finite jointly conservative steady-localization covers in that system, is exactly the required global pseudocompactness condition. The second arrow of (23) realizes $\mathcal{U}_n$ as a steady open substack of $\underline{\mathcal{U}}_{n+1}$, and localization preserves nuclear objects; hence the restriction to every $\mathcal{U}_n$ is nuclear. The second-chart descent condition in A.15, together with A.4, is exactly the required global nuclearity condition. This proves equality with the stated chartwise subcategory [LH26, A.4–A.5, A.10–A.11, and A.15][And21, Theorems 5.42, 5.44, and 5.50].

For general separated T, choose a finite affine Zariski cover (T_a). Huber analytification preserves separatedness, so $T^{\mathrm{ad}/V}$ is quasiseparated. It is the ordinary open gluing of the spaces $T_a^{\mathrm{ad}/V}$ and their multiple intersections. Lemma 4.10 identifies the solid avatar of this gluing with the geometric realization of the corresponding simplicial diagram. Li–Huerta's Theorem A.13 gives the same geometric-realization description of T^{An}. The affine comparisons above are compatible with every face and degeneracy map, so taking geometric realizations proves (20). Formula (21) and the characterization of the pseudocoherent subcategory are local for this Zariski cover and for

rational-affinoid refinements of its multiple intersections; effective descent proves both globally. The affine identity, composition, and higher coherences persist under geometric realization, and a common affine refinement proves independence of the chosen cover. □

Lemma 4.12 (Fixed-base stability under pullback and invertible twists)**.** *Let (R, R^+) be a strongly noetherian complete Tate Huber pair, let $h : S' \to S$ be a morphism of separated finitely presented R-schemes, let*

$$K \in \mathsf{D}^{\square}_{\mathrm{pc}}(S^{\mathrm{An}}),$$

and let L be a line bundle on S'. Then

$$h^{\mathrm{An},*}K \otimes L^{\mathrm{An}} \in \mathsf{D}^{\square}_{\mathrm{pc}}((S')^{\mathrm{An}}).$$

The construction is functorial in h and compatible with composition.

Proof. Use Proposition 4.11 to identify S^{An} and $(S')^{\mathrm{An}}$ with the solid avatars of their Huber analytifications and to identify their pseudocoherent subcategories chartwise. Let

$$W' = \mathrm{Spa}(C', C'^{+}) \subset (S')^{\mathrm{ad}/V}$$

be a strongly noetherian affinoid open. The image of the quasicompact space W' in $S^{\mathrm{ad}/V}$ is covered by finitely many strongly noetherian affinoid opens

$$W_\nu = \mathrm{Spa}(C_\nu, C_\nu^+) \subset S^{\mathrm{ad}/V}.$$

After a finite rational-affinoid refinement $W' = \bigcup_\nu W'_\nu$, with

$$W'_\nu = \mathrm{Spa}(C'_\nu, C'^{+}_\nu),$$

each W'_ν maps to W_ν. By the chartwise characterization in Proposition 4.11, there is an object $P_\nu \in \mathrm{D}_{\mathrm{pc}}(C_\nu)$ and an equivalence

$$K|_{W'^{\square}_\nu} \simeq \mathrm{Cond}_{C_\nu}(P_\nu).$$

Functoriality of the comparison and scalar-extension compatibility of condensification identify the restriction of the pullback with

$$h^{\mathrm{An},*}K|_{(W'_\nu)^{\square}} \simeq \mathrm{Cond}_{C'_\nu}\big(C'_\nu \otimes^{\mathbf{L}}_{C_\nu} P_\nu\big).$$

Ordinary pseudocoherent complexes are preserved by arbitrary derived scalar extension: a bounded-above finite-free resolution remains such after tensoring. Hence

$$C'_\nu \otimes^{\mathbf{L}}_{C_\nu} P_\nu \in \mathrm{D}_{\mathrm{pc}}(C'_\nu).$$

Refine the finite cover once more so that L is trivial on every W'_ν. On this refinement, tensoring by L^{An} does not change membership in the image of $\mathrm{D}_{\mathrm{pc}}(C'_\nu)$. Pseudocoherent complexes satisfy rational descent, so the restrictions glue to show that

$$(h^{\mathrm{An},*}K \otimes L^{\mathrm{An}})|_{(W')^{\square}}$$

belongs to the image of $\mathrm{D}_{\mathrm{pc}}(C')$. Since this holds on every strongly noetherian affinoid open W', Proposition 4.11 proves the asserted global membership.

All comparison maps used above are the canonical associativity maps for derived scalar extension. Their identity, composition, and higher coherences are therefore those of the symmetric monoidal derived categories, which proves the final assertion [And21, Lemma 5.10]. □

Lemma 4.13 (Proper-case stability)**.** *Let Z and Z' be proper and finitely presented over X, let*

$$h : Z' \longrightarrow Z$$

be an X-morphism, let

$$K \in \mathsf{D}^{\square}_{\mathrm{pc}}(Z^{\mathrm{An}}_{\mathcal{X}}),$$

and let L be a line bundle on Z'. Then

$$h^{\mathrm{An},*}K \otimes L^{\mathrm{An}} \in \mathsf{D}^{\square}_{\mathrm{pc}}((Z')^{\mathrm{An}}_{\mathcal{X}}).$$

Here L^{An} denotes the corresponding invertible object on the relative analytic realization.

Proof. Membership in $\mathsf{D}^{\square}_{\mathrm{pc}}$ is detected after restriction to the rational-affinoid basis of $\mathcal{X}$. For

$$V = \mathrm{Spa}(R, R^+) \subset \mathcal{X},$$

put $Z_R = Z \times_X \operatorname{Spec} R$, $Z'_R = Z' \times_X \operatorname{Spec} R$, and let $h_R : Z'_R \to Z_R$ be the base change of h. The restriction of the object in the statement to $(Z'_R)^{\mathrm{An}}$ is

$$h_R^{\mathrm{An},*}(K|_{Z_R^{\mathrm{An}}}) \otimes L_R^{\mathrm{An}}.$$

Lemma 4.12 places this object in $\mathsf{D}^{\square}_{\mathrm{pc}}((Z'_R)^{\mathrm{An}})$ for every V. The rational-base-change coherences identify these local objects with the restrictions of the single global pullback-and-twist, so Proposition 3.5 proves the assertion. □

Lemma 4.14 (A finite fpqc family from Huber charts)**.** *Let $V = \mathrm{Spa}(R, R^+)$ be a strongly noetherian complete Tate affinoid, and let T be a proper finitely presented R-scheme. There are a finite cover by strongly noetherian affinoid opens*

$$T^{\mathrm{ad}/V} = \bigcup_{i=1}^{r} W_i, \qquad W_i = \mathrm{Spa}(B_i, B_i^+),$$

and affine opens $U_i = \operatorname{Spec} A_i \subset T$ with $W_i \subset c_{T/V}^{-1}(U_i)$ such that

$$\coprod_{i=1}^{r} \operatorname{Spec} B_i \longrightarrow T$$

is fpqc.

Proof. The adic space $T^{\mathrm{ad}/V}$ is proper over V by Zavyalov's Remark 6.3 and hence quasicompact [Zav25, Remark 6.3]. Choose a finite affine cover of T and a finite strongly noetherian affinoid cover (W_i) of the analytification subordinate to its inverse image; write $U_i = \operatorname{Spec} A_i$.

We first prove that $A_i \to B_i$ is flat. Let $J \subset A_i$ be a finitely generated ideal. Since A_i is noetherian, J is finitely presented. Choose a finite presentation

$$A_i^r \longrightarrow A_i^s \longrightarrow J \longrightarrow 0.$$

Sheafifying on U_i and pulling back along the analytification morphism gives a right-exact presentation of $c_{T/V}^* \widetilde{J}|_{W_i}$. Zavyalov's Lemma 6.7 says that $c_{T/V}$ is flat on local rings, so pullback is exact on quasi-coherent finite-presentation modules; in particular, pulling back

$$0 \longrightarrow \widetilde{J} \longrightarrow \mathcal{O}_{U_i}$$

gives an injection

$$c_{T/V}^* \widetilde{J}|_{W_i} \longrightarrow \mathcal{O}_{W_i}.$$

On the other hand, applying the same right-exact pullback to the chosen finite presentation shows that the finite free terms pull back to the corresponding finite free B_i-modules. The finite-presentation calculation and Zavyalov's Theorem 4.2 therefore identify $c_{T/V}^* \widetilde{J}|_{W_i}$ with the coherent sheaf associated to $J \otimes_{A_i} B_i$. Under the exact equivalence of Theorem 4.2, the preceding injection is the sheafification of

$$J \otimes_{A_i} B_i \longrightarrow B_i.$$

Hence this module map is injective. The ideal criterion proves that $A_i \to B_i$ is flat [Zav25, Theorem 4.2 and Lemma 6.7].

It remains to prove joint surjectivity. Let x be a closed point of T. The coherent skyscraper $\kappa(x)$ is nonzero. Fixed-base full faithfulness in Proposition 3.13, together with the proper identification $T^{\mathrm{An}} \simeq T^{\mathrm{Alg}}$, makes $\alpha_T^{\mathrm{LH}}\kappa(x)$ nonzero. Proposition 4.11 and the affinoid cover imply that its restriction to some $W_i^{\square}$ is nonzero. Formula (21) identifies that restriction with

$$\mathrm{Cond}_{B_i}\Big(B_i \otimes^{\mathbf{L}}_{A_i} \Gamma\big(U_i, \kappa(x)|_{U_i}\big)\Big).$$

The module $\Gamma(U_i, \kappa(x)|_{U_i})$ is either zero, when $x \notin U_i$, or the residue field $\kappa(x)$, when $x \in U_i$. Full faithfulness of condensification and the nonvanishing above therefore first imply $x \in U_i$ and then give

$$B_i \otimes_{A_i} \kappa(x) \neq 0.$$

A prime in the support of this finite B_i-module lies over the maximal ideal defining x.

For every point $y \in T$, the nonempty noetherian closed subset $\overline{\{y\}}$ contains a closed point x, so y specializes to x. The open U_i is stable under generalization, so $y \in U_i$ whenever $x \in U_i$. If $\mathfrak{p}_y \subset \mathfrak{p}_x \subset A_i$ are the corresponding primes and $\mathfrak{q}_x \subset B_i$ lies over $\mathfrak{p}_x$, flat going-down produces $\mathfrak{q}_y \subset \mathfrak{q}_x$ lying over $\mathfrak{p}_y$. Hence the displayed family is surjective on spectra. It is a finite disjoint union of affine flat morphisms and is therefore fpqc. □

Proposition 4.15 (Corrected fixed-base proper pseudocoherent GAGA)**.** *Let (R, R^+) be a strongly noetherian complete Tate Huber pair and let S be a proper scheme of finite presentation over R. Fixed-base analytification induces an equivalence*

$$\alpha_S^{\mathrm{LH}} : \mathrm{D}_{\mathrm{pc}}(S) \xrightarrow{\sim} \mathsf{D}^{\square}_{\mathrm{pc}}(S^{\mathrm{An}}). \tag{24}$$

This proposition gives a self-contained corrected proof, in the strongly noetherian case required in this paper, of the theorem stated as Li–Huerta A.18. It replaces the printed negative-twist localization telescope by Lemma 4.6, proves the required pullback stability through the chartwise Huber–solid comparison, and detects pseudocoherence of the resulting ordinary algebraization by a finite fpqc family of Huber charts.

Proof. The bounded-chart argument in Li–Huerta A.15 and in the paragraph preceding Theorem A.18 shows that fixed-base analytification sends pseudocoherent complexes to nuclear pseudocompact objects; this forward direction is independent of the boundary-localization telescope [LH26, A.15 and the paragraph preceding Theorem A.18]. Properness identifies S^{An} with S^{Alg} by Theorem A.13, and Proposition 3.13 gives full faithfulness of (24).

Let $C \in \mathsf{D}^{\square}_{\mathrm{pc}}(S^{\mathrm{An}})$. We first construct an ordinary quasicoherent algebraization, without yet asserting pseudocoherence. Fix an affine open

$$j : U = \operatorname{Spec} A \hookrightarrow S$$

and write $q : U \to \operatorname{Spec} R$ for its structural morphism. Apply Lemma 4.9 to obtain

$$p : \widetilde{S} \longrightarrow S, \qquad \widetilde{j} : U \hookrightarrow \widetilde{S}, \qquad s : \mathcal{O}_{\widetilde{S}} \longrightarrow \mathcal{L},$$

where p is proper and finitely presented, is an isomorphism over U, and U is the nonvanishing locus of s. Put $\widetilde{\pi} : \widetilde{S} \to \operatorname{Spec} R$ and

$$\widetilde{C} = p^{\mathrm{An},*}C.$$

Since $\widetilde{S}$ is proper, its algebraic and analytic realizations agree. Lemma 4.12 shows that every

$$\widetilde{C} \otimes \mathcal{L}^{\otimes n}$$

belongs to $\mathsf{D}^{\square}_{\mathrm{pc}}(\widetilde{S}^{\mathrm{An}})$.

Lemma 4.6 gives

$$R\widetilde{j}^{\mathrm{Alg}}_{*}\widetilde{j}^{\mathrm{Alg},*}\widetilde{C} \simeq \operatorname*{colim}_{n\geq 0}\big(\widetilde{C} \otimes \mathcal{L}^{\otimes n}\big).$$

Apply $R\widetilde{\pi}_*$. Corollary 4.2 shows that this proper direct image commutes with the colimit, and Theorem 4.1 shows that

$$E_n := R\widetilde{\pi}_*\big(\widetilde{C} \otimes \mathcal{L}^{\otimes n}\big)$$

lies in the essential image of $\mathrm{Cond}_R : \mathrm{D}_{\mathrm{pc}}(R) \hookrightarrow \mathsf{D}_{\square}(R, R^+)$. The transition maps make $(E_n)_n$ a functor $\mathbf{N} \to \mathsf{D}_{\square}(R, R^+)$. Full faithfulness of Cond_R and the functor-category clause of Lemma 2.6 lift this diagram to $(M_n)_n$ in $\mathrm{D}_{\mathrm{pc}}(R)$. Put

$$M = \operatorname*{colim}_n M_n \in \mathrm{D}(R).$$

Since condensification preserves colimits,

$$R\widetilde{\pi}_* R\widetilde{j}_*^{\mathrm{Alg}} \widetilde{j}^{\mathrm{Alg},*} \widetilde{C} \simeq \mathrm{Cond}_R(M). \tag{25}$$

The morphism p is the identity over U, and $\widetilde{\pi} \circ \widetilde{j} = q$. The two sides of

$$R\widetilde{\pi}_* R\widetilde{j}_*^{\mathrm{Alg}} \simeq Rq_*^{\mathrm{Alg}}$$

are right adjoint to the same composite pullback, so Yoneda gives this canonical equivalence. Thus (25) says that the underlying object of

$$C_U := j^{\mathrm{Alg},*} C \in \mathcal{C}_{\square}(U^{\mathrm{Alg}})$$

belongs to the essential image of Cond_R. Lemma 3.12 and Lemma 3.9 therefore produce $N_U \in \mathrm{D}(A)$ and an equivalence

$$\alpha_A^{\mathrm{Alg}}(N_U) \simeq C_U.$$

Perform this construction on every member U_a of a finite affine cover of S. Since S is separated, all multiple intersections are affine. On each such intersection, the ordinary restrictions of the N_{U_a} have equivalent algebraizations, because both identify with the corresponding restriction of C. Proposition 3.13 lifts these equivalences and all higher cocycle homotopies. Ordinary Zariski descent in D_{qc} therefore gives

$$N \in \mathrm{D}_{\mathrm{qc}}(S), \qquad \alpha_S^{\mathrm{LH}}(N) \simeq C.$$

It remains only to prove that N is pseudocoherent. Choose the finite fpqc family of Huber charts from Lemma 4.14:

$$\coprod_{i=1}^{r} \operatorname{Spec} B_i \longrightarrow S, \qquad W_i = \operatorname{Spa}(B_i, B_i^+) \subset S^{\mathrm{ad}/V},$$

with W_i lying over an affine open $U_i = \operatorname{Spec} A_i \subset S$. Let $s_i : \operatorname{Spec} B_i \to S$ denote the corresponding affine flat morphism and put $N_i = Ls_i^* N$. Proposition 4.11, including formula (21) on the affine neighborhood U_i, identifies the restriction of $\alpha_S^{\mathrm{LH}}(N)$ to $W_i^{\square}$ with

$$\mathrm{Cond}_{B_i}(N_i).$$

On the other hand, the chartwise characterization in the same proposition and the fact that $C \in \mathsf{D}_{\mathrm{pc}}^{\square}(S^{\mathrm{An}})$ give an object $P_i \in \mathrm{D}_{\mathrm{pc}}(B_i)$ and an equivalence

$$C|_{W_i^{\square}} \simeq \mathrm{Cond}_{B_i}(P_i).$$

Full faithfulness of condensification therefore gives

$$N_i \simeq P_i \in \mathrm{D}_{\mathrm{pc}}(B_i)$$

for every i. Pseudocoherence is fpqc local on schemes, so the finite faithfully flat family above implies

$$N \in \mathrm{D}_{\mathrm{pc}}(S)$$

[Stacks, Tags 066D and 068R]. This proves essential surjectivity and the proposition. □

5. Proof of stable pseudocoherent GAGA

5.1. Finite fpqc reflection of pseudocoherence.

Proposition 5.1 (Finite fpqc reflection)**.** *Let $f : Y \to X$ be proper and of finite presentation, let $j : U \hookrightarrow Y$ be an affine open for which the composite $q : U \to X$ is affine, and identify $Y_{\mathcal{X}}^{\mathrm{An}}$ with $Y_{\mathcal{X}}^{\mathrm{Alg}}$ by properness. Let $C \in \mathsf{D}_{\mathrm{pc}}^{\square}(Y_{\mathcal{X}}^{\mathrm{An}})$, and suppose that $N \in \mathrm{D}_{\mathrm{qc}}(U)$ is equipped with an equivalence*

$$\alpha_U(N) \simeq j^{\mathrm{Alg},*}C$$

in $\mathcal{C}_{\square}(U_{\mathcal{X}}^{\mathrm{Alg}})$. Then $N \in \mathrm{D}_{\mathrm{pc}}(U)$.

Proof. Choose the finite fpqc family of Lemma 2.10. Put

$$Y_i = Y \times_X \operatorname{Spec} R_i, \qquad U_i = U \times_X \operatorname{Spec} R_i = \operatorname{Spec} B_i.$$

The last equality holds because q is affine. Base change of s gives an fpqc morphism

$$\coprod_i U_i \longrightarrow U.$$

Since the family is finite and every U_i is affine, this morphism is induced by one faithfully flat ring map

$$B \longrightarrow B' = \prod_i B_i, \qquad U = \operatorname{Spec} B.$$

The scheme Y_i is proper and of finite presentation over the complete Tate ring R_i. By cartesian base change for the relative analytic realization, the pullback of $Y_{\mathcal{X}}^{\mathrm{An}}$ to V_i is Y_i^{An}, and the pullback C_i of C belongs to $\mathsf{D}_{\mathrm{pc}}^{\square}(Y_i^{\mathrm{An}})$. Proposition 4.15 therefore supplies

$$M_i \in \mathrm{D}_{\mathrm{pc}}(Y_i)$$

and an equivalence between its analytic realization and C_i. Restrict the given equivalence $\alpha_U(N) \simeq j^{\mathrm{Alg},*}C$ to V_i. Cartesian base change identifies its left side with the relative algebraization of

$$N_i := N \otimes_B^{\mathbf{L}} B_i,$$

and identifies its right side with the restriction of C_i to U_i^{Alg}. On the other hand, the Li–Huerta algebraization of $M_i|_{U_i}$ is the same restriction of C_i. Composing these two equivalences gives an equivalence between the images of N_i and $M_i|_{U_i}$ in $\mathcal{C}_{\square}(U_i^{\mathrm{Alg}})$. Over the affine Tate base R_i, full faithfulness of condensification together with the module essential-image lemma gives full faithfulness of relative algebraization on U_i. Therefore

$$N_i \simeq M_i|_{U_i} \in \mathrm{D}_{\mathrm{pc}}(B_i).$$

Because the product is finite, derived tensor product commutes with it and gives

$$N \otimes_B^{\mathbf{L}} B' \simeq \prod_i N_i.$$

A finite product of pseudocoherent complexes over the product ring $B' = \prod_i B_i$ is pseudocoherent, since this can be checked componentwise under the equivalence $\mathrm{D}(B') \simeq \prod_i \mathrm{D}(B_i)$. The map $B \to B'$ is faithfully flat. Tag 068R of the Stacks Project states that a complex whose derived pullback along a faithfully flat map is pseudocoherent is itself pseudocoherent. Applying that result to the single map $B \to B'$ proves $N \in \mathrm{D}_{\mathrm{pc}}(B) = \mathrm{D}_{\mathrm{pc}}(U)$. □

5.2. **Algebraization on a relative affine open.**

Proposition 5.2 (Local algebraization)**.** *Let $f : Y \to X$ be proper and of finite presentation, let*

$$C \in \mathsf{D}^{\square}_{\mathrm{pc}}(Y^{\mathrm{An}}_{\mathcal{X}}),$$

and let $j : U = \operatorname{Spec} B \hookrightarrow Y$ be an affine open contained in $f^{-1}(W)$ for a standard affine $W \subset X$. Then there is an object

$$N_U \in \mathrm{D}_{\mathrm{pc}}(U)$$

and an equivalence

$$\alpha_U(N_U) \simeq j^{\mathrm{Alg},*}C$$

in $\mathcal{C}_{\square}(U^{\mathrm{Alg}}_{\mathcal{X}})$.

Proof. By Proposition 3.4, properness identifies $Y^{\mathrm{An}}_{\mathcal{X}}$ with $Y^{\mathrm{Alg}}_{\mathcal{X}}$; throughout the proof we regard C as an object on this common realization. Since X is separated, the inclusion of the affine open W into X is affine. The composite

$$q : U \longrightarrow W \longrightarrow X$$

is therefore affine.

Choose a coherent ideal whose vanishing locus is $Y \setminus U$ and form the blowup

$$p : \widetilde{Y} \longrightarrow Y$$

of Lemma 4.8. Write

$$\widetilde{f} = f \circ p, \qquad \widetilde{j} : U \hookrightarrow \widetilde{Y}, \qquad \mathcal{L} = \mathcal{O}_{\widetilde{Y}}(D),$$

where $D = \widetilde{Y} \setminus U$ is the resulting effective Cartier divisor. The scheme $\widetilde{Y}$ is proper over X, so its analytic and algebraic realizations agree. We use the same symbol $\mathcal{L}$ for the image of $\mathcal{O}_{\widetilde{Y}}(D)$ on this common realization. Put

$$\widetilde{C} = p^{\mathrm{An},*}C.$$

By Lemma 4.13, every twist

$$\widetilde{C} \otimes \mathcal{L}^{\otimes n}$$

belongs to $\mathsf{D}^{\square}_{\mathrm{pc}}(\widetilde{Y}^{\mathrm{An}}_{\mathcal{X}})$.

Lemma 4.7 gives, in the algebraic realization,

$$R\widetilde{j}^{\mathrm{Alg}}_{*}\widetilde{j}^{\mathrm{Alg},*}\widetilde{C} \simeq \operatorname*{colim}_{n\geq 0}\big(\widetilde{C} \otimes \mathcal{L}^{\otimes n}\big).$$

Use the proper identification $\widetilde{Y}^{\mathrm{An}}_{\mathcal{X}} \simeq \widetilde{Y}^{\mathrm{Alg}}_{\mathcal{X}}$ and apply the structural direct image $R\widetilde{f}_{*}$. Proposition 4.4 shows both that this functor commutes with colimits and that each term

$$E_n = R\widetilde{f}_{*}\big(\widetilde{C} \otimes \mathcal{L}^{\otimes n}\big)$$

belongs to $\mathsf{D}^{\square}_{\mathrm{pc}}(\mathcal{X})$. Since p is the identity over U, one has $\widetilde{j}^{*}\widetilde{C} \simeq j^{*}C$ and $\widetilde{f} \circ \widetilde{j} = q$. The direct image $R\widetilde{f}_{*}$ and the algebraic localization right adjoint $R\widetilde{j}^{\mathrm{Alg}}_{*}$ are right adjoints to the corresponding pullbacks. The composite of those pullbacks is the pullback along q. For A on $U^{\mathrm{Alg}}_{\mathcal{X}}$ and B on $\mathcal{X}$, the adjunctions give natural equivalences

$$\operatorname{Map}_{\mathcal{X}}(B, R\widetilde{f}_{*}R\widetilde{j}^{\mathrm{Alg}}_{*}A) \simeq \operatorname{Map}_{U^{\mathrm{Alg}}_{\mathcal{X}}}(q^{*}B, A) \simeq \operatorname{Map}_{\mathcal{X}}(B, Rq^{\mathrm{Alg}}_{*}A).$$

Yoneda therefore gives a canonical equivalence

$$R\widetilde{f}_{*}R\widetilde{j}^{\mathrm{Alg}}_{*} \simeq Rq^{\mathrm{Alg}}_{*}.$$

Applying this equivalence to the localization telescope, replacing $\widetilde{j}^*\widetilde{C}$ by j^*C, and using colimit preservation of $R\widetilde{f}_*$ gives

$$Rq_*^{\mathrm{Alg}} j^{\mathrm{Alg},*} C \simeq \operatorname*{colim}_{n\geq 0} E_n. \tag{26}$$

The transition maps $E_n \to E_{n+1}$ and their composition homotopies make $E_\bullet$ a functor $\mathbf{N} \to \mathcal{C}_\square(\mathcal{X})$. Proposition 2.12 places every value E_n in the essential image of α_X. The functor-category essential-image statement in Lemma 2.6 therefore gives a sequential diagram $M_\bullet$ in $\mathrm{D}_{\mathrm{pc}}(X)$ and an equivalence of diagrams

$$\alpha_X(M_\bullet) \simeq E_\bullet.$$

Put

$$M = \operatorname*{colim}_{n\geq 0} M_n \in \mathrm{D}_{\mathrm{qc}}(X).$$

Theorem 2.8 and (26) give

$$Rq_*^{\mathrm{Alg}} j^{\mathrm{Alg},*} C \simeq \alpha_X(M).$$

Let

$$\mathcal{A} = q_*\mathcal{O}_U \in \mathrm{CAlg}(\mathrm{D}_{\mathrm{qc}}(X)).$$

Because q is affine, Lemma 3.11 identifies $j^{\mathrm{Alg},*}C$ with an $\alpha_X(\mathcal{A})$-module and identifies its underlying object with $Rq_*^{\mathrm{Alg}} j^{\mathrm{Alg},*}C$. The latter is equivalent to $\alpha_X(M)$. Under strong monoidality, the analytic module action is a morphism

$$\alpha_X(\mathcal{A} \otimes M) \longrightarrow \alpha_X(M).$$

Full faithfulness lifts this morphism uniquely to $\mathcal{A} \otimes M \to M$, and the module essential-image lemma lifts the associativity and unit coherences as well. Thus it produces

$$N_U \in \mathrm{Mod}_{\mathcal{A}}(\mathrm{D}_{\mathrm{qc}}(X)) \simeq \mathrm{D}_{\mathrm{qc}}(U)$$

with

$$\alpha_U(N_U) \simeq j^{\mathrm{Alg},*}C.$$

Finally, Proposition 5.1 shows that N_U is pseudocoherent. □

5.3. **The stable equivalence and proper pushforward.**

Lemma 5.3 (Proper pushforward on bounded-above coherent complexes)**.** *Let $h : S \to T$ be a proper morphism of finite presentation between noetherian schemes. Then*

$$Rh_* : \mathrm{D}^-_{\mathrm{coh}}(S) \longrightarrow \mathrm{D}^-_{\mathrm{coh}}(T).$$

Proof. The morphism h is quasi-compact and separated. Choose a finite affine cover (T_a) of T. Each inverse image $S_a = h^{-1}(T_a)$ is quasi-compact and separated, so it admits a finite affine cover by, say, r_a opens. Every finite intersection in that cover is affine because S_a is separated. The associated alternating Čech complex therefore computes the cohomology of every quasicoherent sheaf on S_a and has length at most $r_a - 1$. The same bound remains valid after replacing T_a by any affine open $T' \subset T_a$: the inverse images of the chosen affine opens and of all their intersections are affine, because they are base changes of morphisms between affine schemes. Thus the bound controls the sheaves $R^i h_*$ on an affine basis of T_a. Taking

$$d = \max_a (r_a - 1)$$

shows that Rh_* has cohomological dimension at most d on quasicoherent sheaves.

Let $M \in \mathrm{D}^-_{\mathrm{coh}}(S)$, say $M \in \mathrm{D}^{\leq N}$. Then $Rh_*M \in \mathrm{D}^{\leq N+d}$. Fix an integer n. The cone of

$$\tau_{\geq n-d} M \longrightarrow M$$

has cohomology only in degrees $< n-d$; after applying Rh_* it has cohomology only in degrees $< n$. Hence

$$H^n(Rh_*M) \simeq H^n\big(Rh_*\tau_{\geq n-d}M\big).$$

The truncation $\tau_{\geq n-d}M$ is a bounded coherent complex. Proper coherence therefore implies that its derived pushforward is bounded coherent. Thus $H^n(Rh_*M)$ is coherent for every n, and Rh_*M is bounded above. This proves the assertion. In Stacks Project terminology, the two inputs are finite cohomological dimension and the bounded proper-coherence theorem; compare the results surrounding Lemmas 36.10.3 and 36.11.3 [Stacks]. □

Proof of Theorem 1.2. Use Proposition 3.4 to identify $Y_{\mathcal{X}}^{\mathrm{An}}$ with $Y_{\mathcal{X}}^{\mathrm{Alg}}$; all restrictions to algebraic realizations below are taken through this fixed equivalence. Proposition 3.14 proves full faithfulness of

$$\alpha_Y : \mathrm{D}_{\mathrm{qc}}(Y) \longrightarrow \mathcal{C}_{\square}(Y_{\mathcal{X}}^{\mathrm{An}}),$$

and Proposition 3.15 shows that it restricts to a functor on the pseudocoherent subcategories.

For essential surjectivity, let

$$C \in \mathsf{D}_{\mathrm{pc}}^{\square}(Y_{\mathcal{X}}^{\mathrm{An}}).$$

Choose a finite affine open cover (U_a) of Y such that every U_a is contained in the inverse image of a standard affine open of X. Proposition 5.2 gives

$$N_a \in \mathrm{D}_{\mathrm{pc}}(U_a), \qquad \alpha_{U_a}(N_a) \simeq C|_{U_a^{\mathrm{Alg}}}.$$

On an overlap $U_{ab} = U_a \cap U_b$, the chosen equivalences with C give a specified analytic equivalence

$$\alpha_{U_{ab}}(N_a|_{U_{ab}}) \xrightarrow{\sim} \alpha_{U_{ab}}(N_b|_{U_{ab}}).$$

Let $U_\bullet$ be the full Čech nerve of the cover (U_a). Through the fixed equivalence $Y_{\mathcal{X}}^{\mathrm{An}} \simeq Y_{\mathcal{X}}^{\mathrm{Alg}}$, the restrictions of C form a cartesian section of the cosimplicial diagram of ambient solid quasicoherent categories

$$[n] \longmapsto \prod_{a_0,\dots,a_n} \mathcal{C}_{\square}(U_{a_0\dots a_n}^{\mathrm{Alg}}\mathcal{X}).$$

No preservation of the nuclear-pseudocompact condition under restriction to the nonproper algebraic realizations is asserted or needed here.

For every n, relative algebraization gives a fully faithful functor

$$\prod_{a_0,\dots,a_n} \mathrm{D}_{\mathrm{qc}}(U_{a_0\dots a_n}) \longrightarrow \prod_{a_0,\dots,a_n} \mathcal{C}_{\square}(U_{a_0\dots a_n}^{\mathrm{Alg}}\mathcal{X}),$$

and these functors form a morphism of cosimplicial diagrams by the pullback compatibilities proved in Proposition 3.14. For a tuple $I = (a_0, \dots, a_n)$, the component $C|_{U_I^{\mathrm{Alg}}}$ lies in the essential image of the degree-n comparison: indeed, restriction of the chosen equivalence on U_{a_0} identifies it with

$$\alpha_{U_I}(N_{a_0}|_{U_I}).$$

Thus every component of the cartesian section defined by C lies in the essential image of the corresponding degreewise fully faithful functor. The essential-image clause of Lemma 2.6 now produces a cartesian section of the algebraic Čech diagram whose image is the section defined by C. Concretely, on a double intersection the analytic equivalence between the images of $N_a|_{U_{ab}}$ and $N_b|_{U_{ab}}$ lifts through the equivalence of mapping spaces to an algebraic equivalence θ_{ab}; on triple and higher intersections, the cocycle homotopies of C lift through the same equivalences of mapping spaces. The contractibility of every such fiber makes these lifts compatible with all face and degeneracy maps.

Every object occurring in this algebraic section is a restriction of one of the pseudocoherent complexes N_a and is therefore pseudocoherent. Effective Zariski descent for pseudocoherent complexes now gives an object

$$N \in \mathrm{D}_{\mathrm{pc}}(Y)$$

whose restriction to every U_a is N_a. The degreewise equivalences between the algebraization of this section and the section defined by C are compatible with all Čech structure maps by construction. They therefore form a natural transformation of cosimplicial diagrams, and Lemma 2.6 descends them to a global equivalence $\alpha_Y(N) \simeq C$. This proves essential surjectivity and hence the equivalence.

It remains to prove compatibility with proper pushforward. Let $M \in \mathrm{D}_{\mathrm{pc}}(Y)$. Pullback compatibility gives

$$(f^{\mathrm{An}})^* \alpha_X(Rf_*M) \simeq \alpha_Y(Lf^*Rf_*M).$$

Applying α_Y to the algebraic counit $Lf^*Rf_*M \to M$ and taking the adjoint under $(f^{\mathrm{An}})^* \dashv Rf_*^{\mathrm{An}}$ defines the canonical comparison

$$\gamma_M : \alpha_X(Rf_*M) \longrightarrow Rf_*^{\mathrm{An}}(\alpha_Y M). \tag{27}$$

Lemma 5.3 gives

$$Rf_*M \in \mathrm{D}_{\mathrm{pc}}(X).$$

Forward preservation and Proposition 4.4 give

$$E = Rf_*^{\mathrm{An}}(\alpha_Y M) \in \mathsf{D}^{\square}_{\mathrm{pc}}(\mathcal{X}).$$

By base pseudocoherent GAGA, choose an equivalence $E \simeq \alpha_X(N)$ with $N \in \mathrm{D}_{\mathrm{pc}}(X)$. Under this equivalence, γ_M is a morphism $\alpha_X(Rf_*M) \to \alpha_X(N)$ and hence, by full faithfulness, comes from a unique morphism

$$u_M : Rf_*M \longrightarrow N.$$

For every $P \in \mathrm{Perf}(X)$, consider the map $\mathrm{Map}_X(P, u_M)$. The source and target admit natural identifications

$$\begin{aligned}
\mathrm{Map}_X(P, Rf_*M) &\simeq \mathrm{Map}_Y(f^*P, M),\\
\mathrm{Map}_X(P, N) &\simeq \mathrm{Map}_{\mathcal{X}}(\alpha_X P, E)\\
&\simeq \mathrm{Map}_{Y^{\mathrm{An}}_{\mathcal{X}}}\big((f^{\mathrm{An}})^*\alpha_X P, \alpha_Y M\big)\\
&\simeq \mathrm{Map}_{Y^{\mathrm{An}}_{\mathcal{X}}}\big(\alpha_Y(f^*P), \alpha_Y M\big)\\
&\simeq \mathrm{Map}_Y(f^*P, M).
\end{aligned}$$

By the construction of γ_M as the mate of the algebraic counit, $\mathrm{Map}_X(P, u_M)$ becomes the identity of $\mathrm{Map}_Y(f^*P, M)$ under these identifications. It is therefore an equivalence for every perfect P. Perfect complexes generate $\mathrm{D}_{\mathrm{qc}}(X)$, so u_M is an equivalence. Full faithfulness of α_X then shows that γ_M is an equivalence.

Finally, full faithfulness applied to the unit object gives

$$R\Gamma(Y, M) = \mathrm{Map}_Y(\mathcal{O}_Y, M) \simeq \mathrm{Map}_{Y^{\mathrm{An}}_{\mathcal{X}}}(\mathcal{O}, \alpha_Y M),$$

which is the asserted solid cohomology comparison. □

6. The transported coherent structure and classical GAGA

To recover ordinary GAGA we use the transported coherent heart and bounded part, without identifying the full pseudocoherent subcategory with an ambient analytic $\mathrm{D}^-_{\mathrm{coh}}$. On hearts, "symmetric monoidal" refers to the usual right-exact tensor product, equivalently H^0 of derived tensor.

6.1. **The transported t-structure.** Let $T \to X$ be proper and of finite presentation, and put

$$\mathcal{P}_T = \mathsf{D}^{\square}_{\mathrm{pc}}(T^{\mathrm{An}}_{\mathcal{X}}).$$

By Theorem 1.2, $\alpha_T : \mathrm{D}_{\mathrm{pc}}(T) \to \mathcal{P}_T$ is an equivalence. We equip $\mathcal{P}_T$ with the t-structure transported from the standard coherent t-structure on $\mathrm{D}_{\mathrm{pc}}(T)$. Write $\mathcal{P}_T^{\heartsuit}$ for its heart and $\mathcal{P}_T^{b}$ for its full subcategory of objects bounded on both sides. This is a t-structure on the pseudocoherent stable subcategory only.

Lemma 6.1 (Locality of the transported structure)**.** *Let $V = \mathrm{Spa}(R, R^+) \in \mathrm{RatAff}_X(\mathcal{X})$ and put $T_R = T \times_X \operatorname{Spec} R$. Under rational restriction*

$$\mathcal{P}_T \longrightarrow \mathsf{D}^{\square}_{\mathrm{pc}}(T^{\mathrm{An}}_R),$$

the transported t-structure restricts to the t-structure transported from $\mathrm{D}_{\mathrm{pc}}(T_R)$ through Li–Huerta's equivalence

$$\alpha^{\mathrm{LH}}_{T_R} : \mathrm{D}_{\mathrm{pc}}(T_R) \xrightarrow{\sim} \mathsf{D}^{\square}_{\mathrm{pc}}(T^{\mathrm{An}}_R).$$

The restriction functors for rational localizations are t-exact.

Proof. The rational-base-change construction of Section 3.1 gives a canonical equivalence

$$\rho_V \alpha_T(M) \simeq \alpha^{\mathrm{LH}}_{T_R}(M_R), \qquad M_R = L(T_R \to T)^* M.$$

On affine charts the map from the corresponding schematic coordinate ring to R is flat by Lemma 2.9; hence $M \mapsto M_R$ is t-exact. The same argument applies to a rational localization $R \to R'$. Since the transported t-structures are characterized by the displayed equivalences, restriction is t-exact and the local transported structure is the one claimed. □

6.2. **Bounded and coherent comparison.**

Lemma 6.2 (Left-bounded derived open descent)**.** *Let $(Z, \mathcal{O}_Z)$ be a ringed space, let $(Z_i)_{i \in I}$ be an open cover, and fix an integer a. If*

$$Z_{i_0 \cdots i_n} = Z_{i_0} \cap \cdots \cap Z_{i_n},$$

then restriction induces an equivalence of ∞-categories

$$\mathrm{D}^{\geq a}(\mathcal{O}_Z) \xrightarrow{\sim} \lim_{[n] \in \Delta} \prod_{(i_0, \ldots, i_n) \in I^{n+1}} \mathrm{D}^{\geq a}(\mathcal{O}_{Z_{i_0 \cdots i_n}}).$$

Thus uniformly left-bounded complexes and their mapping spectra satisfy effective open descent.

Proof. Let $H\mathcal{O}_Z$ be the Eilenberg–MacLane sheaf of E_∞-rings on the 1-localic ∞-topos of sheaves on Z. The ∞-category

$$\mathrm{Mod}_{H\mathcal{O}_Z}\big(\mathrm{Shv}(Z, \mathrm{Sp})\big)$$

of sheaves of module spectra satisfies descent for open covers: sheaves of spectra do, and formation of module categories is compatible with limits of symmetric monoidal presentable ∞-categories. Its standard t-structure is detected on the underlying sheaves of spectra and is therefore local. Consequently its full subcategory of objects in cohomological degrees $\geq a$ is the limit of the corresponding subcategories on the Čech nerve.

For a 1-localic topos and a discrete sheaf of rings, Proposition 2.1.8 of Lurie's *Derived Algebraic Geometry VIII: Quasi-Coherent Sheaves and Tannaka Duality Theorems*, translated from its homological convention to our cohomological convention, identifies the ordinary left-bounded derived ∞-category of sheaves of $\mathcal{O}_Z$-modules fully faithfully with the left-bounded objects of this module-spectrum category. Restricting that equivalence to the fixed lower bound a gives the displayed descent equivalence; see [Lur11, Proposition 2.1.8] and [Lur17, Section 4.8.5]. Because the functor is the restriction of an equivalence between the ambient module-spectrum descent categories, it also identifies the mapping spectra inherited from those ambient stable categories. □

Lemma 6.3 (Finite open Čech descent for bounded coherent complexes)**.** *Let $p : Z \to S$ be a quasi-compact and quasi-separated morphism of locally noetherian adic spaces, assume that S is quasiseparated, and let $S = V_1 \cup \cdots \cup V_r$ be a finite cover by quasicompact open subsets. Write*

$$V_{i_0 \ldots i_n} = V_{i_0} \cap \cdots \cap V_{i_n}, \qquad Z_{i_0 \ldots i_n} = Z \times_S V_{i_0 \ldots i_n}.$$

Then restriction induces an equivalence

$$\mathrm{D}^b_{\mathrm{coh}}(Z) \xrightarrow{\sim} \lim_{[n] \in \Delta} \prod_{(i_0, \ldots, i_n)} \mathrm{D}^b_{\mathrm{coh}}(Z_{i_0 \ldots i_n}).$$

The same statement holds for the full subcategories of complexes with cohomological amplitude in a fixed interval $[a, b]$.

Proof. Fix first an interval $[a, b]$. Lemma 6.2, applied with lower bound a, gives effective descent and the mapping-spectrum formula for the ambient categories $\mathrm{D}^{\geq a}$. In a descended object, vanishing of cohomology in degrees $> b$ can be checked on the open cover. Coherence of each cohomology sheaf is likewise local for open covers on a locally noetherian adic space. Hence the full subcategory with coherent cohomology and amplitude in $[a, b]$ is the limit of the corresponding full subcategories on the Čech nerve.

For unrestricted bounded coherent complexes, the higher Čech components are identified by cartesianness with restrictions of the finitely many degree-zero complexes $K_i \in \mathrm{D}^b_{\mathrm{coh}}(Z_i)$. Choose one common interval $[a, b]$ containing all their amplitudes. Every higher component then has the same bound, so the preceding fixed-interval equivalence applies. Full faithfulness follows from the same mapping-spectrum descent after choosing a common lower bound for the two complexes being mapped. □

Lemma 6.4 (Basis descent for category-valued finite stacks)**.** *Let S be a quasicompact quasiseparated space, and let $\mathcal{B}$ be a basis of quasicompact opens such that every finite intersection of members of $\mathcal{B}$ admits a finite cover by members of $\mathcal{B}$. Let $\mathcal{F}$ be a presheaf of ∞-categories on the quasicompact opens of S that satisfies finite Čech descent, including descent of mapping spaces. Then restriction induces an equivalence*

$$\mathcal{F}(S) \xrightarrow{\sim} \mathrm{Cart}_{V \in \mathcal{B}} \mathcal{F}(V).$$

Proof. Let $\mathrm{Op}_{\mathrm{qc}}(S)$ be the category of quasicompact opens and let

$$j : \mathcal{B} \hookrightarrow \mathrm{Op}_{\mathrm{qc}}(S)$$

be the inclusion. The hypotheses say that $\mathcal{B}$ is a basis for the finite-cover topology: every object has a finite $\mathcal{B}$-cover, and the pullback of such a cover along a map from a basis object admits a finite refinement by basis objects.

For a presheaf $\mathcal{G}$ on $\mathcal{B}$, its right Kan extension is

$$(j_* \mathcal{G})(U) = \lim_{(V \subset U),\, V \in \mathcal{B}} \mathcal{G}(V).$$

The finite-refinement condition and finite Čech descent show that, whenever $\mathcal{G}$ satisfies descent on finite basis covers, $j_* \mathcal{G}$ satisfies finite Čech descent on all quasicompact opens. Indeed, choose a finite basis cover of U, refine every finite intersection by basis objects, and compute both sides as the limit over the resulting common finite refinement; a second choice gives the same limit after passage to a common refinement. The same calculation applies to mapping spaces.

Apply this to $\mathcal{G} = \mathcal{F}|_{\mathcal{B}}$. For every quasicompact open U, choose a finite basis cover and use finite descent for $\mathcal{F}$ on that cover and on basis refinements of all its intersections. This identifies the unit map

$$\mathcal{F}(U) \longrightarrow (j_* j^* \mathcal{F})(U)$$

with an equivalence, naturally in U. Hence $\mathcal{F}$ is the right Kan extension of its restriction to $\mathcal{B}$. Evaluating at S gives the asserted equivalence, since the displayed limit is precisely the ∞-category of cartesian sections over $\mathcal{B}$. □

Lemma 6.5 (Rational-basis descent for bounded coherent complexes)**.** *Let $p : Z \to S$ be a quasi-compact and quasi-separated morphism of locally noetherian adic spaces, with S quasicompact and quasiseparated, and let $\mathcal{B}$ be a rational-affinoid basis of S. For $V \in \mathcal{B}$, put $Z_V = Z \times_S V$. Restriction induces an equivalence*

$$\mathrm{D}^b_{\mathrm{coh}}(Z) \xrightarrow{\sim} \mathrm{Cart}_{V\in\mathcal{B}}\big(\mathrm{D}^b_{\mathrm{coh}}(Z_V)\big).$$

Proof. Choose a finite cover $S = V_1 \cup \cdots \cup V_r$ with $V_i \in \mathcal{B}$. Let $(K_V)_{V\in\mathcal{B}}$ be a cartesian section and choose intervals $[a_i, b_i]$ for K_{V_i}. Put

$$a = \min_i a_i, \qquad b = \max_i b_i.$$

We first show that every K_V has amplitude in this same interval. For arbitrary $V \in \mathcal{B}$, quasicompactness of V and quasiseparatedness of S give a finite rational-affinoid refinement

$$V = \bigcup_{i,\nu} W_{i\nu}, \qquad W_{i\nu} \in \mathcal{B}, \quad W_{i\nu} \subset V \cap V_i.$$

Cartesianness identifies $K_V|_{Z_{W_{i\nu}}}$ with $K_{V_i}|_{Z_{W_{i\nu}}}$. Vanishing of cohomology is local, so K_V has amplitude in $[a, b]$.

For a quasicompact open $U \subset S$ admitting a finite cover by members of $\mathcal{B}$, let

$$\mathcal{F}_{[a,b]}(U) = \mathrm{D}^b_{\mathrm{coh}}(Z_U)^{[a,b]}.$$

Lemma 6.3 says that $\mathcal{F}_{[a,b]}$ satisfies finite Čech descent, including descent of mapping spectra. Lemma 6.4 therefore gives

$$\mathcal{F}_{[a,b]}(S) \xrightarrow{\sim} \mathrm{Cart}_{V\in\mathcal{B}}\,\mathcal{F}_{[a,b]}(V).$$

The uniform-bound paragraph places the given cartesian section in the right side of this fixed-interval equivalence, so it descends to an object of $\mathrm{D}^b_{\mathrm{coh}}(Z)$. Conversely, every bounded coherent complex restricts to a cartesian section. The mapping-spectrum part of finite descent, applied after choosing a common interval for two global complexes, proves full faithfulness. □

Proposition 6.6 (Bounded coherent complexes and the solid avatar)**.** *Let Z be a quasicompact quasiseparated locally strongly noetherian analytic adic space. Let $\mathsf{D}^{\square}_{\mathrm{pc}}(Z^{\square})$ denote the full subcategory of $\mathcal{C}_{\square}(Z^{\square})$ whose restriction to every strongly noetherian affinoid open $W = \mathrm{Spa}(A, A^+)$ belongs to the image of*

$$\mathrm{D}_{\mathrm{pc}}(A) \longrightarrow \mathsf{D}_{\square}(A, A^+).$$

Flat rational localization gives this category a chartwise t-structure. Condensification induces a canonical exact symmetric monoidal equivalence on hearts and a canonical exact equivalence on bounded objects

$$\mathrm{Coh}(Z) \xrightarrow{\sim} \mathsf{D}^{\square}_{\mathrm{pc}}(Z^{\square})^{\heartsuit}_{\mathrm{ch}}, \tag{28}$$

$$\mathrm{D}^b_{\mathrm{coh}}(Z) \xrightarrow{\sim} \mathsf{D}^{\square}_{\mathrm{pc}}(Z^{\square})^{b}_{\mathrm{ch}}. \tag{29}$$

These equivalences commute with restriction to analytic affinoid opens.

Proof. On a strongly noetherian affinoid $W = \mathrm{Spa}(A, A^+)$, Zavyalov's Theorem 4.2 gives an exact equivalence

$$\mathrm{Mod}_{\mathrm{fg}}(A) \xrightarrow{\sim} \mathrm{Coh}(W)$$

and affinoid acyclicity. We record explicitly why it extends to the ambient bounded derived categories used here. Write $\mathrm{D}^b_{\mathrm{fg}}(A)$ for the full subcategory of $\mathrm{D}(A)$ consisting of bounded

complexes with finitely generated cohomology. Since A is noetherian, the canonical functor from the bounded derived category of finitely generated A-modules to $\mathrm{D}(A)$ identifies its source with $\mathrm{D}^b_{\mathrm{fg}}(A)$. The sheaf functor $\mathcal{O}_W \otimes_A -$ is left adjoint to global sections on the ambient module categories; on finitely generated modules it is exact by Theorem 4.2. Its derived functor therefore restricts to

$$(-)^\sim : \mathrm{D}^b_{\mathrm{fg}}(A) \longrightarrow \mathrm{D}^b_{\mathrm{coh}}(W),$$

and derived global sections is its right adjoint. If $K \in \mathrm{D}^b_{\mathrm{coh}}(W)$, the hypercohomology spectral sequence

$$H^p\big(W, \mathcal{H}^q(K)\big) \Longrightarrow H^{p+q} R\Gamma(W, K)$$

collapses because every coherent sheaf is acyclic. Consequently

$$H^q R\Gamma(W, K) \simeq \Gamma\big(W, \mathcal{H}^q(K)\big),$$

which is a finite A-module, and Theorem 4.2 identifies the cohomology sheaf of the adjunction counit

$$\big(R\Gamma(W, K)\big)^\sim \longrightarrow K$$

in degree q with the canonical isomorphism

$$\widetilde{\Gamma(W, \mathcal{H}^q(K))} \xrightarrow{\sim} \mathcal{H}^q(K).$$

Thus the counit is an equivalence. Conversely, for $M \in \mathrm{D}^b_{\mathrm{fg}}(A)$, the same spectral sequence and affinoid acyclicity give

$$M \xrightarrow{\sim} R\Gamma(W, \widetilde{M}),$$

so the unit is an equivalence. Hence

$$\mathrm{D}^b_{\mathrm{coh}}(W) \simeq \mathrm{D}^b_{\mathrm{fg}}(A).$$

Andreychev's Theorem 5.9 and Proposition 5.22 make condensification fully faithful and exact on pseudocoherent objects, Lemma 5.10 gives scalar-extension compatibility, and Theorem 5.50 says that nuclear pseudocoherent solid objects are discrete. Li–Huerta's A.15 consequently identifies the nuclear-pseudocompact subcategory of $\mathsf{D}_\square(A, A^+)$ with the image of $\mathrm{D}_{\mathrm{pc}}(A)$ [And21, Theorem 5.9, Lemma 5.10, Proposition 5.22, and Theorem 5.50][LH26, A.15][Zav25, Theorem 4.2]. Since A is noetherian, the chartwise heart and bounded part are exactly the images of finite modules and of $\mathrm{D}^b_{\mathrm{fg}}(A)$.

The local equivalences commute with rational restriction. Because rational localization is flat, the local truncation triangles commute with restriction; ambient descent therefore glues them and gives the chartwise t-structure asserted in the statement, with orthogonality checked affinoid-locally. Lemma 6.5 gives effective descent for the ordinary bounded coherent categories. On the solid side, Proposition 3.6, Andreychev's descent theorems, and the defining chartwise condition give the corresponding cartesian-section description. A finite affinoid cover supplies a uniform cohomological interval for bounded objects. Passing to cartesian sections proves (28)–(29). Exactness and the heart-level monoidal structure are local and therefore descend. □

Proposition 6.7 (Local bounded and coherent GAGA without formal models)**.** *Let $V = \mathrm{Spa}(R, R^+)$ be a strongly noetherian complete Tate affinoid, and let T be a proper finitely presented R-scheme. Through Proposition 4.11, regard $\mathsf{D}^\square_{\mathrm{pc}}(T^{\mathrm{An}})$ as the chartwise pseudocoherent subcategory on $(T^{\mathrm{ad}/V})^\square$. The t-structure transported from $\mathrm{D}_{\mathrm{pc}}(T)$ by Proposition 4.15 agrees with the chartwise t-structure. Consequently there are canonical exact equivalences, the first symmetric monoidal,*

$$\beta^\heartsuit_{T,V} : \mathsf{D}^\square_{\mathrm{pc}}(T^{\mathrm{An}})^\heartsuit \xrightarrow{\sim} \mathrm{Coh}(T^{\mathrm{ad}/V}), \tag{30}$$

$$\beta^b_{T,V} : \mathsf{D}^\square_{\mathrm{pc}}(T^{\mathrm{An}})^b \xrightarrow{\sim} \mathrm{D}^b_{\mathrm{coh}}(T^{\mathrm{ad}/V}). \tag{31}$$

For $\mathcal{F} \in \mathrm{Coh}(T)$ and $M \in \mathrm{D}^b_{\mathrm{coh}}(T)$,

$$\beta^{\heartsuit}_{T,V}(\alpha_T^{\mathrm{LH}}\mathcal{F}) \simeq c^*_{T/V}\mathcal{F}, \qquad \beta^b_{T,V}(\alpha_T^{\mathrm{LH}}M) \simeq Lc^*_{T/V}M.$$

In particular, ordinary analytification induces equivalences

$$\mathrm{Coh}(T) \xrightarrow{\sim} \mathrm{Coh}(T^{\mathrm{ad}/V}), \qquad \mathrm{D}^b_{\mathrm{coh}}(T) \xrightarrow{\sim} \mathrm{D}^b_{\mathrm{coh}}(T^{\mathrm{ad}/V}).$$

Proof. Choose the finite fpqc family of Lemma 4.14. Formula (21) identifies, for $M \in \mathrm{D}_{\mathrm{pc}}(T)$, the restriction of $\alpha_T^{\mathrm{LH}}M$ to $W_i^{\square}$ with

$$\mathrm{Cond}_{B_i}\big(B_i \otimes^{\mathbf{L}}_{A_i} R\Gamma(U_i, M|_{U_i})\big).$$

Since U_i is affine and the maps $A_i \to B_i$ are flat,

$$H^j\big(B_i \otimes^{\mathbf{L}}_{A_i} R\Gamma(U_i, M|_{U_i})\big) \simeq B_i \otimes_{A_i} \Gamma(U_i, \mathcal{H}^j(M)|_{U_i}).$$

Thus the algebraic truncation conditions imply the chartwise truncation conditions. Conversely, the family $\coprod_i \operatorname{Spec} B_i \to T$ is faithfully flat, so vanishing of every coherent cohomology sheaf of M is detected after these pullbacks. Since (W_i) covers $T^{\mathrm{ad}/V}$ and chartwise truncation is local, the transported nonpositive and nonnegative subcategories agree with the chartwise ones. Finiteness of the family gives the same conclusion for bounded objects with a uniform interval.

Proposition 6.6, transported through $\vartheta_{T,V}$ and inverted, gives the two displayed equivalences. For an algebraic coherent sheaf or bounded coherent complex, the formulas in the statement can be checked on (W_i). Formula (21) shows that both sides are obtained there by ordinary derived scalar extension $A_i \to B_i$ followed by the finite-module sheafification functor. Descent gives the global formulas. Exactness and the monoidal comparison on hearts are likewise local. Composing with Li–Huerta's equivalence proves the final two ordinary analytification equivalences. □

Lemma 6.8 (Pseudonaturality of the bounded bridge)**.** *Let $u : V' = \mathrm{Spa}(R', R'^{+}) \to V = \mathrm{Spa}(R, R^{+})$ be a rational localization. The functors in* (30) *and* (31) *commute naturally with restriction to V'. For composable rational localizations, these equivalences satisfy the identity, composition, and higher associativity coherences.*

Proof. Proposition 4.11 and formula (21) are compatible with rational base change. On every affinoid chart, Proposition 6.6 is induced by condensification and ordinary derived scalar extension. Therefore both routes from V to V' are the two reassociations of the same iterated scalar extension. The associativity and pentagon coherences give the asserted transformations and all composition coherences. The formulas of Proposition 6.7 identify these transformations with the canonical base-change transformations for c^* and Lc^*. □

Proposition 6.9 (Global bounded and coherent bridge)**.** *For proper $T \to X$, the local comparisons glue to a canonical exact symmetric monoidal equivalence on hearts and a canonical exact equivalence on bounded coherent complexes*

$$\beta^{\heartsuit}_T : \mathcal{P}^{\heartsuit}_T \xrightarrow{\sim} \mathrm{Coh}(T^{\mathrm{ad}}_{\mathcal{X}}),$$
$$\beta^b_T : \mathcal{P}^b_T \xrightarrow{\sim} \mathrm{D}^b_{\mathrm{coh}}(T^{\mathrm{ad}}_{\mathcal{X}}).$$

For $\mathcal{F} \in \mathrm{Coh}(T)$ and $M \in \mathrm{D}^b_{\mathrm{coh}}(T)$ they satisfy

$$\beta^{\heartsuit}_T(\alpha_T\mathcal{F}) \simeq c^*_T\mathcal{F}, \qquad \beta^b_T(\alpha_T M) \simeq Lc^*_T M.$$

Proof. Lemma 6.8 provides, for every rational inclusion $V' \subset V$, specified natural equivalences between restriction followed by $\beta_{T,V}$ and $\beta_{T,V'}$ followed by restriction, together with all higher cocycles. Thus the local functors define morphisms of the cartesian-section diagrams on the

rational-affinoid basis. Proposition 3.5 identifies $\mathcal{P}_T$ with the ∞-category of its cartesian rational-localization data. Coherent sheaves on a locally noetherian adic space satisfy the analogous rational descent.

We first identify the source heart inside that cartesian-section category. An object of $\mathcal{P}_T$ lies in the transported heart if and only if its algebraic preimage under α_T has no cohomology outside degree zero. This condition may be checked after the finite fpqc base change of Lemma 2.10: after that base change it is precisely the condition that every local restriction belongs to the local transported heart. Indeed, all nonzero-degree cohomology sheaves vanish after the fpqc base change

$$T \times_X \coprod_i \operatorname{Spec} R_i \longrightarrow T,$$

and faithful flatness detects the vanishing of a quasicoherent sheaf. Therefore the heart $\mathcal{P}_T^{\heartsuit}$ is identified with the limit of the local transported hearts. The degreewise equivalences $\beta_{T,V}^{\heartsuit}$ and the pseudonaturality transformations form an equivalence of the corresponding descent diagrams; Lemma 2.6 gives the global heart-level equivalence.

For the bounded statement, apply Lemma 6.5 to the proper morphism $T_{\mathcal{X}}^{\mathrm{ad}} \to \mathcal{X}$ and the basis $\mathrm{RatAff}_X(\mathcal{X})$. It identifies $\mathrm{D}^b_{\mathrm{coh}}(T_{\mathcal{X}}^{\mathrm{ad}})$ with the ∞-category of cartesian rational-localization data

$$\mathrm{Cart}_{V \in \mathrm{RatAff}_X(\mathcal{X})}\big(\mathrm{D}^b_{\mathrm{coh}}(T_R^{\mathrm{ad}/V})\big).$$

The same statement holds on the source. A cartesian section valued in the local transported bounded subcategories is bounded on each member of the finite affinoid family from Lemma 2.10; finiteness gives one common interval. The rational-refinement argument of Lemma 6.5 then gives that same interval on every rational chart. If M is the algebraic preimage of the corresponding object of $\mathcal{P}_T$, its cohomology outside this interval vanishes after the fpqc base change

$$T \times_X \coprod_i \operatorname{Spec} R_i \longrightarrow T.$$

Faithful flatness detects this vanishing degree by degree, so M is globally bounded in the same interval. Conversely, a globally bounded object restricts to a cartesian section of the local bounded subcategories. Hence $\mathcal{P}_T^b$ is the limit of those local subcategories. Lemma 6.5 identifies the target with the corresponding limit of the categories $\mathrm{D}^b_{\mathrm{coh}}(T_R^{\mathrm{ad}/V})$. The pseudonatural local equivalences $\beta_{T,V}^b$ therefore give, by Lemma 2.6, the global equivalence β_T^b.

For an algebraic coherent sheaf or bounded coherent complex, the formula in the statement holds after restriction to every rational-affinoid chart by the definition of the local bridge. The local formulas commute with restriction by Lemma 6.8; descent therefore identifies the global composite with c_T^* or Lc_T^*. Exactness can be tested on the local transported t-structures, so the descended functors are exact. The monoidal comparison maps for the heart-level functors form a compatible natural transformation of the descent diagrams and therefore descend to the global symmetric monoidal structure. We make no monoidal assertion about the bounded coherent categories, which need not be closed under derived tensor product on a singular space. □

Corollary 6.10 (Coherent and bounded classical GAGA)**.** *For proper $T \to X$, ordinary analytification induces an exact symmetric monoidal equivalence*

$$c_T^* : \mathrm{Coh}(T) \xrightarrow{\sim} \mathrm{Coh}(T_{\mathcal{X}}^{\mathrm{ad}})$$

and an exact equivalence of stable categories

$$Lc_T^* : \mathrm{D}^b_{\mathrm{coh}}(T) \xrightarrow{\sim} \mathrm{D}^b_{\mathrm{coh}}(T_{\mathcal{X}}^{\mathrm{ad}}).$$

It also induces equivalences on perfect complexes and vector bundles.

Proof. The first two equivalences are the composites of the restrictions of α_T with Proposition 6.9. The formulas in that proposition identify the composites with ordinary pullback.

The vector-bundle statement follows because finite locally free sheaves are precisely the dualizable objects in the tensor category of coherent sheaves; an exact tensor equivalence preserves and reflects dualizability. Perfectness is local on both the scheme and the adic space. On each affinoid base chart, $Lc^*_{T_R/V}$ preserves perfect complexes. Conversely, suppose that $M \in \mathrm{D}^b_{\mathrm{coh}}(T_R)$ has perfect analytification. The proper adic space $T_R^{\mathrm{ad}/V}$ is quasicompact. Choose a finite affinoid cover on which the analytic perfect complex has Tor-amplitudes $[a_\ell, b_\ell]$, and set

$$a = \min_\ell a_\ell, \qquad b = \max_\ell b_\ell.$$

Tor-amplitude is local for the analytic topology. Hence the finite cover and the common bounds above show that the analytic perfect complex has global Tor-amplitude contained in $[a, b]$. For every coherent N on T_R, flatness gives

$$c^*_{T_R/V} H^j(M \otimes^{\mathbf{L}} N) \simeq H^j(Lc^*_{T_R/V} M \otimes^{\mathbf{L}} c^*_{T_R/V} N).$$

The right side vanishes outside the fixed interval $[a, b]$, independently of N. Coherent GAGA is conservative: if a coherent algebraic sheaf has zero analytification, it is zero. Hence $H^j(M \otimes^{\mathbf{L}} N) = 0$ for $j \notin [a, b]$ and every coherent N. The standard noetherian Tor-amplitude criterion then gives that M has Tor-amplitude contained in $[a, b]$. Since M is pseudocoherent, it is perfect.

This argument is compatible with rational restriction. For a global algebraic bounded coherent complex, pull it back along the finite fpqc cover of Lemma 2.10. If its analytification is perfect, the local argument makes every fpqc pullback perfect. Perfectness is fpqc local, so the original algebraic complex is perfect. Conversely, algebraic pullback preserves perfect complexes, and their analytic realizations are locally finite complexes of finite projectives; rational descent for perfect complexes gives a global analytic perfect complex. Thus the bounded GAGA equivalence restricts to an equivalence on perfect complexes. □

6.3. **Ordinary proper pushforward.**

Lemma 6.11 (Open base change for ordinary derived direct image)**.** *Let*

$$\begin{array}{ccc} Z' & \xrightarrow{j'} & Z \\ {\scriptstyle h'}\downarrow & & \downarrow{\scriptstyle h} \\ V' & \xrightarrow{j} & V \end{array}$$

be a cartesian square of ringed spaces in which j is an open immersion. Then the canonical morphism

$$j^* Rh_* \longrightarrow Rh'_* j'^*$$

is an equivalence on derived categories of modules.

Proof. For an open immersion, extension by zero $j_!$ is exact and left adjoint to the exact restriction functor j^*; similarly for j'. There is a canonical equivalence

$$Lh^* j_! \simeq j'_! L(h')^*,$$

checked on the underlying sheaves on open subsets. For $A \in \mathrm{D}(Z)$ and $B \in \mathrm{D}(V')$, the derived adjunctions give

$$\begin{aligned}\mathrm{Map}_{V'}(B, j^*Rh_*A) &\simeq \mathrm{Map}_V(j_!B, Rh_*A)\\ &\simeq \mathrm{Map}_Z(Lh^*j_!B, A)\\ &\simeq \mathrm{Map}_Z(j'_!L(h')^*B, A)\\ &\simeq \mathrm{Map}_{Z'}(L(h')^*B, j'^*A)\\ &\simeq \mathrm{Map}_{V'}(B, Rh'_*j'^*A).\end{aligned}$$

This equivalence is the mate of the cartesian square and hence is the canonical open-base-change transformation. Yoneda proves the claim. □

Proposition 6.12 (Affinoid proper GAGA with direct images)**.** *Let $V = \mathrm{Spa}(R, R^+)$ be a strongly noetherian complete Tate affinoid, let $h : Z \to \mathrm{Spec}\, R$ be proper and of finite presentation, and let*

$$h^{\mathrm{ad}} : Z^{\mathrm{ad}/V} \longrightarrow V$$

be its relative Huber analytification. For every $M \in \mathrm{D}^b_{\mathrm{coh}}(Z)$, the canonical comparison morphism

$$Lc_V^*Rh_*M \longrightarrow Rh_*^{\mathrm{ad}}Lc_{Z/V}^*M \tag{32}$$

is an equivalence. In particular, its target belongs to $\mathrm{D}^b_{\mathrm{coh}}(V)$. The construction is functorial in M and compatible with rational localization of V, including the higher composition coherences.

Proof. The comparison is the adjoint of the composite

$$L(h^{\mathrm{ad}})^*Lc_V^*Rh_*M \simeq Lc_{Z/V}^*Lh^*Rh_*M \longrightarrow Lc_{Z/V}^*M,$$

where the last arrow is the algebraic counit.

Let $j : V' = \mathrm{Spa}(R', R'^+) \subset V$ be rational, and put

$$Z_{R'} = Z \times_R \mathrm{Spec}\, R', \qquad M_{R'} = M \otimes^{\mathbf{L}}_R R'.$$

Evaluating the restriction of (32) on V' and using algebraic flat base change, coherent affinoid acyclicity on V', and ordinary open base change identifies it with the canonical map

$$R\Gamma(Z_{R'}, M_{R'}) \longrightarrow R\Gamma\big(Z_{R'}^{\mathrm{ad}/V'}, Lc^*_{Z_{R'}/V'}M_{R'}\big). \tag{33}$$

Proposition 6.7 says that ordinary analytification

$$Lc^*_{Z_{R'}/V'} : \mathrm{D}^b_{\mathrm{coh}}(Z_{R'}) \xrightarrow{\sim} \mathrm{D}^b_{\mathrm{coh}}(Z_{R'}^{\mathrm{ad}/V'})$$

is an equivalence. It sends the algebraic unit to the analytic unit. Full faithfulness therefore identifies

$$\mathrm{Map}_{Z_{R'}}(\mathcal{O}, M_{R'}) \xrightarrow{\sim} \mathrm{Map}_{Z_{R'}^{\mathrm{ad}/V'}}(\mathcal{O}, Lc^*_{Z_{R'}/V'}M_{R'}).$$

Under the identifications of these mapping spectra with the two terms of (33), this map sends a section $s : \mathcal{O} \to M_{R'}$ to its derived analytic pullback $Lc^*(s)$. More invariantly, the map of spectra is the composite

$$\begin{aligned}\mathrm{Map}_{Z_{R'}}(\mathcal{O}, M_{R'}) &\xrightarrow{Lc^*} \mathrm{Map}_{Z_{R'}^{\mathrm{ad}/V'}}(Lc^*\mathcal{O}, Lc^*M_{R'})\\ &\simeq \mathrm{Map}_{Z_{R'}^{\mathrm{ad}/V'}}(\mathcal{O}, Lc^*M_{R'}).\end{aligned}$$

After restricting the adjoint defining (32) to V' and applying the algebraic and analytic adjunctions, its map on the same mapping spectra is obtained by applying Lc^* to the algebraic counit $Lh^*Rh_*M_{R'} \to M_{R'}$. This is exactly the displayed composite. Thus the full-faithfulness equivalence

is precisely the canonical map (33), not merely an abstract equivalence between its source and target.

Let Q be the cofiber of (32). The preceding paragraph shows that the value

$$Q(V') = R\Gamma(V', Q|_{V'})$$

vanishes for every member V' of the rational-affinoid basis. The underlying sheaf of spectra of Q therefore has zero value on every member of a basis, and hence is the zero sheaf. Thus $Q = 0$, and (32) is an equivalence. Its source is bounded coherent by algebraic proper coherence and flat pullback, so the target is bounded coherent as well.

For a second rational localization, every comparison is obtained by pulling back the same counit and applying the associativity equivalences for algebraic flat base change, Huber analytification, and ordinary open base change. Naturality of adjunction mates and the pentagon coherence give functoriality and all higher cocycles. □

Proposition 6.13 (Ordinary derived pushforward comparison)**.** *Let $f : Y \to X$ be proper and of finite presentation. For $T = X$, Proposition 3.8 identifies $X^{\mathrm{ad}}_{\mathcal{X}}$ with $\mathcal{X}$ and c_X with λ. For every $M \in \mathrm{D}^b_{\mathrm{coh}}(Y)$, the canonical morphism*

$$Lc_X^* Rf_* M \longrightarrow Rf_*^{\mathrm{ad}} Lc_Y^* M$$

is an equivalence in $\mathrm{D}^b_{\mathrm{coh}}(\mathcal{X})$.

Proof. The global comparison morphism is the mate of the counit

$$L(f^{\mathrm{ad}})^* Lc_X^* Rf_* M \simeq Lc_Y^* Lf^* Rf_* M \longrightarrow Lc_Y^* M.$$

Let $i : V = \mathrm{Spa}(R, R^+) \hookrightarrow \mathcal{X}$ be a member of $\mathrm{RatAff}_X(\mathcal{X})$, contained in $\lambda^{-1}(W)$ for a standard affine $W = \operatorname{Spec} A \subset X$, and form

$$Y_R = Y \times_X \operatorname{Spec} R, \qquad p : Y_R \longrightarrow Y, \qquad M_R = Lp^* M.$$

The ring map $A \to R$ is flat by Lemma 2.9. Algebraic flat base change gives

$$i^* Lc_X^* Rf_* M \simeq Lc_V^* R(f_R)_* M_R. \tag{34}$$

Proposition 6.12 identifies the right-hand side with

$$R(f_R^{\mathrm{ad}})_* Lc^*_{Y_R/V} M_R.$$

On the other hand, Lemma 6.11, applied to the inverse-image open of V in $Y^{\mathrm{ad}}_{\mathcal{X}}$, identifies this complex with

$$i^* Rf_*^{\mathrm{ad}} Lc_Y^* M. \tag{35}$$

The equivalences (34)–(35) identify the restriction of the canonical global comparison morphism with the canonical affinoid comparison of Proposition 6.12: all maps are obtained from the same algebraic counit by algebraic flat base change and ordinary open base change.

Choose a finite rational-affinoid cover $(V_i)_{i=1}^r$ of $\mathcal{X}$. By Proposition 6.12, the restriction of $Rf_*^{\mathrm{ad}} Lc_Y^* M$ to V_i lies in $\mathrm{D}^b_{\mathrm{coh}}(V_i)$; choose an interval $[a_i, b_i]$ containing its cohomological amplitude. With

$$a = \min_i a_i, \qquad b = \max_i b_i,$$

the cohomology sheaves of the global complex vanish outside $[a, b]$, because this can be checked on the jointly covering family (V_i). For $a \le j \le b$, the restriction of $\mathcal{H}^j(Rf_*^{\mathrm{ad}} Lc_Y^* M)$ to every V_i is finite, and the overlap identifications are the restrictions of one global sheaf. Coherence is local on a locally noetherian adic space, so these cohomology sheaves are coherent. Hence $Rf_*^{\mathrm{ad}} Lc_Y^* M$ belongs to $\mathrm{D}^b_{\mathrm{coh}}(\mathcal{X})$.

Let Q be the cofiber of the canonical global comparison morphism. Equations (34)–(35) show that $Q|_{V_i} = 0$ for every i. Derived sheaf descent identifies Q with the limit of its restrictions to the Čech nerve of (V_i); all terms of that diagram are zero. Therefore $Q = 0$, so the canonical global comparison is an equivalence in $\mathrm{D}^b_{\mathrm{coh}}(\mathcal{X})$. □

6.4. **Proof of the classical statements.**

Proof of Theorem 1.3. Corollary 6.10 gives the coherent, bounded-derived, perfect, and vector-bundle equivalences asserted in the theorem.

Apply Proposition 6.13 to $M = \mathcal{F}[0]$. Taking coherent cohomology sheaves gives

$$(R^i f_* \mathcal{F})^{\mathrm{ad}} \xrightarrow{\sim} R^i f_*^{\mathrm{ad}}(\mathcal{F}^{\mathrm{ad}}).$$

By Lemma 5.3 and the finite cohomological-dimension bound in its proof, $Rf_*\mathcal{F}$ lies in $\mathrm{D}^b_{\mathrm{coh}}(X)$. Hence the bounded-complex form of Kedlaya–Liu's coherent hypercohomology comparison applies. Finally,

$$\begin{aligned} R\Gamma(Y, \mathcal{F}) &\simeq R\Gamma(X, Rf_*\mathcal{F}) \\ &\simeq R\Gamma_{\mathrm{ad}}(\mathcal{X}, (Rf_*\mathcal{F})^{\mathrm{ad}}) \\ &\simeq R\Gamma_{\mathrm{ad}}(\mathcal{X}, Rf_*^{\mathrm{ad}}\mathcal{F}^{\mathrm{ad}}) \\ &\simeq R\Gamma_{\mathrm{ad}}(Y_{\mathcal{X}}^{\mathrm{ad}}, \mathcal{F}^{\mathrm{ad}}). \end{aligned}$$

The second equivalence is Kedlaya–Liu coherent cohomology comparison on the base curve, and the third is Proposition 6.13. Taking cohomology proves the last assertion. □

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